

The higher-dimensional Erdős distinct-distances conjecture

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Abstract

For every fixed integer $d \geq 3$, we prove that every set of $n \geq 2$ distinct points in $\mathbb{R}^d$ determines at least $c_d n^{2/d}$ distinct distances, where $c_d > 0$ depends only on d . This resolves the higher-dimensional Erdős distinct-distances conjecture positively.

1 Introduction

For a finite set $P \subset \mathbb{R}^d$, write

$$\Delta(P) = \{|p - q| : p, q \in P, p \neq q\}.$$

The distinct-distances problem asks how small $|\Delta(P)|$ can be among all sets of a prescribed cardinality. Erdős introduced the problem in 1946 [13]. It relates a simple metric statistic to the geometric structure of a finite set: when distances are few, many pairs repeat their lengths, but the arrangement of those repetitions is constrained by Euclidean geometry. The higher-dimensional conjecture predicts a constant multiple of $n^{2/d}$ distances among any n points of $\mathbb{R}^d$, for every fixed $d \geq 3$; see, for example, [4, 26]. We prove this prediction.

Theorem 1.1. *For every integer $d \geq 3$, there is a constant $c_d > 0$ such that every finite set $P \subset \mathbb{R}^d$ of $n \geq 2$ distinct points satisfies*

$$|\Delta(P)| \geq c_d n^{2/d}.$$

This resolves the higher-dimensional Erdős distinct-distances conjecture positively. No assumption is made on the position, spacing, or concentration of the points. The power of n is optimal: the integer grid $\{1, \dots, t\}^d$ has t^d points, and its squared distances are integers between 1 and $d(t-1)^2$. In the plane, the square-grid construction gives $O(n/\sqrt{\log n})$ distances [13]; the planar problem consequently has a different conjectural order.

1.1 Historical context and the methods of the proof

Higher-dimensional distance estimates developed in part through incidence bounds for curves and spheres. Clarkson, Edelsbrunner, Guibas, Sharir, and Welzl bounded the number of pairs realizing a fixed positive distance in three dimensions, giving a distinct-distance lower bound at the square-root scale, up to a slowly growing factor [9]. Aronov, Pach, Sharir, and Tardos improved the three-dimensional lower bound to $\Omega_\varepsilon(n^{77/141-\varepsilon})$ for each $\varepsilon > 0$ and propagated improvements to higher dimensions [3]. Their three-dimensional estimate is *pinned*: it finds a point of the given set from which that many distinct distances occur.

Solymosi and Vu developed two recursive inequalities connecting distance bounds in lower and higher dimensions [21, Theorems 2.1–2.2]. The recurrences have separate forms for the total number

of distances and for the maximum number measured from one point of the set. We use the forms for total distances in the comparisons below. Their original theorem gives $\Omega(n^{0.5643})$ distances in dimension three and $\Omega_d(n^{2/d-2/(d(d+2))})$ for every fixed $d \geq 4$ [21, Theorem 1.1].

A different approach counts repeated distances through rigid motions matching subsets of the point sets. Elekes and Sharir developed this viewpoint for the planar problem [11]. Guth and Katz reduced the resulting planar motion count to incidences of points and lines in three-dimensional space and combined polynomial partitioning with ruled-surface geometry to prove the planar bound $\Omega(n/\log n)$ [16, Theorem 1.1 and Sections 2–4]. That bound supplies the two-dimensional input to our induction. Bardwell-Evans and Sheffer extended the rigid-motion approach to higher dimensions through a reduction to incidences of $(d-1)$ -flats in $\mathbb{R}^{2d-1}$ [4]. Their work identifies structural restrictions on the resulting flats and the need to control their concentration in algebraic varieties.

Combining the Guth–Katz planar bound with the first Solymosi–Vu recurrence gives $\Omega(n^{3/5}/(\log n)^{2/5})$ distances in dimension three [21, Theorem 2.1]. More recently, Tidor, Yu, and Zakharov proved $n^{2/3-o(1)}$ distinct distances for arbitrary point sets in $\mathbb{R}^3$ [26, Theorem 1.1]. Inserting their three-dimensional distance bound into the Solymosi–Vu recurrences gives $n^{8/17-o(1)}$ in dimension four and $n^{3/8-o(1)}$ in dimension five [21, Corollaries 2.3 and 2.6]. The $o(1)$ terms here absorb the subpolynomial loss in that input. The leading exponents in these two deductions still fall below the conjectured value $2/d$. Bounds for points on prescribed surfaces, or for distances from a single point, have different hypotheses or conclusions.¹

Tidor, Yu, and Zakharov’s proof develops rigid-motion flat geometry, regular approximate complete intersections, and degree-sensitive incidence estimates. These methods provide close antecedents for several parts of the argument below: the physical and opposite-orientation flat families, the recovery of actual families of contained flats from tangent data, and the geometric treatment of projected curves [26, Sections 3–6 and 10.2–10.3].

The algebraic argument draws on Hilbert function estimates of Chardin and Chardin–Philippon [6, 7, 8], and on Walsh’s work on approximate complete intersections and algebraic concentration [27, 28]. Hilbert functions measure how many independent polynomial conditions a set can impose. Approximate complete intersections cut out an irreducible variety as a component of the common zero set of equations whose product of degrees is controlled by the variety’s degree. We use these two ideas to keep track of concentration at several polynomial degree scales. The uniform concentration theorem for evaluation flats of skew forms, including its exceptional parameter families, is proved in full in Appendix A. The real incidence arguments, including the planar distinct-distance bound, are supplied in Appendix B; their relation to the work of Szemerédi–Trotter, Beck, Elekes–Sharir, and Guth–Katz is recorded where each argument is used.

1.2 Why the constant factor matters

An estimate $|\Delta(P)| \geq n^{2/d-o(1)}$ does not by itself imply Theorem 1.1. The ratio $|\Delta(P)|/n^{2/d}$ could still tend to zero more slowly than every negative power of n . To obtain a constant factor, the proof must exclude every such sequence, without first fixing a power saving.

We therefore argue in the least dimension $d \geq 3$ in which the theorem could fail, and use the notation

$$N = |P|, \quad B = N^{1/d}, \quad A = B^2, \quad M = 1 + |\Delta(P)|.$$

Failure would give a sequence with $N \rightarrow \infty$ and $M/A \rightarrow 0$. The lower-dimensional bounds limit the number of points in any proper affine flat. They do not prevent substantial concentration on more

¹An earlier manuscript of Aksoy Yazici claimed the constant-factor bound in all dimensions $d \geq 3$, but was withdrawn by the author because the proof of its Theorem 1.2 is not correct [2].

general varieties, nor do they identify a single polynomial degree at which all of that concentration can be studied. The multiscale argument below addresses this second difficulty.

1.3 Overview of the proof

1. Select directions that admit interpolation. For distinct points p, q , let $\pi_p(q) = [q - p] \in \mathbb{P}^{d-1}$ denote their projective direction. We seek a positive fraction of the unordered pairs of points such that, at each center p , homogeneous polynomials of degree $\lfloor A \rfloor$ can prescribe arbitrary values at the selected displacement vectors $q - p$. Equivalently, the corresponding evaluation vectors are linearly independent. The selection must be symmetric: a pair is retained or removed at both endpoints.

The principal geometric obstruction would be a collection C_p of projective curves at each center, each curve spanning at most a projective plane, with total degree $o(N/A)$, such that almost every pair $\{p, q\}$ satisfies

$$\pi_p(q) \in C_p \quad \text{or} \quad \pi_q(p) \in C_q.$$

The union of the corresponding lines through p is a cone; the small total degree explains the term *sparse cones*. Theorem 6.1 rules out this coverage, including the choice of either endpoint for each pair. Once it is excluded, greedy removal of curves containing too many directions leaves a positive fraction of pairs with controlled concentration on every projective variety. A maximal symmetric selection then gives the interpolation property (Propositions 2.2 and 2.3). This last linear-algebra argument costs only a fixed fraction of pairs.

2. Exclude sparse cones at all degree scales. The sparse-cones theorem occupies the central part of the paper. Section 5 divides the point set according to how successive polynomial equations cut down the dimension of the varieties containing its points. It tests degrees in windows $[Be^{-s}, Be^s]$ with several width parameters s , instead of fixing a single power of N . Points isolated by equations of sufficiently small degree form a core that is handled directly. Each remaining piece carries a fixed chain of proper hypersurface cuts, ending in isolated points. A cut is proper when its equation does not vanish identically on the component being cut. The testing degrees at these cuts form a *scale profile*. Between consecutive cuts, the dimension stays constant, and Hilbert-function estimates give a lower bound on the growth of polynomial restriction rank. Bézout’s theorem bounds the degrees and point counts of the intervening components.

The comparison uses labels $(p, \pi_p(q))$, which record both the center and the direction of an ordered pair. At each fixed center, sparse-cone coverage puts the directions on curves of small total degree, limiting their polynomial rank. The profiles give an incompatible rank lower bound for a positive fraction of pairs. A complication is that radial projection can identify several components, or several points on one component. Section 4 controls this behavior by secant geometry. For separated windows, Subsection 6.3 bounds the resulting loss of degree at the actual centers.

Two limiting cases leave no strict margin in the general rank comparison. At the outermost degree window, common equations with independent differentials on most components allow a direct geometric comparison of their projected images. At the last one-dimensional stage for narrower center windows, the argument identifies the projection of the assigned terminal curve with a curve in the covering family. A global estimate has already removed all such incidences at a cost of $o(N^2)$ pairs (Proposition 4.3). These two arguments in Section 6 supply the endpoint control needed for an arbitrarily slow limit $M/A \rightarrow 0$.

3. Turn equal distances into intersections of flats. Take a generic rotated copy Q of P and transport the selected pairs to Q . For $(p, q) \in P \times Q$, consider the affine flat

$$F_{p,q} = \{(Z, c) : Z(p+q)/2 + c = (p-q)/2\}, \quad Z^\top = -Z, \quad c \in \mathbb{C}^d.$$

The coordinates (Z, c) identify with skew forms on $\mathbb{C}^{d+1}$ by adjoining c as the last column and $-c^\top$ as the last row, with last diagonal entry zero. Join (p, q) to (p', q') when both physical pairs have been selected in Step 1 and $|p - p'| = |q - q'|$. Cauchy–Schwarz gives $\Omega(N^4/M)$ edges. On $F_{p,q}$, the intersection with a neighbor is the flat defined by

$$Z(u+w) = u-w, \quad u = p' - p, \quad w = q' - q.$$

The interpolation property ensures that these local flats are distinct and that their parameters have polynomial separators of degree $O(A)$: one parameter can be distinguished from all the others by a polynomial of that degree.

Appendix A proves concentration bounds for these evaluation flats, both in the full skew-form space and inside one fixed pair flat. In the local application an exceptional parameter family corresponds to many matches under one rigid motion. For $d \geq 4$, its required size bound follows from the total number of points. For $d = 3$, Section 7 bounds $\sum_g |gP \cap Q|^2$ over motions with at least CA matches by $O(N^4/A)$, with both orientations included. Removing the corresponding edges costs $o(N^4/M)$. After this removal and pruning vertices of small degree, the minimum number of neighbors, divided by A^{d-1} , tends to infinity. The local concentration bound says that a nonzero polynomial of degree at most A on a pair flat can vanish on at most $O(A^{d-1})$ of its neighbor flats.

4. Sample once and compare polynomial ranks. The remaining contradiction is short. Sample the retained pair flats independently with a small fixed probability. With positive probability, the sample is small enough that the global concentration bound forces the whole family to impose more independent polynomial conditions in degree $\lfloor A \rfloor$ than the sample. Simultaneously, every retained flat has more than CA^{d-1} sampled neighbors. Thus a polynomial of degree at most $\lfloor A \rfloor$ vanishes on every sampled flat but is nonzero on at least one retained flat F . The restriction of the polynomial to F then violates local concentration. The sample uses one fixed probability, and all concentration constants are uniform; this is where the preceding quantitative estimates produce the positive constant in Theorem 1.1.

Organization and prerequisites. Section 2 carries out Steps 1, 3, and 4, assuming the sparse-cones, concentration, and rich-motion theorems. It can therefore be read first as the formal assembly of the argument. Section 3 proves the polynomial interpolation, regularity, and multiplicity estimates. Sections 4 through 6 establish the geometric and multiscale arguments of Step 2. Section 7 proves the rich-motion estimate, and Appendices A and B provide the concentration and real incidence foundations. Figure 1 summarizes the principal dependencies. We use characteristic-zero algebraic geometry, including projective degree and Bézout’s theorem, generic linear sections, regular local rings and Koszul resolutions, and projective-bundle intersection formulas. The precise forms needed are specified where they are used. Constants may depend on the fixed ambient dimension; any additional dependence is indicated explicitly.

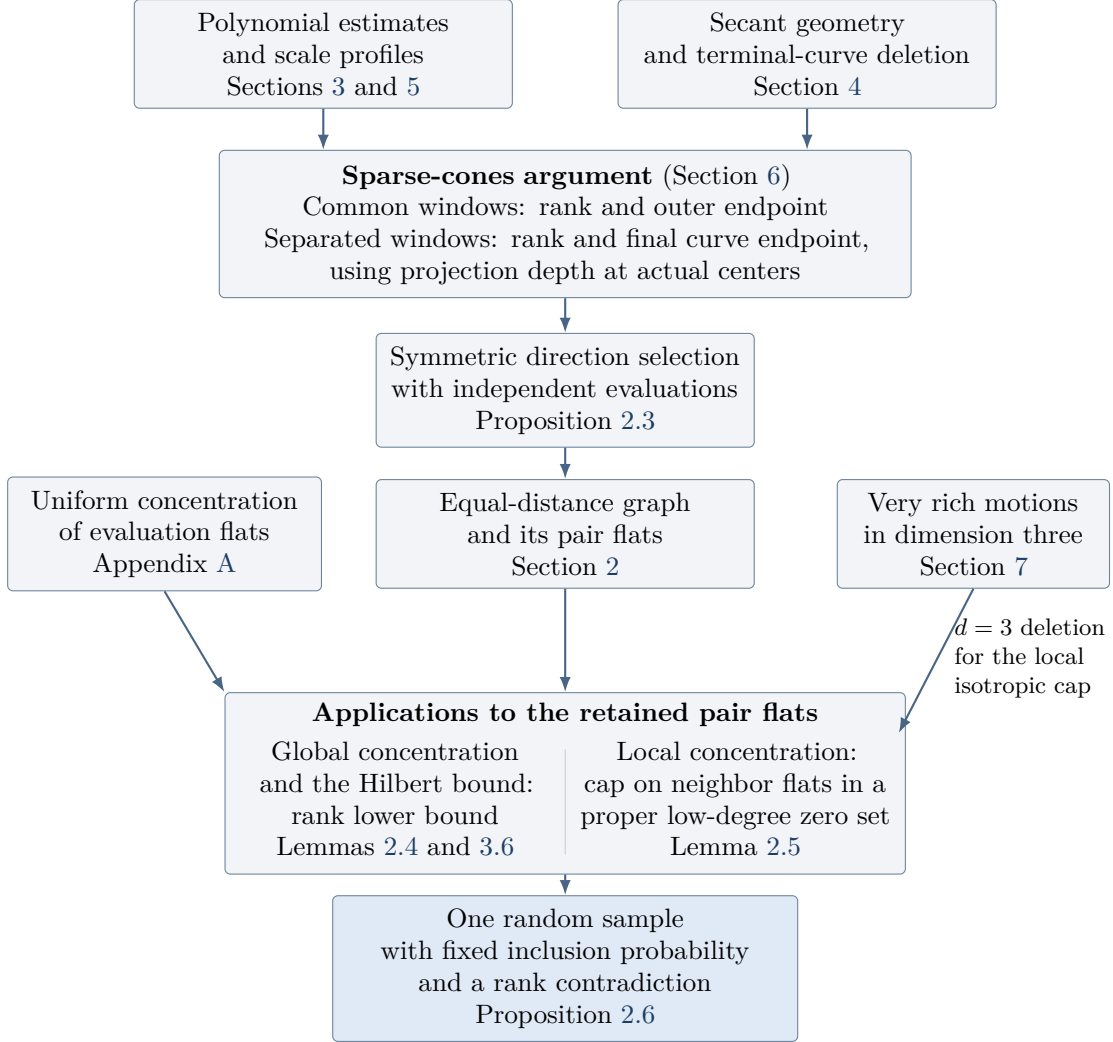


Figure 1: Principal dependencies. Both endpoint arguments belong to the sparse-cones theorem. The separated-window argument uses projection depth at actual centers. Global concentration combines with the component Hilbert bound; the rich-motion bound permits the $d = 3$ edge deletion that supplies the local isotropic cap. The contradiction uses one sample with a fixed inclusion probability. Appendix B supplies the planar induction base and the real incidence inputs to the geometric and motion arguments.

Contents

1	Introduction	1
1.1	Historical context and the methods of the proof	1
1.2	Why the constant factor matters	2
1.3	Overview of the proof	3
2	The reduction to sparse cones and concentration	7
2.1	The contradiction regime	7
2.2	A symmetric selection of directions	8
2.3	The equal-distance graph and its pair flats	10
2.4	Global and local concentration	12
2.5	One random sample completes the proof	13
3	Polynomial interpolation, regularity, and multiplicity	14
3.1	Separators supplied by the distance set	15
3.2	Interpolation at isolated points	15
3.3	Component Hilbert ranks	17
3.4	Common regular equations and pooled proper cuts	20
3.5	Dividing a degree bound by a multiplicity	22
3.6	Projection degrees and local multiplicities	23
4	Secants and terminal curves	27
4.1	The secant tangent calculation	27
4.2	A generic count for families of carriers	28
4.3	Deletion of assigned terminal curves	30
5	Polynomial scale profiles	33
5.1	Conventions and the partition	34
5.2	Degree budgets and propagation of rank	37
5.3	Testing individual components and products	39
5.4	Selecting the two windows	41
6	The sparse-cones assertion	42
6.1	Product images and rank in a common window	43
6.2	The outermost endpoint	47
6.3	Projection depth for separated windows	51
6.4	Rank propagation for separated windows	57
6.5	The final curve plateau for narrower centers	59
7	Very rich rigid motions in three dimensions	61
7.1	Two elementary consequences of planar incidence bounds	62
7.2	A local bound for motions without a rich matched line	64
7.3	A spatial subdivision with uniform overlap	68
7.4	Counting prescribed images of rich lines	70
7.5	Proof of the very-rich-motion estimate	71

A	Uniform concentration of evaluation flats	74
A.1	Approximate complete intersections	75
A.2	Separators and regular equations	78
A.3	Recovering actual rulings from tangent directions	80
A.4	Degree bounds and exceptional parameter families	82
A.5	Closing the estimates with uniform constants	88
B	Real incidence estimates	90
B.1	The planar incidence bound	91
B.2	Polynomial bisection and flat lines	92
B.3	The higher-richness line theorem	93
B.4	A ruled-surface criterion with a degree bound	95
B.5	Intersections among generators	97
B.6	The two-rich line theorem	99
B.7	The planar distinct-distance estimate	100

2 The reduction to sparse cones and concentration

This section proves Theorem 1.1 from the sparse-cones theorem, the rich-motion theorem, and the uniform concentration theorem. Their proofs occupy the later sections and appendices. We include the selection and sampling arguments here, since keeping their constants uniform is essential.

2.1 The contradiction regime

Suppose that Theorem 1.1 fails, and take the least dimension $d \geq 3$ in which it fails. There is then a sequence of finite sets $P \subset \mathbb{R}^d$ for which, with

$$N = |P|, \quad M = 1 + |\Delta(P)|, \quad B = N^{1/d}, \quad A = B^2, \quad N \rightarrow \infty, \quad M/A \rightarrow 0. \quad (2.1)$$

Here and below a statement along the sequence is understood after discarding finitely many terms, if necessary. The fact that $N \rightarrow \infty$ follows because a set of at least two points determines at least one distance. Adding 1 to the number of distances does not affect the vanishing ratio.

For $3 \leq u < d$, the assertion in dimension u is available by minimality of d . We also use the planar result proved in Theorem B.12. These give the following bounds.

Lemma 2.1 (Occupancy of proper flats). *In the regime (2.1), a real affine u -flat contains at most*

$$\begin{cases} 1, & u = 0, \\ 2M, & u = 1, \\ CM \log(2M), & u = 2, \\ C_u M^{u/2}, & 3 \leq u < d \end{cases}$$

points of P . A circle contains at most $2M$ points. In particular there is a fixed $c_0 > 0$ such that every proper real affine flat contains at most NB^{-c_0} points for all sufficiently large terms of the sequence. The same assertion holds for a proper complex affine flat when only its real sample points are counted. Planes have the more precise bound $B^{2+o(1)}$.

Proof. From a point on a line, any positive distance has at most two realizations on that line. For a circle, use a sample point on the circle: the intersection with a circle of prescribed positive radius about that point has at most two points. These observations include the sample point itself after using $M = 1 + |\Delta(P)|$.

The planar bound is Corollary B.13. The other bounds follow by applying the lower-dimensional assertion to the points in the flat, whose pairwise distances are among the distances of P . Since $M \leq B^2$, the planar bound is $O(B^2 \log B)$, while a proper flat of dimension $u \geq 3$ contains $O(B^u)$ points. Thus, for example, $c_0 = 1/2$ works after advancing along the sequence. A complex affine flat containing real points contains their real affine span, whose real dimension is at most its complex dimension. Applying the real bound to that span proves the last assertion. $\square$

We use complex varieties for polynomial arguments. Degree means the degree of the projective closure; for a family of components of a fixed dimension it means the sum of their degrees. A projective direction from p is denoted by

$$\pi_p(q) = [q - p] \in \mathbb{P}^{d-1}.$$

Restriction of polynomials of degree at most D to an affine set T has rank $H_T(D)$. In projective space we use homogeneous polynomials of degree exactly D . The degree- D envelope $\text{Env}_D(T)$ is the common zero locus of all such polynomials vanishing on T . Its rank at that degree equals $H_T(D)$.

Here is the form of the sparse-cones theorem that we will use.

In the regime (2.1), one cannot assign to each $p \in P$ a union C_p of irreducible projective curves, each spanning a projective space of dimension at most two, with

$$\deg C_p \leq \delta N/A, \quad \delta \rightarrow 0,$$

such that all but $o(N^2)$ unordered pairs $\{p, q\}$ satisfy $\pi_p(q) \in C_p$ or $\pi_q(p) \in C_q$.

This is Theorem 6.1. Both the alternative between the two endpoints and the absence of a prescribed rate for δ are important.

2.2 A symmetric selection of directions

Proposition 2.2. *After passing to a subsequence, there is an undirected graph $\mathcal{G}$ on P with at least cN^2 edges such that, for every vertex p and every irreducible projective variety $Y \subset \mathbb{P}^{d-1}$ of dimension a ,*

$$\#\{q : \{p, q\} \in \mathcal{G}, \pi_p(q) \in Y\} \leq C \deg(Y) A^a. \quad (2.2)$$

The positive constants c, C are fixed along the subsequence.

Proof. First discard pairs on lines containing more than H sample points, where H will be a sufficiently large fixed constant. A generic planar projection preserves the relevant distinct points, lines, and incidences. The rich-line consequence of Theorem B.2 therefore bounds the number of physical lines with at least t points by

$$C(N^2/t^3 + N/t), \quad t \geq 2.$$

If $L_{\max} = o(N)$ is the maximum occupancy, a dyadic sum gives

$$\#\{\text{discarded pairs}\} \leq CN^2/H + CNL_{\max} = O(N^2/H) + o(N^2).$$

At each p , independently of the other stars, greedily remove directions on any irreducible curve spanning at most a projective plane that currently contains more than KA times its degree many partners. Each removal contains at least one previously unremoved partner, so the process terminates. Record the curves chosen in that star. Their total degree is at most $N/(KA)$, because their charged partner sets are disjoint. Now keep an unordered pair only when it survived both endpoint procedures.

There must be fixed H, K and a subsequence on which a positive fraction of pairs survives. Otherwise, a diagonal choice with $H, K \rightarrow \infty$ sufficiently slowly would make the surviving fraction tend to zero, while the first, rich-line discard also has vanishing fraction. The recorded curves would then cover all but $o(N^2)$ pairs from one endpoint or the other and have total degree at most $N/(KA) = o(N/A)$ in every star. This contradicts Theorem 6.1. Fix such H, K and a positive-density subsequence.

We prove (2.2) by induction on a . For $a = 0$ it is the light-line bound. Fix a variety Y of dimension $a \geq 1$, a center p , and the selected partners counted on Y . Let U be the real linear span of their displacements from p , and put $\ell = \dim U$. There is nothing to prove if this collection is empty.

Consider a full radius class about p , meaning all points of P at one prescribed distance from p . Suppose it has $L \geq 2M$ distinct orthogonal projections to U , and choose physical points $z_1, \dots, z_L \in P$ in that class representing those projections. For each counted partner q , the distances $|q - z_i|$, including zero when it occurs, take at most M values. Consequently at least

$$L^2/M - L \geq L^2/(2M)$$

ordered pairs $i \neq j$ have equal distance to q . As the z_i lie in a common radius class,

$$|q - z_i| = |q - z_j| \implies (q - p) \cdot (z_i - z_j) = 0.$$

The projection of $z_i - z_j$ to U is nonzero. The resulting hyperplane section is proper on Y : if its linear equation vanished on Y , it would vanish on all counted displacements and hence on U , which is impossible for its nonzero normal in U . Bézout's theorem and the induction hypothesis bound the number of partners on each section by $C \deg(Y) A^{a-1}$. Double counting gives

$$\#\{q \text{ counted}\} \frac{L^2}{2M} \leq CL^2 \deg(Y) A^{a-1}.$$

This is (2.2), since $M \leq A$.

In the remaining case, each of the at most M radius classes has fewer than $2M$ projections to U . Thus the full sample has $O(M^2)$ fibers on which both the radius about p and the projection to U are fixed. A fiber lies on a sphere in an affine space of dimension $d - \ell$. If $\ell \geq 4$ and $d - \ell \geq 3$, lower-dimensional induction bounds each fiber by $O(M^{(d-\ell)/2})$. If $d - \ell = 2$, the circle bound is $O(M)$; if $d - \ell \leq 1$, the fiber has at most two points. In all cases the total is $O(M^{d/2})$, contradicting $N/M^{d/2} \rightarrow \infty$. Thus this alternative is impossible when $\ell \geq 4$.

It remains to consider $\ell \leq 3$. For $a \geq 2$, Lemma 2.1 bounds the counted partners by $O(A^2)$, which is sufficient. If the span is the entire space, then $d = 3$ and the trivial bound $N = A^{3/2}$ is sufficient instead. For $a = 1$, either $Y \subset \mathbb{P}(U_{\mathbb{C}})$, in which case the greedy curve bound applies, or its intersection with that projective subspace is proper and consists of at most $\deg Y$ points. The light-line bound handles the latter case. This completes the induction. Fixed increases in the constants at the finitely many dimensions are harmless. $\square$

Proposition 2.3 (Independent evaluations in every star). *A further symmetric selection retains at least $c'N^2$ pairs and has the following property: the direction-evaluation vectors of the partners in each star are linearly independent for homogeneous polynomials of degree $D = \lfloor A \rfloor$. In particular each selected star admits homogeneous separators of degree D .*

Proof. First consider any subset S of the partners in one star, counting physical partners even when their projective directions agree. Form the projective envelope of those directions at degree D . Assign each direction to an irreducible component containing it, and group these components by dimension. For a group of total degree a_j and dimension j , Proposition 2.2 bounds the assigned physical partners by $C a_j A^j$. The component Hilbert lower bound in Lemma 3.6, applied at the defining degree D , gives

$$H_S(D) = H_{\text{Env}_D(S)}(D) \geq c_d a_j D^j.$$

There are at most d groups, and $D \asymp A$. Therefore

$$H_S(D) \geq c_1 |S| \tag{2.3}$$

with a fixed $c_1 > 0$, uniformly for every subset S .

Take a maximal undirected subset of the edges from Proposition 2.2 whose evaluations are independent at both endpoints. Each omitted edge is blocked at an endpoint: its evaluation vector lies in the span of the selected vectors there. Charge it to one such endpoint. If the selected degree at p is s_p , all vectors charged to p together have rank at most s_p . Equation (2.3) bounds their number by s_p/c_1 . Summing over vertices bounds the number of omitted edges by $2/c_1$ times the number of selected edges. Positive density follows.

Independent evaluation vectors mean that the evaluation map onto the coordinates indexed by the star is surjective. Taking the coordinate basis in its image gives the asserted homogeneous separators. $\square$

Write $\mathcal{S}$ for this final selection. A selected star has at most one partner in any projective direction, since two proportional evaluation vectors could not be independent.

2.3 The equal-distance graph and its pair flats

Take $Q = RP$, where R is a sufficiently generic real rotation, and transport $\mathcal{S}$ to Q . On the N^2 vertices $(p, q) \in P \times Q$, put an edge between (p, q) and (p', q') exactly when

$$\{p, p'\} \in \mathcal{S}, \quad \{q, q'\} \in R\mathcal{S}, \quad |p - p'| = |q - q'|.$$

If s_ρ is the number of selected physical pairs of length ρ , the number of these edges is $2 \sum_\rho s_\rho^2$. Cauchy–Schwarz gives

$$2 \sum_\rho s_\rho^2 \geq \frac{2|\mathcal{S}|^2}{M} \geq c_2 N^4 / M. \tag{2.4}$$

For $d = 3$, let $g(x) = Sx + h$, with $S \in O(3)$ and $h \in \mathbb{R}^3$, range over spatial isometries, and define its set of matched graph vertices by

$$V_g = \{(p, g(p)) : p \in P, g(p) \in Q\}, \quad k_g = |V_g|.$$

The map $g \mapsto R^{-1} \circ g$ preserves match counts and identifies these motions with motions from P to itself. Since $M \leq A$ eventually, Theorem 7.1 shows that the motions with $k_g \geq C_{\text{rich}} A$ form a finite

family, with both orientations included. Delete every existing graph edge whose endpoints both belong to one of these sets V_g . The number of deleted edges is at most

$$\sum_{k_g \geq C_{\text{rich}} A} \binom{k_g}{2} \leq \frac{1}{2} \sum_{k_g \geq C_{\text{rich}} A} k_g^2 \leq \frac{K_{\text{rich}}}{2} \frac{N^4}{A} = o(N^4/M).$$

Overlapping sets V_g can only decrease the actual deletion count. After this deletion, no vertex can have at least $C_{\text{rich}} A$ remaining neighbors which, together with that vertex, are matches of one spatial isometry. Such a motion would have at least $C_{\text{rich}} A$ matches, and all those incident edges would have been deleted.

In every dimension, let E_0 be the edge count at the start of pruning, and repeatedly delete vertices of degree below $E_0/(2N^2)$. The total number of deleted edges is less than $E_0/2$, so at least one edge survives. The resulting nonempty graph $\mathcal{H}$ has minimum degree

$$\delta(\mathcal{H}) \geq c_3 N^2/M, \quad \delta(\mathcal{H})/A^{d-1} \longrightarrow \infty. \quad (2.5)$$

Only subsets of the original stars have been retained, so their independence and separators remain valid.

Identify skew forms on $\mathbb{C}^{d+1}$ with matrices

$$\begin{pmatrix} Z & c \\ -c^\top & 0 \end{pmatrix}, \quad Z^\top = -Z, \quad c \in \mathbb{C}^d.$$

The pair (p, q) defines an affine flat

$$F_{p,q} = \{(Z, c) : Zv_0 + c = y_0\}, \quad v_0 = (p+q)/2, \quad y_0 = (p-q)/2. \quad (2.6)$$

Its dimension is $s_0 = d(d-1)/2$. Distinct pairs give distinct flats: at $Z = 0$ one recovers y_0 , and equality for every skew Z then recovers v_0 .

Each $F_{p,q}$ is the complexification of the corresponding real point-pair flat in the construction of Bardwell-Evans and Sheffer before their generic section [4, Theorem 6.1].

The parameter in the split quadric of $\mathbb{P}(\mathbb{C}^{d+1} \oplus (\mathbb{C}^{d+1})^*)$ is

$$[v : y] = [(v_0, 1) : (y_0, -v_0 \cdot y_0)]. \quad (2.7)$$

Indeed, on $F_{p,q}$,

$$\begin{pmatrix} Z & c \\ -c^\top & 0 \end{pmatrix} \begin{pmatrix} v_0 \\ 1 \end{pmatrix} = \begin{pmatrix} Zv_0 + c \\ -c \cdot v_0 \end{pmatrix} = \begin{pmatrix} y_0 \\ -v_0 \cdot y_0 \end{pmatrix},$$

where the last coordinate uses $v_0 \cdot Zv_0 = 0$. Conversely, the first d coordinates of this evaluation equation give (2.6). Thus $F_{p,q}$ is exactly the evaluation flat for this parameter. This parameter satisfies $y(v) = 0$ and $v \neq 0$. We next check every hypothesis needed for the two different versions of Theorem A.1.

2.4 Global and local concentration

Lemma 2.4 (Global concentration). *For the retained pair flats, an irreducible variety V of dimension $s_0 + h$ in skew-form space contains at most*

$$C \deg(V) A^h$$

of the flats, for $0 \leq h \leq d$, where the endpoint $h = d$ denotes the ambient space.

Proof. In Theorem A.1, take $e = d + 1$, so $r = d$. Physical separators from Lemma 3.1, one in p and one in q , have a product of degree $O(M)$. Express them in the coordinates v_0, y_0 and homogenize with the last coordinate of v , which is 1 on every parameter. They give positive-degree homogeneous separators of degree $O(M) \leq O(A)$.

Suppose parameters lie in one projective maximal isotropic plane. Their original real representatives in (2.7) belong to its underlying isotropic vector space. Polarization gives, for any two of them,

$$(v_0 - v'_0) \cdot (y_0 - y'_0) = 0, \quad |p - p'|^2 = |q - q'|^2.$$

The first physical coordinate p is therefore injective on this set of parameters, because $p = p'$ forces $q = q'$. There are at most $N \leq A^{d-1}$ such parameters. The total number of parameters is at most $N^2 = A^d$. Thus Specification (a) of Theorem A.1 applies, with one fixed constant absorbing the separator degree.

For $h = 0$, an irreducible variety containing a same-dimensional flat equals that flat. For $h = d$, the total-count bound suffices. These give the two endpoints. $\square$

Fix a retained vertex (p, q) . On its flat, use Z as coordinates, so $c = y_0 - Zv_0$. A neighbor (p', q') , with $u = p' - p$ and $w = q' - q$, cuts out

$$Z(u + w) = u - w. \tag{2.8}$$

Here u, w are nonzero real vectors with $|u| = |w|$. The rotation R may be chosen to ensure $u + w \neq 0$ simultaneously for the finite set of relevant displacement pairs.

Lemma 2.5 (Local concentration). *Every nonzero polynomial of degree at most A on $F_{p,q}$ contains in its zero set at most CA^{d-1} of the neighbor flats in (2.8).*

Proof. The local skew-form model has $e = d$, $r = d - 1$, and parameters

$$[v : y] = [u + w : u - w].$$

They lie on the split quadric, with nonzero v . They are distinct. Indeed, proportional parameters give $(u', w') = \lambda(u, w)$. Since the vectors are real and nonzero, λ is real; uniqueness of a selected partner in its projective direction first gives $u' = u$, then $\lambda = 1$ and $w' = w$.

Multiply a homogeneous direction separator for u by one for w and substitute $u = (v + y)/2$, $w = (v - y)/2$. The product has degree $O(A)$ and separates the corresponding neighbor parameter from all others. No physical scale ambiguity remains, because each selected star contains at most one physical partner in each projective direction.

For parameters in a common maximal isotropic plane, polarization on the original real representatives gives

$$(u_i - w_i) \cdot (u_j + w_j) + (u_j - w_j) \cdot (u_i + w_i) = 0,$$

or equivalently

$$u_i \cdot u_j = w_i \cdot w_j.$$

The rule $u_i \mapsto w_i$ is a well-defined real isometry on their spans: every linear relation is preserved by equality of the Gram matrices and positive definiteness of the real norm. Extend it to some $S \in O(d)$. Every neighbor in this isotropic plane, together with the center, is a match for $x \mapsto q + S(x - p)$.

For $d \geq 4$, the number of such neighbors is at most $N = A^{d/2} \leq A^{d-2}$. For $d = 3$, a collection with at least $C_{\text{rich}}A$ neighbors would, together with the center, violate the survivor property established when the rich match edges were deleted. The surviving cap is therefore $O(A)$. This establishes precisely the isotropic hypothesis $O(A^{r-1})$ in every dimension. In particular no restriction to orientation-preserving motions was made.

Factor the polynomial on $F_{p,q}$ into irreducible hypersurfaces. Each component has excess dimension $h = d - 2 = r - 1$ above a neighbor flat, and lies in a hypersurface of degree at most A . Specification (b) of Theorem A.1 applies to it; that specification has no total-count hypothesis. Summing its bound $C \deg(V) A^{d-2}$ over components of total degree at most A proves the result. $\square$

2.5 One random sample completes the proof

Proposition 2.6. *The graph $\mathcal{H}$ cannot satisfy both bounds in Lemmas 2.4 and 2.5.*

Proof. Let $\mathcal{F}$ be the family of retained global pair flats and $m = |\mathcal{F}|$. At degree $D = \lfloor A \rfloor$, form the envelope of their union. Each flat is contained in one of its irreducible components. Assign it to one such component and group components by their dimensions $s_0 + h$, $0 \leq h \leq d$. If a group has total degree a_h , global concentration bounds its assigned flats by $Ca_h A^h$. The Hilbert lower bound at the defining degree gives

$$H_{\bigcup \mathcal{F}}(D) \geq ca_h D^{s_0+h}.$$

There are only $d + 1$ groups. Since $D \asymp A$,

$$H_{\bigcup \mathcal{F}}(D) \geq c_4 m A^{s_0}. \quad (2.9)$$

This includes the case in which the envelope is the entire ambient space.

Choose each flat independently with a small fixed probability $\theta > 0$. With probability at least $1/2$, the number selected is at most $2\theta m$, by Markov's inequality. On that event their restriction rank is at most

$$2\theta m \binom{D + s_0}{s_0} \leq C_4 \theta m A^{s_0}.$$

Choose θ so small that this is less than the right side of (2.9).

At any vertex the sampled neighbors have binomial distribution of mean at least $\theta \delta(\mathcal{H})$. The probability of fewer than half that mean is at most $\exp(-c\theta \delta(\mathcal{H}))$. There are at most N^2 vertices, and $\delta(\mathcal{H}) \gg A^{d-1}$, a fixed positive power of N . A union bound shows that, with probability tending to one, every vertex has at least $\theta \delta(\mathcal{H})/2 \gg A^{d-1}$ sampled neighbors. Both events thus hold for some choice of sample.

Its rank is strictly smaller than that of all flats. Hence a polynomial f of degree at most D vanishes on every sampled flat but does not vanish identically on some $F \in \mathcal{F}$. For each sampled neighbor F' , it vanishes on the neighbor flat $F \cap F'$ inside F . These neighbor flats are distinct by the local parameter check above. There are more than CA^{d-1} of them, contradicting Lemma 2.5 applied to the nonzero restriction $f|_F$. $\square$

Proposition 2.6, together with the three ingredients proved below, rules out (2.1) in the least possible failing dimension. Thus there is no failing dimension. Equivalently, in each fixed dimension the ratio $|\Delta(P)|/|P|^{2/d}$ has a positive infimum: if it did not, a sequence with ratio tending to zero would give (2.1). This proves Theorem 1.1, including its constant-factor strength.

3 Polynomial interpolation, regularity, and multiplicity

We record the algebraic facts used below, with their degree dependence. The ambient dimension in this section is denoted by n and is fixed. Constants written C_n depend only on n ; they may be increased between statements. Varieties are reduced unless a local quotient or an intersection scheme is explicitly specified. The degree of an affine variety is the degree of its projective closure. For a family of distinct irreducible components of one dimension, its degree is the sum of their degrees. Components are counted only once, even if they occur in more than one defining system.

The section supplies interpolation from defining equations, restriction-rank bounds from component degrees, common equations with controlled exceptional degree, and tests for high multiplicity and projection loss. We first prove isolated-point interpolation. Applying it on generic fibers will then give the rank and regularity estimates.

For a subset $T \subset \mathbb{C}^n$, put

$$H_T(D) = \dim_{\mathbb{C}}(\mathbb{C}[x_1, \dots, x_n]_{\leq D} / I(T)_{\leq D}), \quad D \in \mathbb{Z}_{\geq 0}.$$

Here $I(T)$ consists of the polynomials vanishing on T . The degree- D envelope is

$$\text{Env}_D(T) = \{x : f(x) = 0 \text{ for every } f \in I(T)_{\leq D}\}.$$

Its defining collection can be replaced by a finite basis of $I(T)_{\leq D}$. Directly from the definition,

$$H_{\text{Env}_D(T)}(D) = H_T(D). \quad (3.1)$$

For a projective set we use restrictions of homogeneous polynomials of exact degree D . Dehomogenizing in any affine chart meeting every component gives the same rank: homogenization identifies polynomials of degree at most D with homogeneous polynomials of degree D , and a polynomial vanishes on a component if and only if it vanishes on a dense affine part of that component. We always choose charts generically for the finitely many points and components under discussion.

We use the following forms of the elementary intersection facts. A proper hypersurface section of a pure k -dimensional variety of degree a , by a polynomial of degree q , has reduced support of pure dimension $k - 1$ and degree at most aq ; the section may be empty in an affine chart. Successive proper sections obey the corresponding product bound. More generally, the degrees, with local intersection multiplicities, of isolated components of an intersection are at most the Bézout product, even if the same equations have larger components elsewhere. One obtains this latter form by taking generic linear combinations locally at the isolated components, perturbing to proper intersections, and specializing: the isolated intersection multiplicities are positive and persist in the specialization, whereas intersections on other components can only add to the total. We will never replace an isolated component by a subvariety lying inside a larger component.

These intersection bounds are the componentwise refined Bézout inequality [14, Theorem 12.3 and Example 12.3.1]. For isolated complete intersections of hypersurfaces in a regular ambient space, intersection multiplicity equals local quotient length. The hypersurface factors are Cohen–Macaulay, as required in [14, Example 12.3.7(ii)–(iii)].

3.1 Separators supplied by the distance set

Lemma 3.1 (Physical separators and sample bounds). *Let $P \subset \mathbb{R}^n$ be finite and let $M = 1 + |\Delta(P)|$. Each $p \in P$ has a polynomial s_p of degree at most $2(M - 1)$ such that*

$$s_p(p) = 1, \quad s_p(q) = 0 \quad (q \in P \setminus \{p\}).$$

If $V \subset \mathbb{C}^n$ is a pure u -dimensional variety of degree a , then

$$|P \cap V| \leq C_n a M^u. \quad (3.2)$$

Moreover, P is precisely the common zero set of a system of polynomials of degree at most $2M + 1$. In particular, all its points are isolated components of that system.

Proof. For $p \in P$ define

$$s_p(x) = \prod_{\rho \in \Delta(P)} \frac{\sum_{j=1}^n (x_j - p_j)^2 - \rho^2}{-\rho^2}.$$

Every denominator is nonzero. The stated values on the real sample follow even though we subsequently regard s_p as a polynomial over $\mathbb{C}$.

We prove (3.2) by induction on u and add over the irreducible components of V . For $u = 0$ the number of points is at most a . Suppose V is irreducible of positive dimension and contains a sample point p . The polynomial s_p does not vanish identically on V , because it is nonzero at p , and it vanishes at every other sample point of V . Its proper section has degree at most $2Ma$. The induction hypothesis therefore gives

$$|P \cap V| \leq 1 + C_n(2Ma)M^{u-1}.$$

Increasing the dimension-dependent constant proves the claim. This argument requires no smoothness or general-position assumption on the sample points.

Finally use the polynomials

$$1 - \sum_{p \in P} s_p(x), \quad (x_j - p_j)s_p(x) \quad (p \in P, 1 \leq j \leq n). \quad (3.3)$$

They vanish at every point of P . At any common zero, the first equation forces some s_p to be nonzero, and the corresponding n coordinate equations then force $x = p$. Thus there are no additional zeros. $\square$

3.2 Interpolation at isolated points

The next lemma concerns the isolated points of a system, not a system whose entire zero set is necessarily finite. This distinction will be needed for envelopes and for intermediate cuts.

Lemma 3.2 (A determinant in a local parameter quotient). *Let F be an algebraically closed field, let $R = F[Z_1, \dots, Z_c]_{(Z_1, \dots, Z_c)}$, and suppose $f_1, \dots, f_c \in R$ generate an ideal of finite colength. If*

$$(f_1, \dots, f_c)^t = Q(Z)(Z_1, \dots, Z_c)^t,$$

then the class of $\det Q$ in $R/(f_1, \dots, f_c)$ is nonzero.

Proof. The coordinate parameters form a regular sequence. The regular local ring R is Cohen–Macaulay, so the c elements f_i , being a system of parameters, also form a regular sequence [25, Tags 00OQ, 00NQ, and 02JN]. Their Koszul complexes are therefore free resolutions of $R/(f)$ and $R/(Z) = F$, respectively [25, Tag 062F]. The matrix Q induces a chain map from the first Koszul resolution to the second, lifting the quotient map $R/(f) \rightarrow F$; its map in top degree is multiplication by $\det Q$.

For completeness, the vanishing needed to see that this top map is nonzero is especially simple. The coordinate Koszul resolution gives

$$\mathrm{Ext}_R^i(F, R) = 0 \quad (i < c), \quad \mathrm{Ext}_R^c(F, R) = F.$$

Every finite-length R -module has a composition series with factors F . The long exact sequence for Ext therefore gives $\mathrm{Ext}_R^i(K, R) = 0$ for every finite-length K and $i < c$. Apply this to the kernel K of $R/(f) \rightarrow F$. The long exact sequence supplies an injection

$$\mathrm{Ext}_R^c(F, R) \hookrightarrow \mathrm{Ext}_R^c(R/(f), R).$$

Under the identifications provided by the two Koszul resolutions, this map sends the class of 1 to the class of $\det Q$, up to an irrelevant sign. Its image is nonzero. $\square$

Lemma 3.3 (Isolated-point interpolation, including first derivatives). *Let F be algebraically closed, and let $S \subset F^c$ be any selection of distinct isolated points of the common zero set of polynomials of degree at most L , where $L \geq 1$. There are polynomials of degree at most $c(L - 1)$ separating the points of S . In fact a separator for one point can be chosen to vanish at every other common zero of the original system, including its positive-dimensional components.*

Arbitrary values and all first partial derivatives can be prescribed simultaneously on S by a polynomial of degree at most $C_c L$. Furthermore, S itself is the exact zero set of equations of degree at most $C_c L$.

Proof. Fix an isolated point z of the original system. Its local defining ideal is primary to the maximal ideal at z . We may choose c constant linear combinations $f_1, \dots, f_c$ of the given equations whose local ideal has finite colength. To see this, choose the combinations successively, avoiding the finitely many minimal primes of the preceding proper intersections other than the maximal ideal. The original ideal is not contained in any of those primes. Over an infinite field the forbidden coefficient vectors form a finite union of proper linear subspaces, so each choice is possible. Each f_i still has degree at most L . The same avoidance can be imposed simultaneously at any finite list of isolated points. Thus a generic choice of c combinations is a local parameter sequence at every point of that list, although the argument below only needs this property at z .

Introduce two sets of variables Y, Z . Telescoping one coordinate at a time gives a divided-difference matrix $Q(Y, Z)$ such that

$$f(Y) - f(Z) = Q(Y, Z)(Y - Z). \tag{3.4}$$

Every entry has degree at most $L - 1$ in Y , and hence $\det Q(Y, Z)$ has degree at most $c(L - 1)$ in Y . Take the coefficients in the local Artin algebra

$$A_z = F[Z]_{(Z-z)} / (f_1(Z), \dots, f_c(Z)).$$

If $y \neq z$ is any common zero of the original system, then $f(y) = 0$ and one coordinate of $y - Z$ is a unit in A_z . Multiplying $Q(y, Z)(y - Z) = 0$ by the adjugate of $Q(y, Z)$ shows that

$$\det Q(y, Z) = 0 \quad \text{in } A_z.$$

This works for every such y , whether or not it is isolated.

On setting $Y = z$ in (3.4), we obtain $f(Z) = Q(z, Z)(Z - z)$. Lemma 3.2, after translation, says that $\det Q(z, Z)$ is nonzero in A_z . Choose an F -linear functional $\lambda : A_z \rightarrow F$ which is nonzero on this element. Applying λ coefficientwise to $\det Q(Y, Z)$ and normalizing produces a polynomial s_z of the required degree, with $s_z(z) = 1$ and $s_z(y) = 0$ at every other common zero.

Here are explicit first-order interpolants, which also show that the number of points does not enter the degree bound. The polynomial

$$u_z = 3s_z^2 - 2s_z^3$$

has value 1 and all first derivatives zero at z , and has value and all first derivatives zero at the other selected points. The polynomial

$$v_{z,j} = (Y_j - z_j)s_z^2$$

has zero value at every selected point, derivative 1 in coordinate j at z , and all other prescribed first derivatives zero. Linear combinations of the u_z and $v_{z,j}$ give arbitrary first-order data. Their degrees are at most $\max\{3c(L-1), 2c(L-1)+1\} \leq C_c L$.

The exact-isolation Equations (3.3), with these normalized s_z and the set S , now have degree at most $c(L-1)+1$ and have precisely S as their common zero set. The empty selection is isolated by the equation $1 = 0$. $\square$

Remark 3.4. The local quotient A_z need not be reduced. The determinant in Lemma 3.2 records its nonzero top class, and a linear functional extracts a polynomial which separates the underlying point. Thus Lemma 3.3 does not assume that the original equations have independent differentials, and it makes no assertion about the thickness of other components.

Lemma 3.5 (A finite union of isolated-point lists). *Suppose S is the union of finitely many lists $S_\nu \subset F^c$, where S_ν consists of isolated points of a system of degree at most $L_\nu \geq 1$. Put $L_* = \sum_\nu L_\nu$. Values and first derivatives on the distinct points of S can be interpolated at degree at most $C_c L_*$.*

Proof. For a point $z \in S$, treat each list separately. If $z \in S_\nu$, take its normalized separator in that list. If $z \notin S_\nu$, the exact-isolation part of Lemma 3.3 supplies an equation for S_ν which does not vanish at z ; normalize its value there to 1. The product over the lists is 1 at z and zero at every other point of their union. Its degree is at most $C_c L_*$. The first-order construction in the preceding proof costs a further fixed factor only. $\square$

3.3 Component Hilbert ranks

We now relate polynomial restriction rank to total component degree. For the lower bound, a generic projection reduces the family to finitely many points over the function field of the base. Interpolation on that fiber supplies independent transverse polynomials; multiplying them by base monomials produces the required rank.

The upper estimate below is Chardin's Hilbert-function bound [6]. For components of a single equation system, the shifted lower estimate follows from the isolated-component bound of Chardin–Philippon [7, Corollary 3]; see also their erratum [8]. We give the proof, including the extension to several equation systems used here.

Lemma 3.6 (Upper and lower component ranks). *Let $V \subset \mathbb{C}^n$ be a finite union of distinct irreducible varieties of pure dimension k and total degree a . For every $D \geq 0$,*

$$H_V(D) \leq a \binom{D+k}{k}. \quad (3.5)$$

Suppose in addition that every member of this family is an irreducible component of a common locus defined by equations of degree at most $L \geq 1$. Other components of that locus may have any dimension. There is an integer $q \leq C_n L$ such that, for $D \geq q$,

$$H_V(D) \geq a \binom{D-q+k}{k}. \quad (3.6)$$

In particular,

$$H_V(L) \geq c_n a L^k. \quad (3.7)$$

The same assertions hold for projective varieties, with homogeneous restriction rank.

More generally, the lower assertions hold with L replaced by $L_ = \sum_\nu L_\nu$ if the family is a union of lists, the members of the ν th list being irreducible components of a system of degree at most L_ν . A component in several lists is counted once.*

Proof. We first justify the upper bound with its precise leading coefficient. On a projective closure, polynomial restrictions inject into the global sections of $\mathcal{O}_V(D)$. A general hyperplane avoids the associated components and has reduced pure-dimensional intersection with a reduced variety; this is the reduced hyperplane-section form of Bertini's Theorem in characteristic zero. To see the reduced-scheme assertion, the universal hyperplane incidence is a projective-space bundle over the reduced variety, hence is reduced. Its generic fiber over the dual projective space is reduced by localization and is geometrically reduced in characteristic zero. The openness of reduced fibers [25, Tag 0578] gives the assertion for a general hyperplane. Avoiding the components gives pure codimension one, and the section has degree a . The restriction exact sequence gives

$$h^0(V, \mathcal{O}_V(D)) \leq h^0(V, \mathcal{O}_V(D-1)) + h^0(V \cap H, \mathcal{O}_{V \cap H}(D)).$$

For dimension zero, the last space has dimension equal to the number of reduced points, namely the degree. In every dimension, $h^0(V, \mathcal{O}_V) \leq a$: it is at most the number of irreducible components, since a regular function on an irreducible projective variety is constant. Induction on dimension, followed by summation in D , gives

$$h^0(V, \mathcal{O}_V(D)) \leq a \sum_{j=0}^D \binom{j+k-1}{k-1} = a \binom{D+k}{k}$$

for $k \geq 1$. This proves (3.5); alternatively one may add the bound for the irreducible components. Affine homogenization gives exactly the same restriction rank.

For the lower bound, choose generic linear coordinates

$$(t_1, \dots, t_k, y_1, \dots, y_{n-k})$$

such that projection to the t coordinates is finite of degree $\deg V_i$ on every irreducible component V_i of the family. This is the usual generic linear normalization: in the projective closure the projection center is disjoint from V_i , and a general fiber is a complementary linear section. In characteristic

zero the generic fibers are reduced. Put $K = \mathbb{C}(t_1, \dots, t_k)$ and let F be an algebraic closure of K . The geometric generic fiber of the selected family consists of a distinct points of F^{n-k} .

These points are isolated in the generic fiber of the *entire* given system. Indeed, if W is any different irreducible component of that system, then $V_i \cap W$ is a proper subvariety of V_i . Its projection to the k -dimensional base is not dominant. Thus no point of the generic fiber of V_i lies on W . This argument also handles components W of dimension larger than k ; their generic fibers may be positive-dimensional, but they do not contain the selected sheets.

The equations on the generic fiber have degree at most L in y . Lemma 3.3 says that evaluation on its a selected points is surjective from $F[y]_{\leq q}$ for some $q \leq C_n L$. Since ordinary monomials span this space, we can select a monomials $m_1(y), \dots, m_a(y)$ of degree at most q whose evaluation columns are linearly independent over F . For each i , multiply m_i by all base monomials t^α with $|\alpha| \leq D - q$. The resulting global polynomials have total degree at most D and are linearly independent on V . In fact a relation over $\mathbb{C}$ would give

$$\sum_i \left(\sum_\alpha c_{i,\alpha} t^\alpha \right) m_i|_{\text{generic fiber}} = 0.$$

Independence over F forces each coefficient polynomial in t to vanish, and algebraic independence of the t_j forces every $c_{i,\alpha}$ to be zero. Counting these monomials proves (3.6). Notice that no rational coefficient has been cleared: only ordinary transverse monomials were selected.

We explain the passage to degree L , since the lower bound above initially starts at a constant multiple of L . In a monomial order refining total degree, the standard monomials of $I(V)$ form a set closed under division, and their number of degree at most D is $H_V(D)$. Fix an integer $b \geq 1$. The map

$$x^\alpha \longmapsto x^{\lfloor \alpha/b \rfloor}$$

sends standard monomials of degree at most bL to standard monomials of degree at most L . Each image has at most b^n preimages, by the possible coordinatewise remainders. Hence

$$H_V(bL) \leq b^n H_V(L). \quad (3.8)$$

Take a fixed b large enough that $bL - q \geq L$ and use (3.6). This proves (3.7).

For the union-of-systems assertion, take a generic base projection simultaneously for the finitely many systems and components. On its geometric generic fiber, Lemma 3.5 supplies the same surjectivity at degree $C_n L_*$. The monomial argument and (3.8) are unchanged. This proof does not claim that a component of one system stays an isolated component of the union of all the original loci; it uses the selected finite lists on the generic fiber instead. $\square$

Corollary 3.7 (Mixed-dimensional envelopes). *Let Z be a locus of equations of degree at most $L \geq 1$, and let a_k be the total degree of any chosen subfamily of its k -dimensional irreducible components. For their union V ,*

$$H_V(L) \geq c_n \sum_{k=0}^n a_k L^k.$$

For an arbitrary reduced union with dimension-by-dimension degrees a_k , without a hypothesis on its equations,

$$H_V(D) \leq \sum_{k=0}^n a_k \binom{D+k}{k}.$$

In particular these statements apply to the components of $\text{Env}_L(T)$.

Proof. For the lower bound, $H_V(L)$ is at least the rank of each pure dimensional subfamily. Lemma 3.6 bounds each of those below by $c_n a_k L^k$; taking a maximum loses at most the fixed factor $n + 1$. For the upper bound, restriction to the entire union injects into the direct sum of its restrictions to the pure-dimensional subfamilies. Apply (3.5) to each. $\square$

3.4 Common regular equations and pooled proper cuts

The rank bounds have two uses. They give common equations with independent differentials on most components, and they combine separate proper cuts into one polynomial. Both conclusions measure the exceptional family by its degree, so that later mass bounds can control the points lying on it.

Tidor, Yu, and Zakharov formulate regular approximate complete intersections whose Jacobian recovers the tangent space generically [26, Theorem 4.1]. The next lemma has a different family-level interface: its equations vanish on the whole component family, and one degree cutoff controls an explicit exceptional-degree loss. We prove this interface below.

Lemma 3.8 (Regularity for a whole component family). *Let V be a pure k -dimensional family of total degree a as in the lower-bound part of Lemma 3.6, with equation bound L . The union-of-systems version is allowed, with $L = \sum_\nu L_\nu$. If $D \geq C_n L$, then, except for components of total degree at most*

$$C_n a L / D, \tag{3.9}$$

the equations of degree at most D vanishing on the whole family have Jacobian rank $n - k$ generically on every component.

There is a finite set of these equations and a polynomial of degree at most $C_n D$, obtained as a linear combination of their maximal Jacobian minors, which is nonzero on each retained component. At points of V where this polynomial is nonzero, their common zero set is locally a smooth k -fold, equal locally to V . In particular distinct members of the family cannot meet there.

Proof. There is nothing to prove when $k = n$: the only possible nonempty family is the full ambient space, with codimension zero. Otherwise choose the generic base and transverse coordinates used in Lemma 3.6, and keep the notation $K = \mathbb{C}(t)$ and $F = \overline{K}$. Let b be the total degree of the components on which the asserted generic Jacobian rank fails. Their geometric generic fibers contribute b of the a sheets.

We give the vector-space argument carefully. Write

$$W_D = \mathbb{C}[t, y]_{\leq D}, \quad E_D : W_D \longrightarrow F^a, \quad f \longmapsto (f(t, z_i))_{i=1}^a.$$

These are a vector space and a linear map over $\mathbb{C}$, and

$$\ker_{\mathbb{C}} E_D = I(V)_{\leq D}, \quad \text{rank}_{\mathbb{C}} E_D = H_V(D).$$

No linearity of this truncated kernel over K is asserted.

At a failing sheet z_i , the common kernel of the ambient gradients of the equations in $I(V)_{\leq D}$ has dimension at least $k + 1$ over F . Its intersection with the $(n - k)$ -dimensional vertical space therefore contains a nonzero vector v_i . The tangent space of the component projects isomorphically to the base at a generic sheet, so this vertical vector is transverse to the component. Define a first derivative functional

$$J_i(f) = v_i \cdot \partial_y f(t, z_i).$$

Its coefficients may be algebraic functions of t . They are coefficients of this functional, not functions to be differentiated. By construction J_i annihilates $\ker_{\mathbb{C}} E_D$. If J collects these b functionals, then

$$\text{rank}_{\mathbb{C}}(E_D, J) = \text{rank}_{\mathbb{C}} E_D = H_V(D). \quad (3.10)$$

First-order interpolation on the geometric generic fiber gives surjectivity onto these a values and b transverse derivatives from $F[y]_{\leq q}$, where $q \leq C_n L$. Select $a + b$ ordinary y -monomials m_i of degree at most q with linearly independent augmented columns $w_i \in F^{a+b}$. Vertical differentiation gives the identity

$$(E_D, J)(t^\alpha m_i) = t^\alpha w_i \quad (|\alpha| \leq D - q).$$

All these input polynomials lie in W_D . A relation over $\mathbb{C}$ among their columns becomes an F -linear relation among the w_i , with coefficients that are polynomials in t . Independence over F , followed by algebraic independence of t over $\mathbb{C}$, makes every coefficient zero. Thus

$$(a + b) \binom{D - q + k}{k} \leq H_V(D) \leq a \binom{D + k}{k}. \quad (3.11)$$

For $D \geq 2q$, the ratio of the two binomial coefficients is at most $1 + C_n q/D$; this follows by multiplying the k factors $(D + i)/(D - q + i)$. Rearranging (3.11) proves (3.9). When $k = 0$ the binomial coefficients are both 1 and the argument gives $b = 0$.

Choose a basis $f_1, \dots, f_s$ of $I(V)_{\leq D}$. All $(n - k)$ -row Jacobian minors have degree at most $(n - k)(D - 1)$. On each retained component at least one is nonzero. A generic constant linear combination g of all these minors is consequently nonzero on every retained component, since the excluded coefficient vectors are a finite union of proper linear subspaces. Wherever $g \neq 0$, some $n - k$ of the f_i have independent differentials. Their zero locus is a smooth k -fold locally by the complex implicit function theorem. The pure k -dimensional set V is contained in it and has the same local dimension, so its reduced germ is that smooth germ. All remaining f_i vanish on this germ as well. Two distinct irreducible components cannot share this germ: an open common germ is Zariski dense in either component and would make them equal. $\square$

We have obtained regular equations vanishing on the entire family. We next combine different proper cuts on its components, controlling those components on which the combined polynomial vanishes identically.

Lemma 3.9 (Pooling different proper cuts). *Let $V = \bigcup_i V_i$ be a pure k -dimensional family of total degree a satisfying the lower-bound hypotheses of Lemma 3.6, with bound L . On each V_i let h_i be a polynomial of degree at most $L' \geq 1$ which does not vanish identically on V_i . If $D \geq C_n \max\{L, L'\}$, there is a polynomial g of degree at most D which vanishes on every proper cut $V_i \cap \{h_i = 0\}$ and is not identically zero on any member except possibly a subfamily of total degree at most*

$$C_n a L' / D. \quad (3.12)$$

The union-of-systems version of the family is again allowed.

Proof. For $k = 0$ all the cuts are empty, and take $g = 1$. Suppose $k \geq 1$ and let C be the union of their reduced supports. Bézout gives $\deg C \leq a L'$ in dimension $k - 1$, so

$$H_C(D) \leq a L' \binom{D + k - 1}{k - 1}.$$

Let $U_D = I(C)_{\leq D}$, and call a component V_i forced if every polynomial in U_D vanishes identically on it. If their union has degree b , the kernel inclusion $I(C)_{\leq D} \subset I(\bigcup_{i \text{ forced}} V_i)_{\leq D}$ implies

$$H_{\bigcup_{i \text{ forced}} V_i}(D) \leq H_C(D).$$

The lower bound of Lemma 3.6 applies to this subfamily with the same L . For $D \geq C_n L$ it bounds the left side below by $c_n b D^k$, proving (3.12).

For each unforced component, the polynomials in U_D which vanish identically on it form a proper linear subspace of U_D . Choose g outside the finite union of these subspaces. On increasing the fixed lower bound for D/L' , the degree estimate makes $b < a$ for a nonempty family, so such a g is a nonzero polynomial. It has precisely the required properties. $\square$

3.5 Dividing a degree bound by a multiplicity

We use multiplicity of a polynomial at a point in the elementary sense of order of vanishing: all terms below that total degree vanish after translation to the point. The next argument supplies an actual polynomial of smaller degree, rather than an estimate for the size of the set of multiple zeros. The following matrix lemma supplies such vanishing orders from a rank deficit. Together the two statements turn an exceptional rank condition into a lower-degree equation that remains nonzero at a specified test point.

Lemma 3.10 (Multiplicity compression). *Let $S \subset \mathbb{C}^n$ be finite and let $x_* \in \mathbb{C}^n$. Suppose a polynomial f of degree at most E has order at least $r \geq 1$ at every point of S , and $f(x_*) \neq 0$. There is a polynomial g such that*

$$\deg g \leq (n+1)E/r, \quad g|_S = 0, \quad g(x_*) \neq 0.$$

The degree bound has no dependence on $|S|$.

Proof. An empty set S is handled by $g = 1$. Otherwise $E \geq r$, because f is a nonzero polynomial vanishing to order at least r somewhere. First work over an algebraically closed field of characteristic p , where $p \geq r$. Set

$$m = \left\lceil \frac{np}{r} \right\rceil.$$

The Frobenius map is surjective on this field, so grouping monomials by their exponent vectors modulo p gives a unique expression

$$f(X)^m = \sum_{0 \leq i_1, \dots, i_n < p} X^i h_i(X)^p. \quad (3.13)$$

Every nonzero h_i has

$$\deg h_i \leq mE/p \leq nE/r + E/p \leq (n+1)E/r.$$

Fix $z \in S$. The order of f^m at z is at least $mr \geq np$. Every monomial in $X - z$ of total degree at least np has some coordinate exponent at least p . It follows that

$$f^m \in J_z = ((X_1 - z_1)^p, \dots, (X_n - z_n)^p).$$

Modulo J_z , the identity $h_i(X)^p \equiv h_i(z)^p$ holds. The monomials X^i , with every digit below p , form a basis of the quotient by J_z : translating the usual basis $(X - z)^i$ is an invertible triangular change of basis. Reducing (3.13) modulo J_z therefore gives $h_i(z) = 0$ for every i . This applies at every

point of S . Evaluation at x_* shows that at least one $h_i(x_*)$ is nonzero, because $f(x_*)^m \neq 0$. That h_i proves the assertion in characteristic p .

We include a concrete specialization argument to transfer this fixed degree assertion to $\mathbb{C}$. Put $q = \lfloor (n+1)E/r \rfloor$ and suppose that no degree- q polynomial separates the given $S = \{z_1, \dots, z_s\}$ from x_* . In the finite-dimensional space of polynomials of degree at most q , this says that evaluation at x_* belongs to the linear span of the evaluations at the z_j . Thus there are $\lambda_1, \dots, \lambda_s \in \mathbb{C}$ with

$$x_*^\alpha = \sum_{j=1}^s \lambda_j z_j^\alpha \quad (|\alpha| \leq q). \quad (3.14)$$

Let $R \subset \mathbb{C}$ be the finitely generated $\mathbb{Z}$ -algebra generated by the coefficients of f , all coordinates of these points, the λ_j , and $f(x_*)^{-1}$. All hypotheses are polynomial identities over R : order at least r is expressed by the vanishing of the coefficients of total degree below r in $f(z_j + Y)$. The inverse in R preserves $f(x_*) \neq 0$ under every specialization. The identities (3.14) also hold in R .

Such an R has specializations to algebraically closed fields of arbitrarily large positive characteristic. Here is the required algebraic justification. Noether normalization for the finitely generated domain $R \otimes_{\mathbb{Z}} \mathbb{Q}$ gives algebraically independent elements $t_1, \dots, t_b$ over which it is finite. After inverting one nonzero integer a , these elements belong to $R[1/a]$ and the finitely many generators of $R[1/a]$ satisfy monic equations over $\mathbb{Z}[1/a, t_1, \dots, t_b]$. Hence $R[1/a]$ is finite integral over that polynomial ring. For every prime $p \nmid a$, the lying-over property supplies a maximal ideal above $(p, t_1, \dots, t_b)$. Its residue field is a finite extension of $\mathbb{F}_p$, and embeds in $\overline{\mathbb{F}}_p$.

Choose such a prime $p \geq r$. Specializing the coefficients, points, and identities gives characteristic- p data satisfying the hypotheses and (3.14). The characteristic- p construction produces a degree- q polynomial which vanishes at all the specialized z_j but not at the specialized x_* . This contradicts (3.14). Repeated specialized points cause no problem: the evaluation identity and the compression construction remain valid. The contradiction proves the complex assertion. $\square$

Lemma 3.11 (Rank loss gives ambient multiplicity). *Let $A(x)$ be a matrix of polynomials on $\mathbb{C}^n$, and let $F(x)$ be a q -row, q -column minor. If $\text{rank } A(p) \leq q - h$, then $\text{mult}_p F \geq h$. In particular, if F is nonzero at a test point, has degree at most E , and the indicated rank loss occurs at finitely many points, Lemma 3.10 gives a polynomial of degree at most $(n+1)E/h$ vanishing at those points but not at the test point.*

Proof. Expand the determinant using columns from $A(p)$ and from $A(x) - A(p)$. A term with fewer than h latter columns uses more than $q - h$ constant columns and is zero, because those columns have rank at most $q - h$. Each remaining term belongs to the h th power of the maximal ideal at p . This is an assertion in the ambient polynomial ring; no regularity of a subvariety containing the points is needed. $\square$

3.6 Projection degrees and local multiplicities

We next control the degree lost by projection from a point. The exact loss is the local multiplicity of the variety there. Projection to a hypersurface will relate this geometric multiplicity to polynomial vanishing order, allowing the preceding compression lemma to be applied.

For an integral variety X of dimension k and $p \in X$, we write $\text{mult}_p X$ for its Hilbert–Samuel multiplicity. Thus, with $\mathfrak{m}_p$ the maximal ideal of its local ring,

$$\dim_{\mathbb{C}} \mathcal{O}_{X,p} / \mathfrak{m}_p^t = \frac{\text{mult}_p X}{k!} t^k + O(t^{k-1}).$$

Equivalently, this is the degree of its tangent cone, with the scheme multiplicities of that cone included. We set $\text{mult}_p X = 0$ for $p \notin X$. For a hypersurface this multiplicity is the order of its local defining polynomial; for a smooth point it is 1.

Lemma 3.12 (Projection degree and multiplicity). *Let $X \subset \mathbb{P}_{\mathbb{C}}^n$ be irreducible of dimension $k \geq 1$ and degree a .*

(i) *If projection from a point p has image Y of dimension k and generic degree t , then*

$$t \deg Y = a - \text{mult}_p X. \quad (3.15)$$

(ii) *If $k < n$, there is a generic linear projection to $\mathbb{P}^{k+1}$ which is finite and birational on X . Its image is an irreducible hypersurface of degree a . Given finitely many specified points $p_i \in X$, the projection can simultaneously be chosen so that their fibers are singletons and*

$$\text{mult}_{\pi(p_i)} \pi(X) \geq \text{mult}_{p_i} X. \quad (3.16)$$

Given in addition a point $x_ \notin X$, it can also be required that $\pi(x_*) \notin \pi(X)$.*

The assertions apply to affine varieties through their projective closures and generic affine charts.

Proof. For (i), if $p \notin X$, projection is a morphism and the pullback of a target hyperplane is a hyperplane section of X . Intersecting k general target hyperplanes and counting the t points over a general image point gives $t \deg Y = a$.

Suppose $p \in X$. We give the intersection calculation that also justifies the formula when X is singular or not Cohen–Macaulay. For the blowup construction and the degree formula for a linear system with base locus, see [14, Appendix B.6 and Proposition 4.4]. Let $b : \tilde{X} \rightarrow X$ be the blowup of the ideal of p , and let E be its exceptional Cartier divisor. The blowup is the relative projective spectrum of the Rees algebra $\bigoplus_{j \geq 0} \mathcal{I}_p^j$, where $\mathcal{I}_p$ is the ideal sheaf of p ; it is the closure of the graph of projection from p . Write H for the pullback of the hyperplane divisor of X . The linear forms through p generate the ideal of p twisted by $\mathcal{O}_X(1)$, so the resolved projection $f : \tilde{X} \rightarrow Y$ satisfies

$$f^* \mathcal{O}_Y(1) = \mathcal{O}_{\tilde{X}}(H - E).$$

Its top self-intersection is consequently $t \deg Y$, by taking general hyperplanes on the target.

We recall each term in this self-intersection. The divisor H restricts trivially to E , since E lies over the point p . Therefore

$$H^{k-j} E^j = 0 \quad (1 \leq j < k).$$

Moreover

$$E = \text{Proj} \left(\bigoplus_{j \geq 0} \mathfrak{m}_p^j / \mathfrak{m}_p^{j+1} \right), \quad \mathcal{O}(E)|_E = \mathcal{O}_E(-1).$$

The Hilbert polynomial of this associated graded ring shows that the degree of E with respect to $\mathcal{O}_E(1)$ is $\text{mult}_p X$. Thus $E^k = (-1)^{k-1} \text{mult}_p X$. Also $H^k = a$, as can be computed using general hyperplanes avoiding p . Expanding gives

$$t \deg Y = (H - E)^k = a + (-1)^k E^k = a - \text{mult}_p X.$$

Only Cartier-divisor intersection numbers on the integral blowup are used here. In particular, the calculation does not identify multiplicity with the length of an arbitrary local linear section.

For (ii), the case $n = k+1$ is the identity projection. Otherwise choose a linear center $\Lambda \simeq \mathbb{P}^{n-k-2}$. For a fixed point q , let $J(q, X)$ be the closure of the union of the lines joining q to points of X different from q . Its dimension is at most $k+1$. Choose one smooth point $q_0 \in X$. A general Λ is disjoint from each of

$$X, \quad J(p_i, X), \quad J(q_0, X), \quad T_{q_0}X, \quad \text{and, if specified, } J(x_*, X).$$

There are finitely many conditions, and each is a nonempty open condition on the Grassmannian: the dimension of the relevant closed set plus $n - k - 2$ is strictly less than n .

The resulting projection defines a morphism $\pi : X \rightarrow \mathbb{P}^{k+1}$. It is finite. Indeed, a fiber lies in a projective linear space in which Λ is a hyperplane; a positive-dimensional projective fiber would meet that hyperplane, contradicting $\Lambda \cap X = \emptyset$. Avoidance of the joins makes the fibers over the specified p_i , and over q_0 , singletons. Avoidance of $T_{q_0}X$ makes the differential injective at q_0 , so the fiber at q_0 is a reduced point. More explicitly, its Artinian local ring has zero cotangent space and hence zero maximal ideal by Nakayama's Lemma. The generic rank of a finite module is at most the dimension of any fiber, again by Nakayama's Lemma. The generic degree is therefore one. The image is an integral k -fold in $\mathbb{P}^{k+1}$, hence a hypersurface, and its degree equals a by pulling back k general hyperplanes. The additional join condition ensures $\pi(x_*) \notin \pi(X)$.

It remains to verify the local multiplicity assertion without a smoothness assumption at p_i . Write

$$A = \mathcal{O}_{\pi(X), \pi(p_i)}, \quad B = \mathcal{O}_{X, p_i},$$

with maximal ideals $\mathfrak{m}$ and $\mathfrak{n}$. Because the fiber is a singleton, B is a finite local A -algebra; because the projection is birational, it has rank one and $A \subset B$. The quotient B/A is annihilated by a nonzero element of A and therefore has dimension at most $k-1$. The Artin–Rees property gives a fixed c such that

$$\mathfrak{m}^j A \subseteq \mathfrak{m}^j B \cap A \subseteq \mathfrak{m}^{j-c} A \quad (j \geq c).$$

Using the exact sequence

$$0 \rightarrow A/(\mathfrak{m}^j B \cap A) \rightarrow B/\mathfrak{m}^j B \rightarrow (B/A)/\mathfrak{m}^j(B/A) \rightarrow 0$$

shows that $A/\mathfrak{m}^j A$ and $B/\mathfrak{m}^j B$ have the same leading coefficient of degree k in their lengths. Since $\mathfrak{m}B \subseteq \mathfrak{n}$,

$$\dim_{\mathbb{C}} B/\mathfrak{m}^j B \geq \dim_{\mathbb{C}} B/\mathfrak{n}^j.$$

Comparison of the leading coefficients proves (3.16). $\square$

Corollary 3.13 (A bounded-degree test for high multiplicity). *Let $X \subsetneq \mathbb{C}^n$ be irreducible of degree a , and let $S \subset X$ be finite with $\text{mult}_p X > a/2$ for every $p \in S$. For any specified $x_* \notin X$, there is a polynomial of degree at most $2(n+1)$ which vanishes on S and is nonzero at x_* . In particular the specified high-multiplicity centers lie in a proper hypersurface of degree bounded in terms of the ambient dimension alone.*

Proof. For a zero-dimensional irreducible variety, X is one reduced point, and a linear polynomial suffices. In positive dimension apply Lemma 3.12(ii) to the projective closure, the finite set S , and x_* . Compose a homogeneous defining equation of the image with the linear forms of the projection and dehomogenize in the original affine chart. The resulting polynomial has degree at most a and is nonzero at x_* . Choose the target chart to contain the finitely many specified images. Its dehomogenization differs locally from this polynomial by a unit, so its order at every $p \in S$ is at least $\text{mult}_p X$: composition does not decrease order, and hypersurface multiplicity equals the order of its equation. Apply Lemma 3.10 with $r = \lfloor a/2 \rfloor + 1$. $\square$

Corollary 3.14 (A tangent hyperplane controls the projection loss). *Let an irreducible affine curve C lie in a variety X smooth at $p \in C$. Suppose C has a point z for which the direction $z - p$ is not in $T_p X$. Then*

$$\text{mult}_p C \leq \frac{1}{2} \deg C.$$

Consequently, if projection of C from p is nonconstant, its image degree times its generic map degree is at least $\frac{1}{2} \deg C$.

Proof. Choose an affine hyperplane through $p + T_p X$ which does not contain z . Its linear equation ℓ restricts to an element of $\mathfrak{m}_{X,p}^2$, because its value and differential on X both vanish at p . On the curve it therefore belongs to $\mathfrak{m}_{C,p}^2$, and it is not identically zero.

The one-dimensional reduced local ring of the curve is Cohen–Macaulay. Thus, writing $R = \mathcal{O}_{C,p}$, the local intersection number with the hyperplane is $\text{length}(R/(\ell))$, and $\text{length}(R/(\ell^j)) = j \text{length}(R/(\ell))$. Since $(\ell^j) \subseteq \mathfrak{m}_{C,p}^{2j}$, comparison with the Hilbert–Samuel leading term gives

$$\text{length}(R/(\ell)) \geq 2 \text{mult}_p C.$$

Bézout bounds this local intersection number by $\deg C$. The final assertion follows from (3.15). $\square$

Lemma 3.15 (The plane-curve polar bound). *Let $C \subset \mathbb{P}_{\mathbb{C}}^2$ be a reduced plane curve of degree a . If $m_q = \text{mult}_q C$, then*

$$\sum_{q: m_q \geq 2} m_q(m_q - 1) \leq a(a - 1), \quad \sum_{q: m_q \geq 2} m_q^2 \leq 2a(a - 1). \quad (3.17)$$

The sums include points on any chosen line at infinity.

Proof. Write $C = \{F = 0\}$ with F homogeneous and squarefree. A general first polar

$$G = \lambda_0 \frac{\partial F}{\partial X_0} + \lambda_1 \frac{\partial F}{\partial X_1} + \lambda_2 \frac{\partial F}{\partial X_2}$$

has degree $a - 1$ and has no irreducible factor in common with F . Indeed, for every irreducible factor F_i some derivative of F_i is nonzero modulo F_i in characteristic zero. Modulo F_i , the corresponding derivative of F is this derivative times the product of the other factors. A generic coefficient vector avoids vanishing identically on each of the finitely many components.

In an affine chart at q , write f for the local equation of the curve. The three dehomogenized derivatives of F are, after renaming the coordinates,

$$f_x, \quad f_y, \quad af - xf_x - yf_y.$$

Each has order at least $m_q - 1$, so the local equation of G has order at least $m_q - 1$. The elementary local intersection inequality gives

$$I_q(F, G) \geq m_q(m_q - 1).$$

One can see that inequality directly on the reduced local branches: a branch of multiplicity b has a primitive parametrization whose two coordinates have minimum order b , and substituting into a polynomial of order c gives order at least bc . Add over the branches to obtain the displayed bound. Since F and G have no common component, Bézout gives total intersection number $a(a - 1)$ and in particular makes the set of points with $m_q \geq 2$ finite. This proves the first inequality. The second follows from $m^2 \leq 2m(m - 1)$ for every integer $m \geq 2$. $\square$

4 Secants and terminal curves

We work in the contradiction regime (2.1). In particular, $N = B^d$, $A = B^2$, and $M = o(A)$. We use the flat bounds of Lemma 2.1: for some fixed $c_0 > 0$ every proper complex affine flat contains at most NB^{-c_0} sample points, a plane contains at most $B^{2+o(1)}$ sample points, and a line contains at most $2M$ sample points. The complex-flat assertion follows by taking the real affine span of its real points. Constants throughout this section depend only on d . For an affine point p , write

$$\pi_p(y) = [y - p] \in \mathbb{P}^{d-1}, \quad y \neq p.$$

All varieties in the geometric arguments are reduced complex varieties; their degrees are degrees of projective closures.

4.1 The secant tangent calculation

The generic conclusion is a form of generalized trisecant geometry; compare [19, Section 3.2]. We give the tangent calculation because the assertion for a particular pair is also needed.

Lemma 4.1 (Secants and a particular-pair version). *Let $X, Y \subset \mathbb{A}^d$ be irreducible varieties of dimensions $k, l \geq 1$. Suppose that for a general pair $(x, y) \in X \times Y$ the connecting direction is in neither $T_x X$ nor $T_y Y$. Let Y' be another irreducible l -dimensional variety, possibly equal to Y . If a general such line has a further point $z \in Y' \setminus \{x, y\}$, then $k \leq l$. If $k = l$, then X and Y lie in a common affine $(k + 1)$ -flat. More generally, when $k = l$, equality of the projected tangent spaces at a general pair already implies this common-flat conclusion, without an assumption about a further root.*

There is also the following local assertion. Suppose $p \in X$ and $q \in Y$ are smooth, and the map

$$\psi(x, y) = (x, \pi_x(y))$$

has differential of rank $k + l$ at (p, q) . Suppose a smooth local variety W of dimension $k + l$ at $(p, \pi_p q)$ contains both product images $\psi(X \times Y)$ and $\psi(X \times Y')$ locally at the relevant pairs. If $z \in Y' \setminus \{p, q\}$ has $\pi_p z = \pi_p q$, then

$$\overline{T_p X} \subseteq \overline{T_q Y} \quad \text{in } \mathbb{C}^d / \mathbb{C}(q - p). \quad (4.1.1)$$

Here bars denote the images of vector tangent spaces. In particular, if these images have dimensions $k = l$ and are unequal, no such z exists. The same source component X must occur at both roots.

Proof. Choose a direction chart by requiring one coordinate of the direction to equal one. Write $y - x = au$ and $z - x = bu$ in that chart; the two nonzero longitudinal parameters a, b are distinct. For a source velocity $\xi \in T_x X$, variation of x with the other endpoint fixed has direction derivative $-\bar{\xi}/a$ at y , and $-\bar{\xi}/b$ at z . The first coordinates of these two image tangent vectors are both ξ . Their difference is therefore vertical and has direction component $(b^{-1} - a^{-1})\bar{\xi}$.

At a general pair the product image has dimension $k + l$: its first coordinate records the k source directions, and injectivity of the target tangent modulo the connecting line supplies l vertical directions. If the further root exists generally, the two irreducible product images coincide. Indeed they have the same dimension $k + l$, and the first is contained in the second on a dense set. At a general smooth image point their tangent spaces agree. The vertical kernel of the first product image is exactly $\overline{T_y Y}$. The preceding subtraction gives $\overline{T_p X} \subseteq \overline{T_q Y}$ and hence $k \leq l$. This calculation also proves the local assertion: the full-rank first product identifies its tangent image with TW , whose

vertical kernel is $\overline{T_q Y}$. Source variation with the second root held fixed still gives tangent vectors in TW . No assertion that the second root belongs to the finite sample is used.

When $k = l$, the projected tangents agree. Thus

$$T_x X \subseteq T_y Y + \mathbb{C}(x - y), \quad T_y Y \subseteq T_x X + \mathbb{C}(x - y).$$

Fix a general y . Project X projectively away from the affine tangent plane $y + T_y Y$. The first inclusion says that this projection has zero differential wherever it is defined. In characteristic zero a rational map with zero differential has zero-dimensional image: on function fields, every function in its image has zero differential and is constant. Its irreducible image is consequently one point. Thus X lies in an affine $(k + 1)$ -flat containing $y + T_y Y$. The case in which X already lies in the projection center gives the same conclusion. Interchanging X, Y bounds both affine spans by $k + 1$. If one span has dimension $k + 1$, the preceding flat for each general point of the other variety must equal that span, so it contains both varieties. If both spans have dimension k , both varieties are flats, and the displayed tangent inclusions say directly that their affine join has dimension at most $k + 1$. $\square$

4.2 A generic count for families of carriers

The next formulation separates the geometric count from the scale regularization in Section 5.

Lemma 4.2 (Generic projection depth). *Let $\mathcal{X}, \mathcal{Y}$ be finite families of distinct irreducible carriers of dimensions $k \geq l \geq 1$, where $k < d$ and $l < d - 1$. Fix assigned sets*

$$S_X \subseteq P \cap X \quad (X \in \mathcal{X}), \quad T_Y \subseteq P \cap Y \quad (Y \in \mathcal{Y}),$$

pairwise disjoint within each family. The source and target assignments may overlap with each other. If an initial list contains repeated copies of a carrier, pool those copies and their assigned sets before defining these distinct families and checking the following hypotheses. Put

$$n_x = \sum_{X \in \mathcal{X}} |S_X|, \quad n_y = \sum_{Y \in \mathcal{Y}} |T_Y|.$$

The normalized component weights are $|S_X|/n_x$ and $|T_Y|/n_y$, and a component pair has the product of its two weights. Assume $n_x, n_y \geq N \exp(-o(w))$, with $w \rightarrow \infty$ and $w \leq \log B$. Assume that every flat member is real. Suppose, when $k = l$, that for some number e the number of flat members in each family is at most $e \exp(o(w))$, and that the normalized weight of each such member is at most $e^{-1} \exp(o(w))$.

For every fixed $\epsilon > 0$, one can discard component pairs of total product weight $o(1)$, and obtain for each remaining source component X a list $\mathcal{Y}_X$ of target components, such that: for every $Y \in \mathcal{Y}_X$, at a general pair $(x, y) \in X \times Y$ both projected tangents are injective and the number of moving roots on $\bigcup_{Z \in \mathcal{Y}_X} Z$ over π_{xy} is at most $\exp(\epsilon w)$. This counts the generic map degrees of all listed components sharing that projected image, and excludes the undefined root x .

The hypotheses apply to the two carrier configurations used below: in a common window, $k = l$, $w = s$, and $e = e_j$; in separated subpower windows, $w = \log B$, and, when $k = l$, $e = B^{d-l}$.

Proof. First discard pairs for which a general connecting line is tangent at one endpoint. For a fixed target component Y , choose a general smooth $y \in Y$. Every source component tangent in this sense is contained in the fixed proper affine flat $y + T_y Y$. Its total assigned source mass is at most NB^{-c_0} . After division by n_x , this is $B^{-c_0+o(1)}$. Average with target weights. The argument with

source and target interchanged removes tangencies at the other endpoint. The choices of tangent points can be made simultaneously for the finite lists.

For $k = l$, also discard pairs lying in a common $(k + 1)$ -flat when one member is nonlinear. That nonlinear member spans the flat: its span cannot have dimension k unless it is itself a flat. Fixing that member therefore pins down a proper flat containing the other member's assigned points. The same flat bound and weighted averaging show that these pairs have product weight $o(1)$.

Lemma 4.1 now rules out a third moving root except when $k = l$ and the source and first target are flats in a common $(k + 1)$ -flat H . Every further listed target with the same projection must also lie in H : for a general $x \in X$, its affine join with x equals H . It is a flat, since the nonlinear exception was removed. Each such flat contributes exactly one moving root. Thus it remains to bound the number of listed flats in H for most weighted pairs.

The flat members are already distinct within each family. In the slice, identify a flat occurring in both families and add its two weights; this changes the bounds by at most a factor two. Fix a small constant $\eta > 0$. Delete flats of normalized weight less than $e^{-1} \exp(-\eta w)$; their total weight is $\exp(-\eta w + o(w))$. Slice all remaining flats by a common generic real projective subspace of codimension k . Each becomes a point, and each generated $(k + 1)$ -flat becomes a line. For these finite families, genericity ensures that distinct generated flats yield distinct lines and that a sliced point belongs to such a line exactly when the original flat is contained in the corresponding H . The number of sliced points is at most $e \exp(o(w))$.

Let t_H be their number on the line for H . The lower weight cutoff and the physical flat bound give

$$t_H \leq e \exp(\eta w + o(w)) B^{-c_0}.$$

Choose $\eta < c_0/4$. Since $w \leq \log B$, the last bound is at most $e B^{-c_0/2}$ for large B . By the rich-line form of Szemerédi–Trotter, Theorem B.2, the number of these lines at dyadic occupancy t is

$$O\left(\frac{e^2 \exp(o(w))}{t^3} + \frac{e \exp(o(w))}{t}\right).$$

Each has pair weight at most $t^2 e^{-2} \exp(o(w))$. Summing for $t \geq \exp(\epsilon w)$ gives

$$\exp(-\epsilon w + o(w)) + B^{-c_0/2} \exp(o(w)) = o(1).$$

Delete these pairs and, for each source, delete the corresponding targets from its list. A pair which remains belongs to an H with at most $\exp(\epsilon w)$ remaining target flats. Removing targets cannot increase this number. This proves the assertion.

For the announced applications, the degree and per-degree mass bounds of Proposition 5.2 give the common-window hypotheses. In subpower windows all testing degrees are $B^{1+o(1)}$, so the number of l -flat carriers is $B^{d-l+o(1)}$ and their normalized weights are at most $B^{-(d-l)+o(1)}$. These are the second set of hypotheses. Exceptional intersections of distinct projected images are not part of this statement: at a generic direction, only components having the same image contribute. $\square$

4.3 Deletion of assigned terminal curves

Fix, for each $p \in P$, a union C_p of irreducible curves in $\mathbb{P}^{d-1}$, each spanning a projective subspace of dimension at most two, with summed degree at most

$$E = \delta N/A, \quad \delta \rightarrow 0.$$

Identical components are pooled. An assignment below is partial: some sample points may be left unassigned, and no point is assigned twice.

Proposition 4.3 (Terminal-curve deletion). *Assign sample points to finitely many irreducible physical curves Y_i , each containing its assigned points. Pool repetitions of a curve, and require every line member to be real. Then the number of ordered pairs (p, q) , $p \neq q$, for which q is assigned to Y_i and*

$$\overline{\pi_p Y_i} \subseteq C_p$$

is $o(N^2)$. The estimate is uniform in the assignment and the curves.

Proof. Write n_i for the assigned mass of Y_i and $d_i = \deg Y_i$. Lemma 3.1 gives

$$\sum_i n_i \leq N, \quad n_i \leq CM d_i. \quad (4.3.1)$$

We treat lines first, then nonlinear curves.

Assigned lines. Centers on their assigned target line contribute at most $2M \sum_i n_i = o(N^2)$. Otherwise the projected line is itself a line component of C_p , and corresponds to a real plane H containing p and the assigned line. Put

$$S_H = \sum_{Y_i \text{ a line}, Y_i \subset H} n_i, \quad h_H = |P \cap H|.$$

Planes with $S_H < A$ account for at most $NAE = \delta N^2$ pairs, because a center has at most E line components in C_p . For the other planes it suffices to prove $\sum_{S_H \geq A} h_H S_H = o(N^2)$. They form a finite family determined by pairs of distinct assigned lines: one line has weight at most $2M = o(A)$.

If an assigned line in H has weight b , then

$$bh_H \leq CM^2. \quad (4.1)$$

To prove this, choose plane coordinates with that line as the first axis and map a point (x_1, x_2) to $(x_1, x_1^2 + x_2^2)$. Every fiber has size at most two. A distance from an axis point $(a, 0)$ becomes a line equation $v - 2au = r^2 - a^2$ in these coordinates. There are at most bM distinct such lines, with different slopes for different axis points, and the incidence count is at least $bh_H/2$. The point-line estimate gives

$$bh_H \leq C((bh_H M)^{2/3} + h_H + bM).$$

For b above a sufficiently large fixed constant, absorb the h_H term. If $h_H = O(M)$, use $b \leq 2M$; otherwise absorb bM as well and cube the remaining inequality. Both cases give (4.1). For bounded b , the case $h_H \leq 1$ is immediate. Otherwise choose two distinct sample points in H . Every point of $P \cap H$ is an intersection of two circles whose radii belong to the at most M distance values; two distinct-center circles have at most two intersections. Thus $h_H \leq 2M^2$, which proves the bounded- b case too.

Group the planes by dyadic values $h \leq h_H < 2h$ and $s \leq S_H < 2s$, where $h, s \gtrsim A$. For a dyadic line weight b , let N_b be the total assigned mass of all lines with weight in $[b, 2b)$. Equation (4.1) permits only $b \leq L = CM^2/h$. Let F be the number of planes in this (h, s) group. A generic hyperplane slice turns the finite line arrangement into points and the relevant planes into lines, preserving exactly the needed incidences. A further generic planar projection permits use of the real point-line estimate. Applying it to each weight class and multiplying by its weight yields

$$Fs \leq CF^{2/3}T(L) + CN + CLF, \quad T(L) = \sum_{b \leq L} b^{1/3} N_b^{2/3}.$$

Since $L/s \leq CM^2/A^2 = o(1)$, the last term is absorbed, giving

$$F \leq C \left(N/s + T(L)^3/s^3 \right). \quad (4.2)$$

The contribution of N/s to $\sum hsF$ is at most $CNB^{2+o(1)}(1 + \log N)^2 = o(N^2)$, using the plane cap and $d \geq 3$. For the other contribution, Hölder's inequality gives

$$T(L)^3 \leq L \left(\sum_{b \leq L} (b/L)^{1/2} N_b^2 \right) \left(\sum_{b \leq L} (b/L)^{1/4} \right)^2 \leq CL \sum_{b \leq L} (b/L)^{1/2} N_b^2.$$

Consequently

$$\frac{hT(L)^3}{s^2} \leq \frac{CM^2}{s^2} \sum_{b \leq L} (b/L)^{1/2} N_b^2.$$

For each fixed b , summing over dyadic h , equivalently over inverse dyadic $L \geq b$, costs a bounded geometric sum. Then $\sum_b N_b^2 \leq N^2$, and $\sum_{s \gtrsim A} s^{-2} = O(A^{-2})$. The result is $O(N^2 M^2/A^2) = o(N^2)$, completing the line case.

Nonlinear curves when $d \geq 4$. If $\overline{\pi_p Y_i}$ spans at most a projective plane, the affine span of Y_i has dimension at most three. A curve with three-dimensional span requires the center to lie in that fixed three-flat; the flat cap and (4.3.1) in the form $\sum n_i \leq N$ show that these pairs cost $o(N^2)$. For a planar curve, centers in its own plane also cost $o(N^2)$. It remains to consider planar curves viewed from outside their planes. Projection is then a projective isomorphism of the plane onto its image and preserves the curve's degree.

At a center p , group curves by their common nonlinear projection $\gamma \subset C_p$, writing $e_\gamma = \deg \gamma$ and $W_{p,\gamma} = \sum_{i \text{ in the group}} n_i$. Each individual mass is at most $CM e_\gamma$. For two different curves in this group, their planes are distinct and lie, with p , in a projective three-space. Their intersection is a projective line L .

For a fixed ordered pair of such curves, there are at most Ce_γ possible centers. Here is the required stabilizer argument. Each perspective between the two planes fixes L pointwise and maps one curve to the other. Comparing two perspectives gives a projectivity of the first plane preserving its nonlinear curve and fixing L pointwise. In the affine chart with L at infinity, it has the form $u \mapsto \lambda u + b$. A nonidentity translation cannot preserve an irreducible nonlinear curve: the infinite translation orbit of any point lies on a line, whose closure would be contained in that curve. Thus the scalar homomorphism on this stabilizer has trivial kernel; the stabilizer is abelian because every commutator is a translation. If it is nontrivial, a nonidentity element has a unique affine fixed point c , fixed also by every commuting element. The group consists of homotheties about c . The orbit of a generic point is free and lies on its line through c , so Bézout bounds the group order by the

curve degree. This argument also rules out an infinite group. Finally a perspective between distinct planes determines its center uniquely: it is the intersection of the joining lines of corresponding points, two suitably chosen pairs being enough. The claimed bound on centers follows.

Discard groups with $W_{p,\gamma} \leq KMe_\gamma$, for a sufficiently large fixed K . Their total contribution is at most $KNME = o(N^2)$. In every remaining group the off-diagonal ordered weight satisfies

$$\sum_{i \neq j} n_i n_j = W_{p,\gamma}^2 - \sum_i n_i^2 \geq \frac{1}{2} W_{p,\gamma}^2.$$

If the surviving pair mass were at least εN^2 , weighted Cauchy–Schwarz, first in γ and then in p , would give

$$\sum_{p,\gamma} \frac{1}{e_\gamma} \sum_{i \neq j} n_i n_j \geq \frac{\varepsilon^2 N^3}{2E}.$$

For each fixed ordered pair of distinct curves, degree preservation and the center bound give $\sum_p e_\gamma^{-1} \leq C$. The left side is therefore at most $C \sum_{i,j} n_i n_j \leq CN^2$, a contradiction since $E = o(N)$.

Nonlinear curves when $d = 3$. The degree and multiplicity analysis of curves sharing a radial projection here is closely related to the curve–cone estimates of Tidor, Yu, and Zakharov [26, Sections 10.2–10.3].

We first delete centers where $\text{mult}_p Y_i > d_i/2$. There are only boundedly many such centers for each curve. To see this, take a generic birational plane projection, preserving or increasing multiplicities at this finite list of points, as in Lemma 3.12. For a reduced plane curve of degree D , intersection with a general polar derivative gives

$$\sum_{z: \text{mult}_z \geq 2} (\text{mult}_z)^2 \leq 2D(D-1). \quad (4.3)$$

Indeed the intersection multiplicity at a point of multiplicity m is at least $m(m-1)$, and $m^2 \leq 2m(m-1)$ for $m \geq 2$. A nonlinear irreducible degree-two curve has no point of multiplicity greater than one; for all other degrees this bounds the number with multiplicity greater than half the degree by an absolute constant. The deletion therefore costs $O(\sum n_i) = O(N)$ pairs. Projection to a line requires the curve and center to lie in one fixed plane; these pairs cost $o(N^2)$ by the plane cap.

Group the remaining curves at p by their common nonlinear projection γ . Let t_i be the generic degree of $\pi_p : Y_i \dashrightarrow \gamma$ in this group. The point-projection formula, obtained from the linear-system degree formula with its base-point contribution [14, Proposition 4.4], and the multiplicity deletion give

$$d_i - \text{mult}_p Y_i = t_i e_\gamma, \quad d_i \leq 2t_i e_\gamma. \quad (4.4)$$

An ordered triple (i, j, k) in a group is called valid if each index occurs at most its corresponding map degree. We claim that for each fixed ordered triple of curve indices

$$\sum_{p: (i,j,k) \text{ valid at } p} e_\gamma^{-2} \leq C. \quad (4.5)$$

Choose v general on Y_i , simultaneously for the finite list of centers in this sum. Distinct centers give distinct directions from v , since v can avoid all the lines joining pairs of centers. The map degrees t_i, t_j, t_k still refer to projection from each original center p ; projection from v will be used to count the directions of those centers.

We first claim that, viewed from v , each of Y_j, Y_k projects birationally to a plane curve, and that the two plane images are distinct unless $Y_j = Y_k$. A general line joining $v \in Y_i$ to a point of either

target curve is tangent at neither endpoint. Otherwise, after fixing the tangent endpoint, the other nonlinear curve would lie in its tangent line. Thus the non-tangency hypothesis of Lemma 4.1 holds.

Failure of birationality would give a further point on the same target curve over a general joining direction. Equality of the two plane images for $Y_j \neq Y_k$ would give a further point on the other target curve. In either case Lemma 4.1 puts Y_i and the first target curve in a common plane. At any center p contributing to the sum, the curves belong to one original nonlinear group. If the two curves are distinct, this is impossible: projection from a point in their plane has a line image, whereas projection from outside that plane is injective. If the first target is Y_i itself, the conclusion makes Y_i planar. The repeated first index then makes validity at that center require $t_i \geq 2$, whereas a planar curve in a nonlinear group is projected from outside its plane and has map degree one. This proves the claim.

For a valid center p , the line pv consequently gives at least

$$r_j = t_j - \mathbf{1}_{i=j}, \quad r_k = t_k - \mathbf{1}_{i=k}$$

distinct preimages other than v on the respective curves. They can be chosen generic on those curves by the original choice of v . Birationality identifies the normalizations of the projective closures of the curve and its plane image. These distinct generic smooth preimages therefore give distinct normalization points, hence distinct branches at the corresponding point of the plane image. Validity and (4.4) imply

$$r_j \geq t_j/2 \geq d_j/(4e_\gamma), \quad r_k \geq t_k/2 \geq d_k/(4e_\gamma).$$

If the two plane images are distinct, Bézout bounds the sum of their local multiplicity products at these distinct projected centers by $d_j d_k$. If they agree, then $j = k$, and the residual multiplicity is at least two: it is $t_j \geq 2$ when $i \neq j$, and $t_i - 1 \geq 2$ when all indices agree. Equation (4.3) bounds the sum of squared residual multiplicities by $2d_j^2$. Both cases give (4.5).

Discard once more groups of mass $W_{p,\gamma} \leq KMe_\gamma$. Their contribution is $O(NME) = o(N^2)$. In a larger group, an invalid ordered triple must repeat an index whose map degree is at most two. Such an index has mass at most $4CMe_\gamma$ by (4.3.1) and (4.4). The weight of invalid triples is at most

$$3W_{p,\gamma} \sum_{t_i \leq 2} n_i^2 \leq 12CMe_\gamma W_{p,\gamma}^2.$$

Choose K so this is at most half of $W_{p,\gamma}^3$. If the remaining pair mass were at least εN^2 , Hölder in each row and then in the centers would imply

$$\sum_{p,\gamma} \frac{1}{e_\gamma^2} \sum_{(i,j,k) \text{ valid}} n_i n_j n_k \geq \frac{\varepsilon^3 N^4}{2E^2}.$$

Indeed $\sum_\gamma W_{p,\gamma}^3/e_\gamma^2 \geq (\sum_\gamma W_{p,\gamma})^3/E^2$, and the sum of the row cubes is at least their total cubed divided by N^2 . On the other hand, (4.5) bounds this expression by $C \sum_{i,j,k} n_i n_j n_k \leq CN^3$. For $d = 3$ we have $E = \delta N^{1/3}$, so N^4/E^2 exceeds N^3 by an unbounded factor. This contradiction proves the proposition. $\square$

5 Polynomial scale profiles

We work along the contradiction sequence, with

$$N = |P|, \quad B = N^{1/d}, \quad A = B^2, \quad M = o(A).$$

In this section a further parameter $\delta \rightarrow 0$ is given, and $E = \delta N/A$. The parameter will be the total directional degree in the sparse-cones argument. All constants below may depend on d . Our aim is to partition the sample into pieces on which successive polynomial cutting degrees can be recorded uniformly. The resulting profiles will bound component masses from above and polynomial restriction ranks from below.

5.1 Conventions and the partition

For a width $s \rightarrow \infty$, $s \leq \log B$, put

$$T_s(b) = B \exp(bs). \quad (5.1)$$

Degrees are rounded to integers. Throughout, an estimate $U \geq V \exp(-o(s))$ means

$$\liminf \frac{\log(U/V)}{s} \geq 0.$$

Equivalently, every fixed $\varepsilon > 0$ can replace $o(s)$ by εs , for all sufficiently large indices. The analogous convention applies to upper bounds. We only use such errors in the presence of a subsequently specified strict positive margin.

A *piece* below is a subset of the sample, together with one path of polynomial cuts. The entire path, rather than a different path at each point, is attached to the piece. At every point on the piece each successive cut is proper on every component of the preceding local system. We refer to the irreducible components of that system meeting the piece as its *carriers*. Different points can have different carriers. Properness at the counted points will suffice for every degree estimate.

At testing degree D , the normalized time is $\log(D/B)/s$. We record each cut at the testing time when it is made; its actual degree may be smaller. When the d recorded times on a path have subsequential limits $b_1, \dots, b_d$, we call each open interval between consecutive distinct limits a *plateau*. Its number of completed cuts, and hence its local dimension, is constant. We also use initial and final plateaux for the intervals before the first and after the last limiting cut.

Proposition 5.1 (Multiscale partition). *There are $s_* \rightarrow \infty$ and partitions of all but $o(N)$ sample points into a core and non-core pieces with the following properties.*

1. *The core is simultaneously isolated by equations of a degree L such that $EL^2 = o(N)$.*
2. *Each non-core piece has a dyadic width $s = 2^{-a} \log B \geq s_*$ and size $n = N \exp(-o(s))$. Its path has d proper hypersurface cuts. Along any sequence of pieces, after passing to a subsequence, their normalized cut times converge to*

$$-1 \leq b_1 \leq \dots \leq b_d \leq 1, \quad \max_i |b_i| \geq \frac{1}{2}.$$

The degree of cut i is at most $D_i \exp(o(s))$, where $D_i = B \exp(b_i s)$.

3. *Let b lie strictly between consecutive distinct limiting cut times, and let $j = \#\{i : b_i < b\}$ and $l = d - j$. At each point of the piece the current system is pure of local dimension l . Uniformly over polynomials of degree $O(T_s(b))$, the number of points where such a polynomial vanishes but contains none of the local carriers is $o(n)$. The same assertion holds before the first cut whenever $T_s(b) \geq 1$ and $b < b_1$, and it is automatic after the last cut.*

The assertions about error terms and negligible proportions hold along every sequence of pieces. In particular, a violating choice of one piece at each index is not permitted.

Proof. Choose $s_* \rightarrow \infty$ sufficiently slowly that

$$s_* \leq \frac{1}{4} \log B, \quad \delta \exp(6s_*) \rightarrow 0.$$

Start with width $\log B$, halve the width at each step, and stop on entry to the first width in $[s_*, 2s_*]$. On entry to a step of width s , the unresolved set is isolated by equations of degree

$$B \exp((1 + o(1))s). \quad (5.2)$$

This is true initially by Lemma 3.1, since $M = o(B^2)$. Except at the initial step, the unresolved set also has no nonzero polynomial of degree at most $B \exp(-s)$ capturing as many as $N \exp(-\sqrt{2}s)$ points.

At a step which is not the final one, first remove greedily batches of at least $N \exp(-\sqrt{s})$ points on zero sets of nonzero polynomials of degree at most $B \exp(-s/2)$. These batches are sent to the regularization below. From the remaining points remove greedily batches of at least $N \exp(-\sqrt{s})$ points which occur as isolated zero-dimensional components of some system of degree at most $B \exp(s/2)$. Only the union of these latter batches proceeds to the next window. The final remainder is sent to regularization if its size is at least $N \exp(-s^{1/3})$, and otherwise is discarded.

There are at most $\exp(\sqrt{s})$ batches of either kind. Isolated point interpolation in Lemma 3.3 gives equations of degree $O(B \exp(s/2))$ for precisely each batch of the second kind. Indeed, from normalized separators q_i for its points p_i use $1 - \sum_i q_i$ and all coordinates of $(x - p_i)q_i$. Their common zero set is precisely the batch. A union of finitely many such zero sets is cut out by products, taking one equation from each system; the degrees add. Thus the union proceeding to the next step has isolation degree at most

$$B \exp(s/2 + O(\sqrt{s})).$$

This is (5.2) at width $s/2$. The first greedy operation supplies exactly the asserted lower boundary there. Consequently the final core has isolation degree at most $B \exp(3s_*)$ for large indices, and hence $EL^2 = o(N)$.

The shrinking windows have produced the core and the batches that will become non-core pieces. It remains to give each retained batch one path of proper cuts and to ensure that a proper polynomial test between cut times captures a vanishing fraction of the piece. We now enforce this condition while discarding only $o(N)$ points. For each width s use a grid in normalized time from -1 to $1 + o(1)$, containing the upper isolation degree, with mesh tending to zero and $G = O(1 + \sqrt{\log s})$ grid points. Index the potential steps by increasing grid time, and, within one grid time, by decreasing positive dimension. There are $K = O_d(G)$ indices. Choose thresholds $\lambda_i = \exp(-a_i)$ with

$$a_K = s^{1/10}, \quad a_i = 10a_{i+1}. \quad (5.3)$$

The same indexed thresholds are used in all trees of this width.

In a node of dimension k and original size m , repeatedly extract at least $\lambda_i m$ remaining points where a polynomial of the current testing degree vanishes properly on every local component. An extracted set becomes a child, with that polynomial imposed, and is processed in dimension $k - 1$ at the same time. Here m stays the original mass of this node during all its extractions. The unextracted residual advances to the next time, unless its size is less than $\sqrt{\lambda_i} m$, in which case it is discarded. At the last time, at most d generic combinations of the isolation equations finish the path: on any positive-dimensional local component through a counted point, some isolation equation is nonidentically zero, so a generic combination is a proper cut simultaneously on all such components.

Every path visits an index at most once. Extraction decreases dimension; taking a residual increases time. Sibling extractions are alternative branches, not repeated steps on one path. Therefore

$$\sum_i a_i = s^{1/10+o(1)} = o(s^{1/3}), \quad \sum_{r>i} a_r < \frac{a_i}{9}.$$

If a residual survives at step i , every later terminal piece inside it has size at least $m \exp(-a_i/2 - \sum_{r>i} a_r)$. A proper cut available when extraction stopped captured fewer than $m \exp(-a_i)$ residual points. Its captured fraction in that terminal piece is consequently at most

$$\exp\left(-\frac{a_i}{2} + \sum_{r>i} a_r\right) \leq \exp(-7a_i/18) \rightarrow 0. \quad (5.4)$$

This proves the required no-capture assertion on any limiting open plateau. To allow a fixed multiplicative factor in the testing degree, choose a slightly later grid time which is still inside that plateau.

At a fixed index the nodes in all these trees are disjoint. Their discarded residual mass is at most $\exp(-a_i/2)$ times their total mass. Thus the regularization discards at most

$$C_d N(1 + \sqrt{\log s}) \exp(-s^{1/10}/2) \quad (5.5)$$

points at width s . This estimate is summable over windows even if s_* grows arbitrarily slowly. To see this, reverse their order and bound by widths $2^m u$, $m \geq 0$, with $u \geq s_*$. Put $t = u^{1/10}$ and $c = (\log 2)/10$. The inequalities

$$2^{m/10} \geq 1 + cm, \quad \sqrt{\log(2^m u)} \leq \sqrt{\log u} + \sqrt{m \log 2}$$

bound the sum of the fractions in (5.5) by

$$C_d e^{-t/2} \sum_{m \geq 0} (1 + \sqrt{\log u} + \sqrt{m \log 2}) e^{-ctm/2} = O_d((1 + \sqrt{\log u}) e^{-u^{1/10}/2}) = o(1).$$

The separately discarded small remainders have fractions $\exp(-s^{1/3})$ and are summable by the same argument. No factor equal to the number of windows is used.

The discarded mass is therefore negligible across all windows. We check the remaining size and cut-time conditions. Each retained terminal piece has size at least its initial batch size times $\exp(-o(s^{1/3}))$, and hence is $N \exp(-o(s))$. The inherited lower boundary is negligible even relative to the smallest such piece, since

$$\frac{N \exp(-\sqrt{2}s)}{N \exp(-\sqrt{s} - o(s^{1/3}))} = \exp(-(\sqrt{2} - 1)\sqrt{s} + o(s^{1/3})) \rightarrow 0.$$

There are exactly d cuts on a path. Their times have subsequential limits $b_1 \leq \dots \leq b_d$ in $[-1, 1]$. For a first-type batch, some cut occurs by limiting time $-1/2$: otherwise its defining polynomial would capture the entire piece in an initial full-dimensional no-capture interval. For a remainder piece the last cut cannot have limiting time less than $1/2$. Otherwise its path would isolate more than $N \exp(-\sqrt{s})$ points by equations of degree at most $B \exp(s/2)$, contradicting termination of the second greedy operation. This proves the imbalance and all stated properties. The estimates were uniform over nodes and pieces, which also proves the final sequence quantifier. $\square$

5.2 Degree budgets and propagation of rank

The partition supplies paths and intervals on which proper tests capture few points. We now prove that their degree scales are balanced and control both component masses and restriction ranks. In particular, small exceptional component degree will imply small sample mass.

For a limiting profile define

$$D_i = B e^{b_i s}, \quad e_j = \prod_{i \leq j} D_i, \quad e_0 = 1, \quad \sigma(b) = \prod_{i=1}^d \min(T_s(b), D_i). \quad (5.6)$$

On a plateau after j cuts, $\sigma(b) = e_j T_s(b)^{d-j}$. Figure 2 illustrates this relation.

Proposition 5.2 (Budgets and balanced profiles). *For every non-core profile,*

$$\sum_{i=1}^d b_i = 0, \quad b_1 < 0 < b_d, \quad b_1 + b_2 < 0. \quad (5.7)$$

On a plateau after j cuts, its carrier family has degree at most $e_j \exp(o(s))$. Every subcollection of total degree a meets the piece in at most

$$a \left(\prod_{i>j} D_i \right) \exp(o(s)) \quad (5.8)$$

points. This counts all points touching those components, whether or not they have been assigned to them. Finally, for every fixed $\alpha > 0$, every subset S of at least αn piece points has

$$H_S(T_s(b)) \geq \sigma(b) \exp(-o(s)) \quad (5.9)$$

at each interior testing time, and with the usual limiting interpretation at a cut time.

Proof. Apply Bézout successively, retaining the proper intersections through counted points. It gives the carrier degree bound and, after applying the later cuts to a specified carrier subcollection, (5.8). Applied to the entire path it gives $n \leq (\prod_i D_i) \exp(o(s))$. Since $n = N \exp(-o(s))$ and $N = B^d$, this proves $\sum_i b_i \geq 0$.

We prove (5.9) by propagation through the finitely many plateaux; the assertion is always for every fixed positive density subset, including subsets chosen during the proof. Before the first cut, no nonzero polynomial of the testing degree can vanish on S . Its envelope is therefore the ambient space, whose rank is comparable to $T_s(b)^d$.

On a plateau of dimension l , let V be the testing-degree envelope of S . At all but $o(n)$ points of S , at least one local carrier is contained in V . Indeed, if no carrier through a point is contained, a generic linear combination of the envelope equations is proper on all those carriers and vanishes at the point. There are finitely many components and sample points, so the combination can be chosen simultaneously; no-capture then applies.

For this envelope, choose at each such point one contained local carrier, and choose one irreducible component of V containing each distinct carrier so chosen. Let the points inherit the component chosen for their carrier. Group all irreducible components of V by dimension, counting the points by these choices; components receiving no points remain in the degree of their group. Some dimension $h \geq l$ group, of total degree a , covers a fixed positive fraction of S . Writing S' for that fraction and $T = T_s(b)$, the component Hilbert bounds give

$$H_S(T) = H_V(T) \gtrsim a T^h, \quad H_{S'}(T') \lesssim a(T' + 1)^h \quad (T' < T).$$

The lower estimate uses that the group consists of components of the envelope at its defining degree T ; the upper estimate uses only that S' lies on that group. Eliminating a gives

$$H_S(T) \gtrsim H_{S'}(T') \left(\frac{T}{T' + 1} \right)^h.$$

Take $T' = T_s(b')$ at a time b' just before the cut starting the current plateau, and apply (5.9) there to S' . The displayed inequality increases that lower bound by a factor at least $(T/T_s(b'))^h$, to exponential accuracy. Since $h \geq l$, this is at least the growth required by σ on the plateau. Take b' arbitrarily close to the boundary; the logarithmic endpoint error can be made smaller than any prescribed fixed multiple of s . This proves the next plateau, including after a group of simultaneous limiting cuts.

There is one possible boundary convention: if the first cut has testing degree only $\exp(o(s))$, there may be no preceding testing time of degree at least one. Rank one at that boundary is already the required bound to exponential accuracy, even if several first cuts have that same time. Use it as the start of the induction.

The same envelope argument advances across the final cut, with required growth exponent zero. Since rank is at most N , it gives $\prod_i D_i \leq N \exp(o(s))$, proving the reverse inequality $\sum_i b_i \leq 0$. The profile is nonzero by the imbalance in Proposition 5.1. Its monotonicity and zero sum give $b_1 < 0 < b_d$. If $b_1 + b_2 \geq 0$, then $b_2 > 0$ and the $d - 2$ later entries are at least b_2 , contradicting the zero sum because $d > 2$. This proves (5.7). $\square$

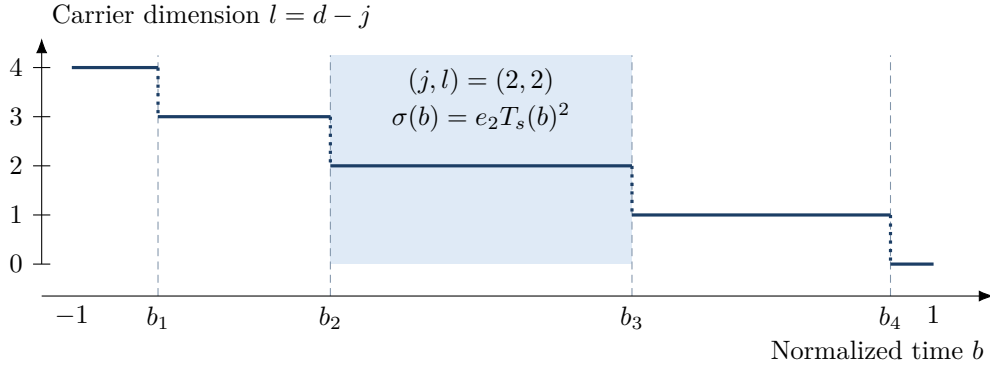


Figure 2: An illustrative limiting profile in dimension four, with fixed window width s and cut times $(b_1, b_2, b_3, b_4) = (-0.8, -0.4, 0.3, 0.9)$. The degree scales are $D_i = T_s(b_i)$, where $T_s(b) = Be^{bs}$, and $e_j = \prod_{i \leq j} D_i$. On the highlighted open plateau (b_2, b_3) , two cuts have occurred and $\sigma(b) = e_2 T_s(b)^2$. The staircase shows carrier dimension; its vertical drops are schematic. General profiles may have coincident limiting cuts.

5.3 Testing individual components and products

Proposition 5.3 (Regularity and individual cuts). *Inside an open plateau, individual proper cuts of degree $O(T_s(b))$ on the different carriers collectively capture only $o(n)$ points. At a slightly later interior testing time there are common equations for the entire carrier family whose Jacobian has full codimension rank at all but $o(n)$ points. At those surviving points the carrier is unique and smooth, and its tangent space is testable by equations and minors of the same interior degree scale.*

The assertions also hold for the union of two such families, and for unions from $\exp(o(s))$ pieces with the same limiting profile and width, provided each piece retains its own assigned masses. Common equations vanish on the whole union. In particular, two different members of that union cannot meet at a surviving point where the common Jacobian has full rank. Flat carriers which are not defined over $\mathbb{R}$ may be discarded with negligible mass.

Proof. Every carrier used here is an irreducible component of its fixed path locus. Indeed, at every point of the piece on that carrier, the successive proper cuts give the current local dimension. A larger irreducible component containing the carrier would pass through that point and contradict this local dimension. The path locus may have larger components elsewhere; they remain in its equations.

Let the plateau start at time $\beta < b$, and choose $b < b' < b''$ in the same plateau. The carrier components are therefore isolated components of their cut system, whose equations have degree at most $B \exp((\beta + o(1))s)$. Lemma 3.9, used at degree $T_s(b')$, pools proper component cuts of degree $O(T_s(b))$ into one polynomial. It is nonidentically zero except on components of total degree at most

$$e_j \exp(-(b' - b)s + o(s)).$$

By (5.8) and $\prod_i D_i = N$, all points of the piece touching those exceptional components have mass $o(n)$. At every remaining counted point the pooled polynomial is proper on every local carrier. No-capture at b' proves the assertion.

Apply Lemma 3.8 at degree $T_s(b')$. Since the cut-system equations have degree at most $B \exp((\beta + o(1))s)$, the components on which the common equations fail to have generic full rank have total degree at most

$$e_j \exp(-(b' - \beta)s + o(s)).$$

The mass budget removes all piece points touching those components at a cost of $o(n)$. On the other components a generic linear combination of maximal minors is nonidentically zero. Its zero set contains every rank failure, and its degree is $O(T_s(b'))$. The preceding individual-cut conclusion applies to this linear combination using the remaining interior margin before b'' , and removes its zeros at another $o(n)$ points. At a surviving point the implicit function theorem gives a smooth locus of the correct dimension; any two carrier components through that point would have the same germ and hence would coincide. The equations vanish on the whole family, including components whose generic regularity failed.

For pooled pieces, use the union-of-systems versions of Lemmas 3.9 and 3.8. Each carrier remains a component of its own path system. On a generic transverse fiber, Lemma 3.5 interpolates the selected isolated lists, including first jets, with an effective bound equal to the sum of the individual system bounds: separators for the other lists are multiplied, using their exact-isolation equations for a test point outside a list. If these bounds are L_ν , then for $\exp(o(s))$ pieces with the common limiting profile,

$$L_* = \sum_{\nu} L_\nu \leq B \exp((\beta + o(1))s).$$

The strict interior gaps above absorb this change in the effective bound and the $\exp(o(s))$ changes in the degree budgets.

Mass accounting for such a union is performed separately in each original piece. Its own later cuts bound all of its points touching exceptional components of its own current system. We do not apply one piece's later cuts to points assigned to another piece. Rank failures on every good assigned component are then discarded by that piece's no-capture assertion. The equations still vanish on the whole union, so full rank at a surviving point also excludes intersections with a carrier from another piece. The sequence quantifier makes the discarded fractions uniform over these pieces; summing their masses creates no loss equal to the number of pieces.

Finally, the real points of a nonreal complex affine flat lie in a proper complex affine linear section of that flat. Choose one such degree-one proper cut for each nonreal flat, and no cut on the other carriers. The individual-cut assertion just proved makes their total assigned mass negligible. $\square$

For the next lemma, write S_x, S_y for the two point sets and $n_x = |S_x|, n_y = |S_y|$. A dense graph is a subset $\Gamma \subset S_x \times S_y$ of size at least $\alpha n_x n_y$ for some fixed $\alpha > 0$. A label map is a rational map ϕ from pairs to a fixed affine space, with bounded-degree coordinate numerators and denominators on the chosen charts. The image assigned to carriers X, Y is the closure of $\phi(X \times Y)$.

Lemma 5.4 (Product testing). *Take two pieces, or the pooled pieces described above, in a common window with the same limiting profile. On a fixed plateau, fix in each set an assignment of every point to one carrier of its original piece that contains it. These assignments precede the tests below and are retained under later restrictions. Let X, Y denote the assigned carriers of a pair, and fix an interior testing time b . A polynomial of degree $O(T_s(b))$ captures only $o(n_x n_y)$ pairs at which it does not vanish identically on $X \times Y$. The same holds for a number of rational tests bounded independently of the sequence, provided their denominators are nonzero at the pairs concerned and their numerator polynomials have degree $O(T_s(b))$ on the chosen charts, with a common implied constant.*

Consequently an equation system containing the labels of a dense graph contains the complete assigned component images at all but a negligible fraction of its pairs, whenever the label map has bounded degree on these charts and the cleared numerator polynomials of all pulled-back equations have degree $O(T_s(b))$ with a common implied constant. The denominators used in these pullbacks must be nonzero at the actual pairs concerned. For any collection of rational rank tests whose size is bounded independently of the sequence and which are generically nonzero on the relevant assigned images, the tests are nonzero at all but $o(n_x n_y)$ corresponding actual pairs, provided their denominators are nonzero there and their cleared pullback numerators obey the same degree bound.

Proof. For an actual source point p , apply the individual-cut conclusion of Proposition 5.3 on every target carrier Y on which the specialization is nonidentically zero. For the other case, fix Y . If the polynomial is not identically zero on $X \times Y$, choose a generic $y \in Y$ for which its specialization is nonidentically zero on each such X . The same individual-cut conclusion on the source bounds points p where the polynomial vanishes identically on Y . Average this bound with the target masses of the Y 's. The fixed assigned sets are disjoint within each factor, so these weighted sums are unambiguous even when carriers meet. For pooled pieces, retain their original mass accounting. All bounds are uniform over the polynomial and the pieces by their sequence formulation.

For a system of equations, pull back each equation and clear its chart denominator separately. The resulting numerator polynomials vanish at the actual system points and have the common degree bound. On every component product carrying a pair on the chart whose image is not contained in the system, at least one numerator is nonidentically zero. A generic linear combination of the numerators is therefore nonidentically zero on every such product simultaneously. It has

the same degree bound: a finite basis of their span suffices, regardless of the number of system equations. Apply the first assertion to this combination.

For rank tests use their cleared pullback numerators. When a rank condition is expressed by minors whose cleared numerators obey the common degree bound, a generic linear combination of those numerators is nonidentically zero on every relevant component product where the rank condition holds generically, and its zero set contains every rank failure. A bounded number of the resulting tests has a union of failure sets of size $o(n_x n_y)$. A slightly later interior time absorbs the uniform constant in the degree bound. This proves the last assertion without assuming that sample points are generic on their carriers. $\square$

5.4 Selecting the two windows

Proposition 5.5 (Selection of profile pairs). *For every $p \in P$, let C_p be a finite family of irreducible projective curves in $\mathbb{P}^{d-1}$, each spanning a projective subspace of dimension at most two, with total degree at most $E = \delta N/A$. Suppose that $1 - o(1)$ of the unordered pairs $\{p, q\}$ satisfy $\pi_p(q) \in C_p$ or $\pi_q(p) \in C_q$. After the global terminal deletion described below, one of the following configurations occurs.*

1. *A core of positive relative size is covered.*
2. *Two sequences of non-core pieces have a common dyadic width s , the same limiting profile, and an ordered product covered in one orientation with positive relative density.*
3. *Two sequences of non-core pieces have widths with ratio tending to infinity, both $o(\log B)$, the same limiting profile, and such a positively covered ordered product.*

In the common outermost window $s = \log B$, the two factors may instead be the same pool of pieces in a shrinking profile bin. That pool has size $n \geq N\sqrt{\delta}$, is the union of $\exp(o(s))$ pieces, and retains all the preceding degree, mass, regularity, and product-testing conclusions to exponential accuracy. All selected incidences may be required to lie on physical lines containing at most $B^{o(1)}$ sample points.

The global terminal deletion means the following. Whenever a target point was assigned to its carrier curve after the $(d-1)$ st cut, its projection from the center is not a curve contained in the center's directional family.

Proof. Assign each point of every non-core piece to one of its carrier curves after the $(d-1)$ st cut. Omit assignments to nonreal lines. Such a line contains at most one real point, so the omitted fraction in a piece is at most

$$\frac{e_{d-1} \exp(o(s))}{n} = D_d^{-1} \exp(o(s)) = o(1),$$

using (5.7). This is uniform over pieces. Every point is assigned at most once; pool repeated curves. Apply Proposition 4.3 once to this global assignment. It removes only $o(N^2)$ ordered incidences, and hence leaves the required almost-complete unoriented coverage. Assignments retain their original meaning if pieces are later pooled.

If the core has positive relative size, the cone over a directional family has degree $O(E)$ and dimension at most two. The core's isolation equations, restricted to its components, and Bézout bound its intersections with the core by $O(EL^2)$ per center. This is $o(N)$, so neither orientation can cover a positive fraction of the core pairs. This resolves the first alternative. Otherwise discard the core.

Here are details of the remaining selection, including the order of the limits. First fix a precision $1/h$, and partition the compact space of normalized drop lists and values of $s/\log B \in [0, 1]$ into finitely many bins of that diameter. Also bin the dyadic index modulo an integer tending to infinity with h . Two independent random retained points fall in one common bin with probability bounded below by a positive constant depending only on h . Thus the relative uncovered probability conditional on this agreement tends to zero for fixed h . Select a pair of pieces with small relative uncovered probability, and pass to an orientation containing at least half the covered pairs. Now let h tend to infinity sufficiently slowly with the sample index. This standard diagonal choice retains conditional coverage while making the profiles and normalized widths converge to the same limits.

The selected dyadic indices are either equal on a subsequence, or their difference tends to infinity, by their congruence condition. In the latter case the width ratio tends to infinity. Their normalized widths have the same limit, so that limit must be zero. This proves the two window alternatives.

There is a useful refinement at the outermost window. If its total mass tends to zero, discard it before the selection. If its mass is bounded below on a subsequence, work only there. For each fixed precision choose a largest profile bin and use its pool as both factors. It has a fixed positive mass at that precision, and its relative uncovered fraction tends to zero. In the diagonal choice increase precision slowly enough that its mass is at least $N\sqrt{\delta}$ and $N \exp(-o(\log B))$. This is possible because $\delta \rightarrow 0$. Each constituent piece has size $N \exp(-o(s))$ uniformly, so there are only $\exp(o(s))$ of them. Their profiles differ by $o(1)$, and the conclusions for pooled families follow from Proposition 5.3 and Lemma 5.4.

Finally, Szemerédi–Trotter, with a generic planar projection, makes the fraction of ordered pairs on lines of occupancy above a threshold H arbitrarily small as $H \rightarrow \infty$: summing its dyadic rich-line estimate gives $O(N^2/H + Nh_{\max})$, and the proper-flat bound gives $h_{\max} = o(N)$. At each fixed selection precision first choose H large enough for its conditioning probability. Increase the precision sufficiently slowly that $H \leq B^{o(1)}$ and all the preceding conditional errors still vanish. This is compatible with the outer-pool mass condition. All constants and precisions are fixed before taking each asymptotic limit; the final diagonal sequence merely combines these finitely specified requirements. $\square$

6 The sparse-cones assertion

We retain the contradiction hypotheses, including the lower-dimensional induction bounds and their proper-flat consequences. For $p \in \mathbb{R}^d$ and $y \neq p$, write $\pi_p(y) = [y - p]$. A directional curve family is a finite family of curves in $\mathbb{P}^{d-1}$; membership means membership in its union, and its total degree is the sum of the component degrees.

Theorem 6.1 (Sparse cones). *Along a sequence with $M = o(A)$, there cannot be families C_p , one for each $p \in P$, of irreducible projective curves, each spanning projective dimension at most two, such that*

$$\deg C_p \leq E = \delta N/A, \quad \delta \rightarrow 0,$$

and $1 - o(1)$ of the unordered pairs $\{p, q\}$ satisfy $\pi_p(q) \in C_p$ or $\pi_q(p) \in C_q$.

Suppose these data exist. Apply the results of Section 5. The positive-mass core alternative is already contradictory. We must treat a common window and two widely separated windows. The selected covered incidences retain the global terminal deletion and the light-line restriction in Proposition 5.5. For the common-window rank estimate we will also use auxiliary graphs of arbitrary ordered pairs (p, q) with $p \neq q$ on the same selected point sets. That estimate will be proved without a coverage or light-line hypothesis.

We use affine charts of the label map

$$\psi(x, y) = (x, \pi_x(y)). \quad (6.1)$$

For each finite family under discussion, these charts may be chosen to contain all actual labels and to meet every component image. The map and its pullbacks have bounded degree on these charts; constant factors in testing degree fit inside a chosen open plateau. A *dense graph* means at least a fixed positive fraction of the ordered product of the two selected pieces. Assertions about dense graphs always include arbitrary further fixed positive-density restrictions and subsequences.

6.1 Product images and rank in a common window

Consider two pieces with width s and the same limiting profile, or the outermost pool allowed by Proposition 5.5. Use D_i, e_j, σ from (5.6). For an ordered graph Γ on these pieces, write $H(T) = H_{\psi(\Gamma)}(T)$ for the restriction rank of its labels.

We first make the passage from carrier pairs to product images precise. The same construction will be used in the rank argument and, with one fixed choice of parameters, at the outermost endpoint. Its last estimate applies to every later choice of image subfamily.

Lemma 6.2 (Retained products and their image degree). *Fix a common positive-dimensional plateau after j cuts, of dimension $k = d - j < d - 1$, and choose an interior time τ on that plateau. Write n_x, n_y for the original source and target counts. Given a fixed $\varepsilon > 0$, one may discard $o(n_x)$ source points and $o(n_y)$ target points. Write m_x, m_y for the remaining counts, so $m_x = (1 - o(1))n_x$ and $m_y = (1 - o(1))n_y$. One may then discard carrier pairs of total assigned product mass $o(m_x m_y)$, with the following conclusions.*

Each remaining point is assigned to its unique smooth carrier in the union of the two physical families. Repeated carriers are pooled within each family, and every flat carrier with positive assigned mass is real. There are common physical equations for the whole original union whose Jacobian has full codimension rank at these points. A single polynomial g_{ph} , a linear combination of maximal Jacobian minors, is nonzero at every retained point. The equations and this test have degrees $O(T_s(\tau))$. When the two factors are the same pool, the point deletion and the assignment may be taken to be the same in both roles.

Write $\mathcal{R}$ for the retained ordered carrier pairs and

$$Z_{X,Y} = \overline{\psi(X \times Y)} \quad ((X, Y) \in \mathcal{R}).$$

Every $Z_{X,Y}$ has dimension $2k$, and at a general pair the connecting direction is transverse to both physical tangent spaces. Let U be the reduced union of the distinct $Z_{X,Y}$. Then

$$\deg U \leq e_j^2 \exp(o(s)). \quad (6.2)$$

For any subfamily $\mathcal{J}$ of these distinct images, including one chosen after further equations have been chosen, its represented carrier pairs have normalized assigned mass at most

$$\frac{1}{m_x m_y} \sum_{\substack{(X,Y) \in \mathcal{R} \\ Z_{X,Y} \in \mathcal{J}}} n_X n_Y \leq \frac{\deg(\bigcup_{Z \in \mathcal{J}} Z)}{e_j^2} \exp(\varepsilon s + o(s)). \quad (6.3)$$

Here n_X, n_Y are the assigned masses after the point deletions.

Proof. Choose fixed times $\tau_{\text{eq}} < \tau_{\text{test}} < \tau$ between the start of the plateau and τ . Use the earlier part of this interval for the equation choices in Proposition 5.3, leaving the strict gap up to τ_{test} for its minor and proper-cut tests at the sample points. Apply its union version to the two physical carrier families. It removes the negligible point sets excluded by its regularity conclusion and supplies common equations for the whole union. At each remaining point there is now a unique smooth carrier. Denote by g_{ph} the common combination of maximal minors whose zeros were removed in this step. Discard every such point whose carrier is a nonreal flat, using the degree-one proper sections in the last paragraph of that proposition. This costs another $o(n_x) + o(n_y)$ points. The source and target assignments are now unambiguous. Each assigned carrier belongs to the point's own original piece path: that path already supplies a carrier through the point, and uniqueness in the full union identifies it with the assigned member. For two copies of the same pool take the union of the two negligible point deletions in both roles; the resulting assignment is the same. Extend the retained assignments by assigning each deleted point to a containing carrier of its original piece before any later product tests, using the same extension in both roles for a common pool. Product testing on the original sets then retains the present assignments under restriction, and its $o(n_x n_y)$ pair loss is also $o(m_x m_y)$. Pool equal retained carriers within each of the two lists and omit members with zero mass. All these choices have been made inside the interval ending at τ , which gives the stated degree bound.

The pooled budgets, with each original piece still supplying the bound for its own points, give

$$\begin{aligned} \max \left\{ \sum_X \deg X, \sum_Y \deg Y \right\} &\leq e_j \exp(o(s)), \\ \frac{n_X}{m_x} &\leq \frac{\deg X}{e_j} \exp(o(s)), \quad \frac{n_Y}{m_y} \leq \frac{\deg Y}{e_j} \exp(o(s)). \end{aligned} \tag{6.4}$$

Indeed, the mass of a carrier in one piece is at most $\deg(X)(N/e_j) \exp(o(s))$, and pooling $\exp(o(s))$ pieces changes this by only $\exp(o(s))$. The remaining point counts are still $N \exp(-o(s))$. A flat has degree one, so its number and its normalized weight satisfy the hypotheses of Lemma 4.2 with $w = s$ and $e = e_j$. The flat members are real by the preceding deletion.

Apply that lemma now with the fixed ε . Let $\mathcal{Y}_X$ be its target list for X , and set $\mathcal{R} = \{(X, Y) : Y \in \mathcal{Y}_X\}$. The removed carrier pairs have normalized product weight $o(1)$, hence account for $o(m_x m_y)$ actual pairs. This deletion includes pairs generically tangent at either endpoint. Thus the target tangent projects injectively at a general pair, and the first coordinate of ψ records all source tangent directions; the product image has dimension $2k$. The preimages of a general $2k$ -fold linear section of that image are isolated away from the base locus of the bounded-degree map on $X \times Y$. The isolated-intersection form of Bézout therefore gives $\deg Z_{X,Y} \leq C_d \deg X \deg Y$, and (6.2) follows from (6.4).

We prove the uniform subfamily estimate while keeping these same lists. Images belonging to different source carriers are distinct: the closure of the first projection of $Z_{X,Y}$ is X . Fix one source X and one of its distinct images Z . To identify its source fibers, take its closure $\widehat{Z} \subset X \times \mathbb{P}^{d-1}$ before restricting the direction coordinate to our affine chart. Localizing the ideal of the rational image at $\mathbb{C}(X)$ identifies the generic fiber with the closure of the projection of Y from the generic source point. This fiber is geometrically integral, since it is the image closure of the geometrically integral variety $Y_{\mathbb{C}(X)}$. After restricting X to a dense open, the projective family is flat with geometrically integral fibers of a constant degree d_Z ; this uses generic flatness, geometric integrality after shrinking, and constancy of the Hilbert polynomial in a flat projective family [25, Tags 052A, 0559, 0C0E, and 0B9T]. We may also require that projection of Y from each x in this open set has dimension k , by generic target transversality. Its closed image is then a k -dimensional irreducible subvariety of the

k -dimensional integral fiber, so it equals the whole fiber. Make these restrictions simultaneously for the finite image and target lists. Thus all targets with $Z_{X,Y} = Z$ have the same complete projected image at a common general $x \in X$. Write $t_x(Y)$ for their generic map degrees. The fixed list bound says

$$\sum_{Y: (X,Y) \in \mathcal{R}, Z_{X,Y}=Z} t_x(Y) \leq \exp(\varepsilon s).$$

For a general x , the point-projection formula gives $\deg Y - \text{mult}_x Y = t_x(Y)d_Z$. The multiplicity is zero unless $X = Y$. In that case x is smooth and the multiplicity is one; a coincident flat pair was deleted as tangential. Consequently $\deg Y \leq 2t_x(Y)d_Z$ for every target in this sum.

Choose k general affine hyperplanes in the source coordinates. They meet X in $\deg X$ points of the preceding open set. Then choose k general hyperplanes in the direction coordinates; on each complete fiber they give d_Z points, with multiplicity. The choices can be made so all these points lie in the selected label chart and avoid intersections between distinct images. The resulting mixed section of Z therefore has $\deg X d_Z$ points. The $2k$ blockwise hyperplanes are also affine linear equations in the full label space. Their homogenizations restrict to linear equations on the ordinary projective closure of Z , where all the counted affine points are isolated intersections. The isolated-intersection form of Bézout bounds their total multiplicity by the ordinary degree, even if the same equations have additional components at infinity. Applying this to a reduced image subfamily and summing the preceding map-degree bound gives

$$\sum_{\substack{(X,Y) \in \mathcal{R} \\ Z_{X,Y} \in \mathcal{J}}} \deg X \deg Y \leq C_d \exp(\varepsilon s) \deg\left(\bigcup_{Z \in \mathcal{J}} Z\right).$$

Multiplying the two weight bounds in (6.4) proves (6.3). No further generic-depth deletion is made when $\mathcal{J}$ is chosen. $\square$

Lemma 6.3 (Common-window label rank). *Let $\Gamma \subset \{(p,q) \in S_x \times S_y : p \neq q\}$ be any dense graph on the selected pieces or pool; its pairs need not be covered by the C_p . For its labels, at an interior testing time before the last cut,*

$$H(T_s(b)) \geq \sigma(b)^2 \min\left(1, \frac{T_s(b)}{D_1 D_2}\right) \exp(-o(s)). \quad (6.5)$$

The convention is a lower limit after dividing the logarithm by s ; in particular it permits any prescribed fixed positive logarithmic tolerance, uniformly for dense subgraphs.

Proof. On the initial full-dimensional interval, Lemma 5.4, or its two successive full-dimensional no-capture arguments, forces the testing-degree envelope of a dense label graph to be the whole label space. Its rank is comparable to T^{2d-1} . Since $D_1 D_2 \geq T^2$ there, this is at least the right side of (6.5).

On a positive-dimensional plateau after $j \geq 1$ cuts put $k = d - j$. Choose a physical regularity time between the start of this plateau and the current testing time b . When $k < d - 1$, apply Lemma 6.2 with any fixed $\varepsilon > 0$. When $k = d - 1$, apply Proposition 5.3 to the union of the two physical families, discard its rank-failure and nonreal-flat points, and assign each survivor to its unique carrier from its original path. Use the same supported extension to the original point sets before product testing, and pool equal retained carriers with their assigned masses. The fixed tangent-flat argument and the proper-flat cap then discard carrier pairs generically tangent at either endpoint. In both cases the point and carrier-pair losses remove only a negligible fraction of the

graph, and every remaining complete product image $\overline{\psi(X \times Y)}$ has dimension $2k$. Product testing forces these complete images into the testing-degree envelope at almost all remaining graph pairs.

For this fixed envelope, choose one irreducible envelope component containing each distinct complete image occurring at those surviving pairs. Let the pairs with that image inherit its choice. Group all envelope components by dimension and count each surviving pair in the group of its chosen component; components receiving no pairs remain in the degree of their group. Every chosen component has dimension at least $2k$, so some dimension $h \geq 2k$ group carries a positive fraction of the graph. Write a for its degree and Γ' for the graph portion it carries.

If $h \geq 2k + 1$, the Hilbert bounds give

$$H(T) \gtrsim aT^h, \quad H_{\psi(\Gamma')}(T') \lesssim a(T' + 1)^h.$$

Borrow (6.5) from a time just before the cut starting this plateau. The required right side grows with exponent at most $2k + 1$ within the plateau, so growth with exponent h suffices. Endpoint approximation costs an arbitrarily small fixed multiple of s in the logarithm.

It remains to bound a when $h = 2k$. Each image assigned to this group is an irreducible $2k$ -fold contained in its chosen irreducible $2k$ -dimensional envelope component, and therefore equals that component. If $k = d - 1$, necessarily $j = 1$, and the projection of a generic target hypersurface from a generic source point fills direction space. For each source carrier X the product image is thus $X \times \mathbb{P}^{d-1}$ in these label coordinates. A positive mass of pairs requires a positive source mass. The carrier mass budget therefore gives

$$a \gtrsim e_1 \exp(-o(s)) = D_1 \exp(-o(s)).$$

The resulting rank $D_1 T^{2d-2} \exp(-o(s))$ is sufficient: on this plateau $T \leq D_2$ to exponential accuracy and

$$D_1^2 T^{2d-2} \min(1, T/(D_1 D_2)) \leq D_1 T^{2d-2}.$$

Now suppose $k < d - 1$. The images assigned to this group represent a positive amount of assigned product mass, since they carry Γ' . Apply (6.3) to this subfamily. Its degree is at most a , so

$$a \gtrsim e_j^2 \exp(-\varepsilon s - o(s)).$$

Since ε is arbitrary, $a \geq e_j^2 \exp(-o(s))$. The Hilbert lower bound now gives $H(T) \geq e_j^2 T^{2k} \exp(-o(s))$, which is stronger than (6.5) on this plateau.

These alternatives propagate through the finitely many distinct cut times. If the first boundary has degree $\exp(o(s))$ and there is no earlier testing time of degree at least one, use rank one there; the required boundary value is itself $\exp(o(s))$. This also handles several simultaneous first cuts. Every borrowed graph portion has fixed positive density after passage to a subsequence, so the induction applies to it as asserted. $\square$

The graph covered in the selected orientation has an independent upper bound, obtained by specializing at each actual center:

$$H(T) \leq CNE(T + 1). \tag{6.6}$$

Near the last cut, $\sigma(b)$ is N within an arbitrarily small logarithmic tolerance. Comparing (6.5) and (6.6), the relevant ratio is, to the same accuracy, at least

$$\min \left(\frac{A}{\delta D_d}, \frac{A}{\delta D_1 D_2} \right). \tag{6.7}$$

The second term has a strict positive s -scale margin because $b_1 + b_2 < 0$. The first does also unless $s = \log B$ and $b_d = 1$: if the common window is not outermost, $s \leq \frac{1}{2} \log B$, whereas in the outermost window its margin is $1 - b_d$. Choosing a fixed testing time sufficiently close to the final cut therefore gives a contradiction in every other case. No comparison between δ and an uncontrolled $\exp(o(s))$ factor has been made.

6.2 The outermost endpoint

Assume now $s = \log B$ and $b_d = 1$. Use the common pool supplied by Proposition 5.5; initially its size n_0 satisfies

$$n_0 \geq N\sqrt{\delta}, \quad n_0 = N \exp(-o(s)). \quad (6.8)$$

Let Γ_0 be its graph covered in the selected orientation. It has at least αn_0^2 pairs for a fixed $\alpha > 0$, and retains the terminal deletion and the light-line restriction. Let k be the dimension of the final open plateau, $j = d - k$, and let $\beta = b_j < 1$ be its starting time. Balance gives

$$1 \leq k < d - 1, \quad e_j = B^{d-2k}, \quad N/e_j = A^k. \quad (6.9)$$

At this endpoint the rank comparison can have zero logarithmic margin. We will instead lift the covering direction curves to physical curves. For this, we need one fixed family of product images and one common smooth system containing the images used by all the lifts.

Choose fixed interior times

$$\beta < \tau < t_0 < t_1 < t_2 < 1, \quad t_0 > \max\{\beta, 1 + b_1 + b_2\}. \quad (6.10)$$

Such choices exist because $b_1 + b_2 < 0$. Reserve the interval (β, τ) for the finitely many physical regularity choices, use t_0 for the image envelope and t_1 for the common label equations, and reserve (t_1, t_2) for the later pullback tests. Now fix

$$0 < \varepsilon < \frac{1}{4}(t_1 - t_0).$$

Also fix the endpoint approximation and the logarithmic tolerances in the rank argument below smaller than a quarter of both positive gaps $t_0 - \beta$ and $t_0 - (1 + b_1 + b_2)$.

Apply Lemma 6.2 to the two copies of this pool, with τ and ε just fixed. Use its same-pool deletion and assignment, and retain its common physical equations and g_{ph} for the full original union of final-plateau carriers.

When $k \geq 2$, make one further point deletion before fixing the final product family. The carriers whose affine spans have dimension at most $k + 1$ number at most $e_j \exp(o(s))$, and each lies in a proper flat containing at most $B^{k+1+o(1)}$ sample points. The points assigned to all these carriers therefore have total mass at most

$$B^{d-2k} B^{k+1+o(1)} = B^{d-k+1+o(1)} = o(n_0).$$

Discard them in both roles. No such deletion is made when $k = 1$.

Write S for the resulting pool, $n = |S|$, and $\mathcal{C}$ for its positive-mass physical carriers. Then

$$n = (1 - o(1))n_0 \geq (1 - o(1))N\sqrt{\delta}, \quad n = N \exp(-o(s)). \quad (6.11)$$

Restrict the returned carrier pairs to $\mathcal{C} \times \mathcal{C}$ and call this fixed set $\mathcal{R}$. Its omitted assigned product mass is still $o(n^2)$. The old assigned weights only decreased in the last point deletion, while the

normalizing point count changed by a factor $1 - o(1)$. Thus (6.3) remains valid, with the same ε and an adjusted $o(s)$, for every subfamily of the distinct images

$$Z_{X,Y} = \overline{\psi(X \times Y)}, \quad (X, Y) \in \mathcal{R}.$$

Let U be their reduced union. This is the final product-image family for this endpoint; we will only take subfamilies of it. In particular $\deg U \leq e_j^2 \exp(o(s))$. Restrict Γ_0 to $S \times S$ and to the carrier pairs in $\mathcal{R}$, and call the result Γ . The point and carrier-pair deletions together cost $o(n_0^2)$ pairs, so $|\Gamma| \gtrsim n^2$.

When $k = 1$, the unique carrier of each surviving target is its globally assigned terminal curve from Proposition 5.5. Indeed, the final plateau is then after the $(d - 1)$ st cut. The curve chosen in that global assignment belongs to the full original carrier union used by the common physical equations, so uniqueness at the target identifies it with the current carrier. If that global assignment was omitted because its chosen curve was a nonreal line, uniqueness would make the current carrier that line, and the nonreal-flat point deletion has removed the point. Thus no globally unassigned target remains in this case.

Lemma 6.4 (Regular product images at the endpoint). *For the fixed $\mathcal{R}$ and U above, there is a subfamily $\mathcal{R}_{\text{sm}} \subseteq \mathcal{R}$ whose omitted assigned product mass is $o(n^2)$, and one system of label equations of degree at most B^{2-c} for some fixed $c > 0$, with the following properties. The equations contain every complete image $Z_{X,Y}$ for $(X, Y) \in \mathcal{R}_{\text{sm}}$. Outside $o(n^2)$ further actual pairs assigned to $\mathcal{R}_{\text{sm}}$, their common zero locus is smooth of dimension $2k$ at the label, the product map has differential rank $2k$, and the connecting direction is transverse to both physical tangent spaces. If $k \geq 2$, the two projected tangent spaces are also unequal there. The equations and every polynomial test used for these actual-pair conclusions have degree at most B^{2-c} .*

Proof. Put $T_i = T_s(t_i)$ and form

$$V = \text{Env}_{T_0}(U).$$

For each distinct image $Z \subset U$, choose once an irreducible component $W(Z)$ of V containing the whole image. Every assigned pair represented by Z inherits that choice. Since $\dim Z = 2k$, a chosen component has dimension at least $2k$; equality implies $W(Z) = Z$.

We first show that the images assigned to larger components represent only $o(n^2)$ product mass. Otherwise, after taking a subsequence, some dimension $h \geq 2k + 1$ group represents a fixed positive normalized mass. Form an auxiliary graph Λ_h from all actual ordered pairs in those assigned products, omitting the diagonal where ψ is undefined. Disjointness of the assignments makes its size equal to that product mass before the omission, and the diagonal has at most n pairs. Thus Λ_h is dense in $S \times S$. This graph is independent of the covered graph Γ ; the arbitrary-graph scope of Lemma 6.3 applies to it. Every label of Λ_h lies in the chosen dimension- h component containing its complete product image.

Write a_h for the total degree of the dimension- h component group of V , including components receiving no images. Its upper Hilbert bound at an earlier degree T' controls $H_{\psi(\Lambda_h)}(T')$, while its lower envelope bound gives $H_V(T_0) \gtrsim a_h T_0^h$. Borrow the common-window label rank at a time just before β , with the fixed endpoint tolerance chosen above. If $\beta = -1$ and no earlier testing degree is at least one, use the rank-one boundary convention in Lemma 6.3; then all j earlier cut times equal -1 and the required boundary rank is one to exponential accuracy. On the final plateau $\sigma(b) = e_j T_s(b)^k$. Propagating with $h \geq 2k + 1$ therefore gives

$$H_V(T_0) \geq e_j^2 T_0^{2k} \min \left\{ \frac{T_0}{T_s(\beta)}, \frac{T_0}{D_1 D_2} \right\} \exp(-\xi s - o(s)),$$

where the fixed $\xi > 0$ can be smaller than both gaps specified after (6.10). The two factors have normalized logarithms $t_0 - \beta$ and $t_0 - (1 + b_1 + b_2)$, respectively, so the lower bound has a strict positive margin. But envelope equality and the upper Hilbert bound on the pure $2k$ -dimensional family U give

$$H_V(T_0) = H_U(T_0) \leq C e_j^2 T_0^{2k} \exp(o(s)).$$

This contradiction proves the claimed product-mass deletion.

Let $\mathcal{I}$ be the distinct images left after that deletion. Each is an actual irreducible $2k$ -dimensional component of the single locus V , defined by equations of degree at most T_0 ; larger components elsewhere in V are kept in those equations. Also $\deg \mathcal{I} \leq \deg U \leq e_j^2 \exp(o(s))$. Apply Lemma 3.8 to this pure component family at degree T_1 . Its equations vanish on the whole family $\mathcal{I}$. The exceptional image subfamily $\mathcal{E}$ has total degree at most

$$C \deg(\mathcal{I}) \frac{T_0}{T_1} \leq e_j^2 \exp(-(t_1 - t_0)s + o(s)).$$

Use (6.3) for this particular subfamily of the already fixed U . Its represented normalized pair mass is at most

$$\exp(-(t_1 - t_0 - \varepsilon)s + o(s)) = o(1).$$

Here the same ε -lists chosen before U are used; no new generic-depth deletion depends on $\mathcal{E}$. Let $\mathcal{R}_{\text{sm}}$ consist of the carrier pairs represented by $\mathcal{I} \setminus \mathcal{E}$. The two image deletions have omitted only $o(n^2)$ assigned product mass.

The regularity lemma supplies a common polynomial g of degree $O(T_1)$, a linear combination of maximal Jacobian minors, which is nonzero on each image in $\mathcal{I} \setminus \mathcal{E}$. The single rational pullback $g \circ \psi$ has a cleared numerator of degree $O(T_1)$ on the chosen charts, nonidentically zero on every carrier pair in $\mathcal{R}_{\text{sm}}$. Product testing, with its internal times in (t_1, t_2) , removes its actual zeros at a cost of $o(n^2)$ pairs. At every surviving label the common equations are locally a smooth $2k$ -fold. They still vanish on all of $\mathcal{I}$, including its exceptional components.

It remains to test the two tangent injectivities at actual pairs. Write f_a for the common physical equations chosen before τ , and $v = q - p$. At a point $r \in S$, their Jacobian has kernel T_r , the tangent space of its unique carrier. Thus $v \notin T_r$ exactly when at least one polynomial $\nabla f_a(r) \cdot v$ is nonzero. These tests have degree $O(T_s(\tau))$ in (p, q) . On every retained carrier pair the source collection and the target collection each contain a generically nonzero test, by the fixed generic-depth deletion. For each endpoint, choose a generic constant linear combination nonzero on every retained product. Product testing removes its actual zeros at cost $o(n^2)$. The target injectivity, together with the source coordinate of ψ , gives differential rank $2k$; the source injectivity is retained separately for the secant step.

If $k \geq 2$, every remaining physical carrier spans dimension at least $k + 2$ by the preceding carrier deletion. Equality of the two projected tangent spaces at a general retained pair would put both carriers in a common $(k + 1)$ -flat by Lemma 4.1, which is impossible. This inequality also has a bounded-degree test using the same physical equations. Put $c' = d - k$ and write $a_i(r) = \nabla f_i(r)$. On the full-rank transverse locus the row vectors

$$(a_i(r) \cdot v) a_j(r) - (a_j(r) \cdot v) a_i(r)$$

span the $(c' - 1)$ -dimensional annihilator of $T_r + \mathbb{C}v$. Indeed, they lie in that annihilator, and choosing one row with nonzero value on v expresses every row combination annihilating v in their span. The two projected tangent spaces are unequal exactly when the combined row spans for $r = p, q$ have rank at least c' . Their $c' \times c'$ minors are polynomials of degree $O(T_s(\tau))$. Choose a

generic combination nonzero on every retained product, and apply product testing once more to remove its actual zeros. All these tests fit in the interval reserved below t_2 . Since $s = \log B$ and $t_2 < 1$, their degrees and the common label-equation degrees are at most B^{2-c} for some fixed $c > 0$. This proves the lemma. $\square$

Restrict Γ by the carrier-pair and actual-pair deletions in Lemma 6.4. It still has $\gtrsim n^2$ pairs.

Fix a center $p \in S$. If a component γ of C_p is not contained in the specialized common label equations, choose one such equation nonzero on γ . Bézout bounds its intersections with γ by $O(\deg \gamma B^{2-c})$. Summing over these components gives $O(EB^{2-c})$ directions. Each supporting physical line has at most $B^{o(1)}$ sample points, by the light-line restriction on Γ . The corresponding target fraction is therefore at most

$$C \frac{EB^{2-c+o(1)}}{n} \leq C\sqrt{\delta} B^{-c+o(1)} = o(1).$$

This is uniform in p , so discarding all these incidences costs $o(n^2)$.

For a remaining component $\gamma \subset C_p$, take a marked pair (p, q) in the graph with $\pi_p(q) \in \gamma$. Let X, Y be its unique physical carriers. Near this label the common equations give a smooth $2k$ -fold W , and $\psi : X \times Y \rightarrow W$ has full differential rank at (p, q) . The analytic inverse function theorem therefore gives a local biholomorphism. On the affine open containing q where π_p is regular, consider the algebraic preimage $Y \cap \pi_p^{-1}(\gamma)$. Because ψ records the source coordinate, its local inverse identifies the reduced analytic germ of this preimage at q with the germ of γ at $\pi_p(q)$. That germ is pure one-dimensional. Algebraic and analytic local dimensions agree over $\mathbb{C}$, so an algebraic irreducible component of the preimage containing any selected inverse branch is a curve through q . Let C be its closure in Y . The selected branch projects nonconstantly to γ ; hence $\overline{\pi_p C} = \gamma$, since γ is irreducible.

If $k = 1$, this curve is the target carrier Y itself. The identity with the global terminal assignment established above now gives $\overline{\pi_p Y} = \gamma \subset C_p$, exactly an incidence removed by Proposition 5.5. This resolves $k = 1$.

Suppose $k \geq 2$. For each remaining marked pair over γ , choose one such irreducible lift, and pool repeated physical curves. Retain its carrier pair $(X, Y) \in \mathcal{R}_{\text{sm}}$. The physical minor test $g_{\text{ph}}(q)$ and the label and tangent tests are nonzero at its marked point, so their restrictions are nonzero on the lift and hold on a nonempty open part of it. Choose one general direction on γ , common to this finite family, avoiding the exceptional loci, branch values, and images of intersections between distinct lifts. Each lift then contributes its generic map degree in distinct roots, and roots from distinct lifts are also distinct.

There can be at most one such root in total. At their common label the same label equations are smooth of dimension $2k$ and contain every complete product image attached to these lifts. All products use the same source carrier X : it is the unique member of the full physical family at p . If two roots q', z' were distinct, the first product would have full differential rank at (p, q') , and the complete second product would also lie in the same local label system. The particular-pair assertion of Lemma 4.1 would then give

$$\overline{T_p X} \subseteq \overline{T_{q'} Y} \quad \text{in } \mathbb{C}^d / \mathbb{C}(q' - p).$$

Both spaces have dimension k by the two tangent-injectivity tests, so they would be equal, contrary to the tangent-inequality test. Hence

$$\sum_{C \text{ distinct chosen lifts over } \gamma} \deg(C \rightarrow \gamma) \leq 1. \quad (6.12)$$

Finally, each lift loses at most half its physical degree under projection. If $p \in C \subset Y$, uniqueness for the full original physical family at p forces $Y = X$. The carrier X is smooth at p , and the marked point $q \in C$ has $q - p \notin T_p X$ by the source tangent test. Corollary 3.14 therefore gives $\text{mult}_p C \leq \deg C/2$. If $p \notin C$, the multiplicity is zero. In both cases the point-projection formula yields

$$\deg \gamma \deg(C \rightarrow \gamma) = \deg C - \text{mult}_p C \geq \frac{1}{2} \deg C.$$

Together with (6.12), this bounds the total degree of the lifts over γ by $2 \deg \gamma$. Every remaining target in the row lies on a chosen lift. The physical curve bound in Lemma 3.1, summed over $\gamma \subset C_p$, bounds their number by CEM . But

$$\frac{EM}{n} \leq C\sqrt{\delta} \frac{M}{A} \rightarrow 0.$$

Every remaining row thus has $o(n)$ partners, contradicting the surviving $\gtrsim n^2$ graph. This resolves the common outermost endpoint.

To treat separated windows, we next bound the projected degree at the actual source centers.

6.3 Projection depth for separated windows

We retain the notation and contradiction regime (2.1). Let P_x, P_y be the two pieces selected by Proposition 5.5, in separated windows of widths v and s , respectively. Thus $v, s \rightarrow \infty$, both are $o(\log B)$, and either $v/s \rightarrow 0$ or $v/s \rightarrow \infty$. The limiting normalized drop profile is the same for both pieces. Before using the target plateaux, omit from P_y the points left unassigned by the global terminal assignment in Proposition 5.5. Its proof bounds their number by $o(|P_y|)$ uniformly over the pieces, so this removes $o(|P_x||P_y|)$ ordered pairs. Continue to write P_y for the retained target set. It has size $N \exp(-o(s))$ and inherits the profile conclusions, since it is a $(1 - o(1))$ fraction of the selected piece. The proposition below uses this retained P_y .

For the target piece write

$$D_i = B \exp(b_i s), \quad e_j = \prod_{i \leq j} D_i, \quad l = d - j,$$

and consider a positive-dimensional plateau after at least one drop. The path equations defining its current system have degrees $B \exp(O(s))$. For this plateau, fix one containing l -dimensional carrier for each point of P_y . On the final one-dimensional plateau use its carrier from the global terminal assignment. Pool repeated carriers and denote the resulting family by $\mathcal{Y}$, with disjoint assigned target sets of masses n_Y . Thus $\sum_{Y \in \mathcal{Y}} n_Y = |P_y|$. All later incidence and list restrictions use this fixed assignment. Our objective is a lower bound for the degree of the distinct projected components carrying any fixed positive fraction of target mass. Several components may share a projected image, and each projection may have degree greater than one. We control this combined multiplicity after deleting negligible incidence mass.

Give a center-component incidence (p, Y) normalized weight $n_Y/(|P_x||P_y|)$. The weight of a collection is thus the probability that a uniformly chosen center and a uniformly chosen assigned target point determine an incidence in that collection.

We use the following precise interpretation of exponential accuracy. An assertion with a factor $\exp(o(s))$ in a lower bound means that for every fixed $\epsilon > 0$ it holds with the smaller factor $\exp(-\epsilon s)$ for all sufficiently large members of the sequence, after the stated vanishing-mass deletions. All finite choices of plateaux, density constants, and positive logarithmic tolerances can be imposed simultaneously.

Proposition 6.5 (Projected degree in separated windows). *For each fixed positive tolerance ϵ , one can discard center–target–component incidences of total normalized weight $o(1)$, and then discard $o(|P_x|)$ centers, so that the following holds. For each remaining center p , every target component Y whose incidence (p, Y) remains satisfies $\dim \pi_p \bar{Y} = l$. For every subcollection of those components carrying at least a fixed positive fraction of $|P_y|$, the distinct projected components have summed degree at least*

$$\frac{e_j}{R} \exp(-\epsilon s). \quad (6.13)$$

The assertion is uniform over those subcollections, including ones chosen after p and after subsequent equations have been chosen. The following values of R are valid:

$$R = D_1 \quad \text{in either orientation;} \quad (6.14)$$

$$R = B \exp \left(-s \sum_{i>j} b_i \right) \quad \text{if } v/s \rightarrow 0; \quad (6.15)$$

$$R = B \exp(-2s) \quad \text{if } v/s \rightarrow \infty, \quad b_{j+1} > 0, \quad l < d-1. \quad (6.16)$$

In (6.13), “fixed positive fraction” means any constant $\rho > 0$ independent of the sequence. After fixing ϵ and any finite batch of plateaux and tolerances, the incidence and center deletions are independent of the later fixed choice of ρ ; only the large-index threshold may depend on ρ .

Proof. We first explain the degree and mass bookkeeping common to all three cases, then prove their depth bounds.

Components, multiplicity, and incidence deletions. Proposition 5.2 gives

$$\sum_{Y \in \mathcal{Y}} \deg Y \leq e_j \exp(o(s)), \quad \sum_{Y \in \mathcal{A}} n_Y \leq \deg(\mathcal{A}) \left(\prod_{i>j} D_i \right) \exp(o(s)) \quad (6.17)$$

for every subcollection $\mathcal{A}$. Since $\sum_i b_i = 0$ and $|P_y| = N \exp(-o(s))$, every subcollection of fixed positive assigned mass has degree at least $e_j \exp(-o(s))$.

Every $Y \in \mathcal{Y}$ is an irreducible component of the fixed target path locus, by the component argument at the start of the proof of Proposition 5.3. Larger components elsewhere remain in the target equations.

Projection from p can drop the dimension of Y only if its generic fiber is a line. For a fixed general smooth $y \in Y$, this puts p in the proper affine tangent flat $y + T_y Y$. The flat cap makes the proportion of such centers at most $B^{-c_0+o(1)}$, uniformly for each Y . Also, Corollary 3.13 puts centers where $\text{mult}_p Y > \deg(Y)/2$ in a hypersurface of bounded degree. The initial full-dimensional no-capture property of the center piece makes their proportion $o(1)$: its initial cutoff is $B^{1+o(1)}$, well above every bounded degree. Average both conclusions with weights $n_Y/|P_y|$. They remove only $o(1)$ of the center–component incidence mass.

At every retained incidence, the point-projection formula gives

$$t_p(Y) \deg \pi_p \bar{Y} = \deg Y - \text{mult}_p Y \geq \frac{1}{2} \deg Y, \quad (6.18)$$

where $t_p(Y)$ is the generic map degree. For a fixed p , group components having the same projected image Z and define their depth to be the sum of their map degrees. If all retained groups have depth at most $R \exp(\eta s)$, then any subcollection of degree a has projected degree at least

$$a/(2R \exp(\eta s)). \quad (6.19)$$

Thus it suffices to bound depth after negligible weighted incidence deletions. Fixed constants and the errors in (6.17) are absorbed by choosing $\eta > 0$ sufficiently smaller than the desired ϵ .

Whenever the deleted incidence mass is $\alpha = o(1)$, Markov's inequality leaves all but $\sqrt{\alpha}$ of the centers with deleted target mass at most $\sqrt{\alpha}$. Every later subcollection of fixed positive mass still has positive mass after this deletion. This is the uniformity mechanism in the proposition. No union bound over the possible subcollections will be needed.

The first-cut bound. Use the system at the first plateau after a drop. Its equations have degrees at most $D_1 \exp(o(s))$, and every current descendant is contained in this first system. The regularity conclusion of Proposition 5.3 makes its relevant carrier unique and smooth at all but $o(|P_y|)$ assigned target points. For each such point q , centers p in its affine tangent flat form a $B^{-c_0+o(1)}$ fraction of the source piece.

If, for a center and a current descendant Y , the generic joining ray were contained in the first system, then the same containment holds for every point of Y where the ray is defined, by polynomial identity. At an assigned regular point q the line would lie locally in its unique first carrier, so its direction would be tangent there. The preceding averaging therefore discards all such center-descendant incidences at negligible mass. On every other generic ray at least one first-system equation restricts nontrivially. Its degree bounds the number of distinct intersections with all current target components by $D_1 \exp(o(s))$. In characteristic zero, the generic fibers defining the map degrees consist of distinct points. This proves the depth bound for (6.14). Only regularity of the carrier is used here; the original path equations need not themselves have full Jacobian rank.

Centers in the narrower window. Suppose $v/s \rightarrow 0$. The center piece can be simultaneously isolated as zero-dimensional components of equations of degree

$$L = B \exp(o(s)), \quad |P_x| = N \exp(o(s)),$$

where the second error is signed. Fix $Y \in \mathcal{Y}$, and choose y general on it, also general for the projections from all finitely many actual centers. For each center, count the additional moving roots $z \neq y, p$ on target components whose images agree with $\overline{\pi_p Y}$. These points are generic on their components. Here this includes avoiding their intersections with the other members of the same finite family $\mathcal{Y}$. Their number is the group depth minus the one fixed root y .

Parametrize such pairs by

$$p = y + t(z - y), \quad z \in \bigcup_{Z \in \mathcal{Y}} Z.$$

Impose the center-isolating equations on this parametrization. At every counted solution the image point p is isolated in the center system. At its root z , the target union is locally just the one member of $\mathcal{Y}$ containing that root, by the preceding genericity condition. The generic ray has finite intersection with that component, so the fiber is locally finite. Thus the solution is isolated on the variety parametrized by (z, t) . This variety has dimension $l + 1$ and degree at most a dimensional constant times $\sum_Z \deg Z$; the pulled-back equations have degree at most $2L$. The isolated-intersection form of Bézout therefore bounds the total number, over all centers, by

$$C e_j \exp(o(s)) L^{l+1}. \tag{6.20}$$

Possible larger components of the center equations away from their isolated sample points do not affect this count.

Divide (6.20) by $|P_x|$. Its value, to the stated accuracy, is

$$\frac{e_j B^{l+1}}{N} = B \exp \left(s \sum_{i \leq j} b_i \right) = B \exp \left(-s \sum_{i > j} b_i \right) = R.$$

The fixed root adds only one; $R = B^{1+o(1)} \rightarrow \infty$. For every fixed logarithmic slack $\eta > 0$, Markov's inequality bounds the proportion of centers whose depth at this Y exceeds $R \exp(\eta s)$ by $\exp(-\eta s + o(s))$. Weighted averaging in Y , followed by the row deletion already described, proves (6.15).

Centers in the wider window: the generic lists. Suppose $v/s \rightarrow \infty$, $b_{j+1} > 0$, and $l < d - 1$. Choose an interior center plateau just to the right of zero, before the first strictly positive center drop. Its carrier dimension is

$$k = \#\{i : b_i > 0\} \geq l, \quad k < d.$$

All componentwise proper cuts of degree $B \exp(Cs)$, with C fixed, are negligible for the center piece. To verify the degree comparison, choose a fixed positive interior testing time τ on this center plateau. Since $s = o(v)$, $B \exp(Cs) = o(B \exp(\tau v))$, and leave a second interior time for the pooling step of Proposition 5.3.

For each $p \in P_x$, fix one carrier $X(p)$ of this center plateau which contains it. The distinct values form a source family $\mathcal{X}$, with disjoint assigned sets $S_X = \{p \in P_x : X(p) = X\}$. Apply the nonreal-flat deletion in Proposition 5.3 to this source assignment and to the fixed target assignment on $\mathcal{Y}$. Use it to omit every nonreal flat member from both input families, and omit any member of zero assigned mass. The removed assignments have source mass $r_x = o(|P_x|)$ and target mass $r_y = o(|P_y|)$. Discard every incidence whose center belongs to the removed source set or whose target component is removed. Their total normalized weight is at most

$$\frac{r_x}{|P_x|} + \frac{r_y}{|P_y|} = o(1).$$

Call the surviving distinct families $\mathcal{X}_0, \mathcal{Y}_0$, keeping their assigned sets, and put

$$n_x = \sum_{X \in \mathcal{X}_0} |S_X| = (1 - o(1))|P_x|, \quad n_y = \sum_{Y \in \mathcal{Y}_0} n_Y = (1 - o(1))|P_y|.$$

Both are at least $N \exp(-o(\log B))$, and every flat member of these families is real. When $k = l$, the degree and assigned-mass budgets, using $v, s = o(\log B)$, give at most $B^{d-l+o(1)}$ flat members in each family and normalized weight at most $B^{-(d-l)+o(1)}$ for each one. Thus Lemma 4.2 applies with error scale $\log B$ and, when $k = l$, with $e = B^{d-l}$. Its removed product weight has normalized incidence weight

$$\frac{n_x n_y}{|P_x| |P_y|} o(1) = o(1)$$

in the incidence normalization fixed above.

The lemma supplies a list $\mathcal{Y}_X \subseteq \mathcal{Y}_0$ for each $X \in \mathcal{X}_0$ such that for every $Y \in \mathcal{Y}_X$, a general line through $x \in X, y \in Y$ has at most

$$q = B^{1/2}$$

moving roots on that list. Using a smaller fixed exponent in the lemma, if necessary, ensures this displayed bound for all sufficiently large B . This is a bound on all possible generic list meetings,

not just on the roots visible in the finite sample. For a fixed general x , a proper subvariety of an l -dimensional listed target has projected dimension less than l , so cannot supply roots over the generic direction image of Y . For a retained actual center p , use only the list indexed by its fixed carrier $X(p)$, further restricted by the earlier incidence deletions at p .

We prove that excessive depth at an actual center is detected by a proper cut on X of degree $B \exp(O(s))$. Put

$$r = \lceil B \exp(-2s) \rceil.$$

Because $s = o(\log B)$, we have $q = o(r)$ and $r = B^{1+o(1)}$. Fix X and $Y \in \mathcal{Y}_X$ for the following construction.

The generic list bound does not yet control actual centers. We encode intersections with a joining line by two binary forms. Excessive retained depth will force a large rank decrease in their Sylvester matrix. Multiplicity compression will then place these centers in a proper cut of degree fitting the center plateau.

A fixed generic target point and two binary forms. Write the current target path locus as

$$Z = V(f_1, \dots, f_m), \quad \deg f_i \leq L_0 = B \exp(O(s)).$$

This is the fixed full path locus, including components omitted from the generic-depth input family. Every retained target is an irreducible component of Z , as proved above. Choose $y \in Y$ general with two sets of conditions. First, the generic list-root bound and the non-tangency conditions must hold for generic $x \in X$. A dense open subset of $X \times Y$ has a dense open fiber over general y , so this is possible. Second, for each of the finitely many retained actual centers p assigned to X and each listed target Y' with l -dimensional image equal to $\overline{\pi_p Y}$, require all points of the generic fiber over $\pi_p y$ to be affine, distinct, different from p , and outside every other irreducible component of Z . The generically finite map from Y' has a dense open image locus with these properties: every excluded proper subset of Y' has image of dimension at most $l - 1$. Pulling back these open sets to Y and intersecting the finite list preserves a nonempty open set. Also avoid the finitely many actual centers. This fixes one y for all the specializations below.

Choose an integer $L \geq \max(L_0, B)$ with $L = B \exp(O(s))$. Homogenize each path equation to its own degree and take all its homogeneous monomial multiples to degree L . Their span has projective base locus exactly $V(f_1^h, \dots, f_m^h)$: at every projective point some monomial of each prescribed degree is nonzero. This base locus may contain extra components at infinity; we retain all of them. The closure of each retained affine target is still a component, since any larger component containing it would meet the affine chart and contradict the component property.

Parametrize the line through x and y by

$$\iota_x([U : V]) = [U : Uy + V(x - y)].$$

For $x \neq y$, this identifies $\mathbb{P}^1$ with the line. The parameters of y, x , and infinity are respectively $[1 : 0], [1 : 1], [0 : 1]$, independently of x . Every restricted equation is a binary form of degree L , whose coefficients are polynomials of degree at most L in x . The general line is not contained in the base locus. At the chosen smooth point y , outside its other components, such containment would put its direction in $T_y Y$, contradicting the retained non-tangency condition.

Over $K = \mathbb{C}(X)$, let H be the greatest common divisor of the restricted system and put $g = \deg H \leq L$. After division by H the binary linear system has no base point. Two generic members of it have no common zero: choose one nonzero member, then choose the second to avoid its finitely many zeros. Consequently two generic constant complex linear combinations of the

original degree- L forms restrict to binary forms F_x, G_x whose generic gcd is exactly H . The good choices form a nonempty algebraic open set over K and admit constant complex choices: expanding the coefficients of a nonzero polynomial over K in a complex-linearly independent basis shows that it cannot vanish on every tuple of complex constants. This also covers $g = L$, when the residual forms are constants.

The Sylvester matrix and specialization of its divisor. Let $S(x)$ be the $2L$ -square matrix of

$$K[U, V]_{L-1} \oplus K[U, V]_{L-1} \longrightarrow K[U, V]_{2L-1}, \quad (a, b) \longmapsto aF_x + bG_x.$$

Its entries have ambient degree at most L in x , and its generic rank on X is $\rho = 2L - g$. Indeed, after writing $F_x = Hf$ and $G_x = Hh$ with f, h coprime, the kernel consists of $(hQ, -fQ)$ with $\deg Q = g - 1$, and has dimension g (zero if $g = 0$).

Consider an actual center p assigned to X at which the retained depth over $\overline{\pi_p Y}$ exceeds r . By the choice of y , at least $r - O(1)$ distinct common roots of F_p, G_p are affine points of retained components, different from p, y , and each lies on no other component of the fixed base locus. Call these distinguished roots.

The generic gcd divisor specializes with its full length, even when X is singular at p . To see this explicitly, choose an irreducible algebraic curve through p whose general point lies in the open set where the preceding generic properties, including the gcd degree, hold. To obtain it, take successive general linear sections through p until the local dimension is one, avoiding every positive-dimensional component of the excluded closed set at each step. That closed set then has local dimension at most zero. Choose an irreducible curve component through p ; it meets the good open set. Normalize it and use the discrete valuation ring at a point above p . Over its fraction field choose the gcd form, and scale it so its coefficients are integral and at least one is a unit. Its reduction H_0 is a nonzero homogeneous form of degree g . It divides both specialized forms. In fact, the quotients are integral: the content law for polynomials over a discrete valuation ring says that a primitive factor times a quotient of negative coefficient valuation cannot have all coefficients integral. Reducing the integral factorizations proves divisibility. Thus H_0 defines a common limiting divisor D_0 of length g on $\mathbb{P}^1$.

At most q distinguished roots lie in the support of D_0 . For this assertion, pass to a finite extension splitting the generic gcd and a valuation ring above the chosen one. Each distinct generic root has a unique projective specialization, obtained by scaling its two coordinates to be integral with one a unit. Each physical root belongs to some fixed irreducible component of the projective base locus. Closedness forces its specialization to remain in that component. A distinguished point lies only on its own retained component, so a branch specializing to it must already have come from that retained component. It cannot be either immobile endpoint, whose fixed parameters specialize to y and p , nor a root fixed at infinity. The generic list bound permits at most q remaining distinct roots. Their specializations have at most q support points, regardless of the multiplicities carried by the gcd divisor.

If F_p, G_p are not both zero, their common divisor therefore has degree at least

$$g + r - O(1) - q.$$

This includes the case that exactly one form is zero: its common divisor with the other is the other degree- L form, and the matrix has rank L . In all these cases

$$\rho - \text{rank } S(p) \geq r - q - O(1) \geq r/2$$

for large B . If both forms vanish, $S(p) = 0$ and the rank decrease is $\rho \geq L \geq r$, giving the same conclusion. Working throughout with binary forms of degree L records any loss of an affine leading coefficient as a root at infinity; hence that phenomenon introduces no exception to the comparison.

We have shown that depth exceeding r forces a matrix rank decrease of at least $r/2$. It remains to turn this into a proper polynomial cut of degree $B \exp(O(s))$, to which center no-capture applies.

Multiplicity compression and the center no-capture bound. Choose a ρ -square minor $Q(x)$ nonzero generically on X . It has degree at most $2L^2$ and has ambient multiplicity at least $r/2$ at every excessive actual center. For clarity, at such a center the selected submatrix has rank at most $\rho - r/2$. Constant invertible row and column operations put its value into rank normal form. At least $\lceil r/2 \rceil$ rows then have every entry in the ambient maximal ideal at p , so every determinant term belongs to its $\lceil r/2 \rceil$ -th power. No smoothness of X , or agreement between ambient generic rank and generic rank on X , is needed for this conclusion.

Choose a test point of X at which Q is nonzero. The multiplicity compression lemma, Lemma 3.10, produces a polynomial which vanishes at all these excessive centers, is nonzero at the test point, and has degree

$$O(L^2/r) = B \exp(O(s)).$$

It is consequently a proper cut on X . For each fixed Y , these cuts can be chosen separately on the center components X with $Y \in \mathcal{Y}_X$. Their degree bound is uniform. The componentwise pooling and no-capture statement on the chosen center plateau makes the captured center mass $o(|P_x|)$. This is uniform in the choice of Y : a contrary sequence of targets would contradict the same no-capture statement, which allows arbitrary polynomials of the specified degree. Averaging with the normalized weights $n_Y/|P_y|$ therefore removes only $o(1)$ of all incidences.

On the remaining incidences, every group in the retained part of $\mathcal{Y}_{X(p)}$ at p has depth at most r . Deleting further targets cannot increase depth. Apply the row deletion and (6.19). Since rounding r costs a factor $1 + o(1)$, this proves (6.16) and completes the proof of the proposition. $\square$

Remark 6.6. The proof bounds incidence mass before discarding centers. It does not claim a uniform depth estimate for every individual original target at every center. The conclusion needed later is that, after one fixed deletion, every target subcollection of fixed positive row mass satisfies the projected-degree bound. This permits adaptive choices such as the family on which later regularity equations fail. It also explains why no bound on the number of target components enters a union bound.

6.4 Rank propagation for separated windows

Let v be the source width and s the target width, with $v/s \rightarrow 0$ or $v/s \rightarrow \infty$. Both are $o(\log B)$. Their limiting normalized profile (b_i) is the same, but T_s, D_i, e_j, σ below refer only to the target window. The source and target piece sizes are respectively $N \exp(-o(v))$ and $N \exp(-o(s))$; we do not replace the former error by $o(s)$ when $v \gg s$.

Use the retained target set and fixed plateau assignments in Proposition 6.5. Its final one-dimensional assignment is the persistent global terminal assignment, after the negligible omission of globally unassigned targets. Apply the proposition for the finite batch of plateaux and logarithmic tolerances needed in an argument, take the union of its negligible incidence deletions, and apply row Markov once. Call the remaining source centers typical. This set is chosen independently of the later fixed density threshold $\rho > 0$; only the large-index threshold may depend on ρ . Thus the same typical centers serve every later subcollection of fixed positive target mass, including one chosen after the center and a testing-degree envelope. On a plateau of dimension $l = d - j$, every retained individual carrier projects with dimension l , and such a subcollection has distinct projected degree at least

$$(e_j/R) \exp(-o(s)). \tag{6.21}$$

The permitted depth denominators are

$$R = D_1 \quad \text{always,} \quad (6.22)$$

$$R = B \exp \left(-s \sum_{i>j} b_i \right) \quad \text{if } v/s \rightarrow 0, \quad (6.23)$$

$$R = B \exp(-2s) \quad \text{if } v/s \rightarrow \infty, \ b_{j+1} > 0, \ l < d-1. \quad (6.24)$$

Here a row consists of the targets paired with one fixed center p . For a chosen target subset S in that row, $H(T)$ now denotes $H_{\pi_p(S)}(T)$, the restriction rank of its directions.

Lemma 6.7 (Strict rank margin on target plateaux). *At typical centers, ranks of directions to arbitrary dense remaining target subsets satisfy*

$$H(T_s(b)) \geq \frac{\sigma(b)T_s(b)}{A} \exp(cs) \quad (6.25)$$

with a fixed $c > 0$ through all plateaux up to the final endpoint, except possibly the final one-dimensional plateau when the center window is narrower. The constant may be decreased finitely many times during propagation. The assertion is needed only at fixed interior testing times and their one-sided limiting endpoints.

Proof. On the initial interval target no-capture, applied to the pullback of a direction polynomial, gives rank comparable to T^{d-1} . Its ratio to $\sigma(b)T/A$ is

$$\frac{A}{T^2} = \exp(-2bs).$$

This has a strict positive margin near the first cut, since $b_1 < 0$.

On a plateau of dimension l , consider the testing-degree envelope of the directions to the current row subset. For each fixed assigned target carrier whose projection is not contained in this envelope, a generic linear combination of its pulled-back equations is a proper cut on that carrier. The combination can be chosen simultaneously for the finite carrier list. Use a slightly later interior time for the fixed pullback and pooling costs. Every counted target on such a carrier lies in the cut. The individual-cut conclusion of Proposition 5.3 therefore removes only $o(|P_y|)$ such targets, uniformly in the center and the envelope. Hence the complete projection of the assigned carrier is contained in the envelope at almost all remaining counted targets.

For this fixed envelope, choose one irreducible component containing each distinct projected carrier occurring at those targets, and let the targets inherit the choice for their projected carrier. Group all envelope components by dimension, counting targets by these choices; components receiving no targets remain in the degree of their group. Some group of dimension $h \geq l$ carries a fixed positive fraction of the target mass.

If $h > l$, that group borrows the preceding bound with growth exponent $h \geq l+1$. This preserves the required growth exponent of σT on the plateau. Approximate the preceding endpoint closely enough that the lost logarithmic tolerance is a small fraction of the existing c .

If $h = l$, each projected carrier assigned to this group equals its chosen envelope component, since both are irreducible of dimension l . The projected degree lower bound (6.21) and the envelope Hilbert bound give

$$H(T) \gtrsim T^l e_j / R \exp(-o(s)).$$

Relative to $\sigma T/A = e_j T^{l+1}/A$, this supplies the ratio $A/(RT)$, to exponential accuracy. It suffices to have a strict negative normalized logarithm for RD_{j+1}/A , the worst point at the next endpoint. The three choices give, respectively,

$$b_1 + b_{j+1}, \quad - \sum_{i>j+1} b_i, \quad b_{j+1} - 2. \quad (6.26)$$

The first is strictly negative on the hypersurface plateau, by $b_1 + b_2 < 0$, and whenever $b_{j+1} \leq 0$. For wider centers all remaining cases have $b_{j+1} > 0$ and $l < d - 1$, so the third choice applies and has margin at least one. For narrower centers and $b_{j+1} > 0$, the second choice has a strict margin when $l \geq 2$, since its sum includes a positive later entry. Only $l = 1$ gives zero rather than a strict margin.

There are at most d distinct cut times. Choose all endpoint approximations and fixed logarithmic tolerances smaller than a sufficiently small fraction of the finitely many positive margins just displayed. Decrease c when necessary. This completes the propagation. The degree bounds and prior rank statements quantify over every dense restriction, so choosing a different dimension group or dense subset for different centers does not change the argument. $\square$

For directions covered by C_p the upper bound is

$$H(T) \leq CE(T + 1). \quad (6.27)$$

If the strict margin reaches the last endpoint, choose a fixed time just below it with $\sigma(b)/N \geq \exp(-cs/2)$ to exponential accuracy. Equations (6.25) and (6.27) then have a ratio tending to infinity, at least $\delta^{-1} \exp(cs/3)$ for large indices. This contradicts the positive density of covered rows.

6.5 The final curve plateau for narrower centers

At the remaining zero-margin endpoint, all but $o(|P_y|)$ of the covered targets in each typical row will have their fixed assigned terminal curve project onto a component of the covering family C_p . These are exactly the incidences excluded by the global terminal deletion.

The only remaining case has $v/s \rightarrow 0$ and a final one-dimensional target plateau, $b_{d-1} < b_d$. The preceding plateau has the strict margin in Lemma 6.7. Choose a fixed time $b_0 < b_d$ sufficiently close to b_d within this last plateau, and then choose fixed times

$$b_0 < b_{\text{eq}} < b_{\text{test}} < b_d.$$

Put $\eta = b_{\text{eq}} - b_0 > 0$, $T_0 = T_s(b_0)$, and $T_1 = T_s(b_{\text{eq}}) = T_0 \exp(\eta s)$. Fix $0 < \varepsilon < \eta$. Include this tolerance for the final plateau in the finite depth batch, and use the typical-center set for the combined batch. Its deletion is made before the envelopes below are chosen. Apply the preceding rank argument with this combined finite batch; its bounds then hold for every fixed positive-density restriction. Only now, for each typical center, let

$$V_p = \text{Env}_{T_0}(C_p)$$

be the envelope of the entire directional family, not just of its observed incidences. Its testing rank is at most $CE(T_0 + 1)$.

First, a component group of V_p of dimension $h \geq 2$ cannot contain covered targets of mass at least $\rho|P_y|$ for any fixed $\rho > 0$. Indeed take that actual subset and borrow its strict rank bound from just before the final plateau. The Hilbert bounds for the envelope group propagate this

with exponent $h \geq 2$, exactly the required exponent of σT on a one-dimensional plateau. At b_0 sufficiently close to b_d it contradicts (6.27). If their target mass were not $o(|P_y|)$ uniformly over the typical centers, a subsequence would supply one such fixed ρ and contradict this bound on the same typical-center set. Thus we may discard all actual labels on the larger components at a cost of $o(|P_y|)$ per row, including labels at their intersections with curves; we do not try to cut those intersections by equations of possibly excessive degree.

Let $\mathcal{F}_p$ be the family of every one-dimensional irreducible component of V_p , whether or not it is already represented by an assigned target projection. We will impose common equations on this whole family so that they also contain any covering component through a surviving label. The envelope Hilbert lower bound for its entire one-dimensional component group and the rank upper bound imply

$$\deg \mathcal{F}_p = O(E). \quad (6.28)$$

We next place the projection of the fixed assigned terminal curve in this family. For each such carrier Y whose projection is not contained in V_p , a generic linear combination of the pulled-back envelope equations is a proper cut on Y . Choose the combination simultaneously for the finite carrier list, and use a slightly later interior time for the fixed pullback and pooling costs. Every covered target assigned to one of these carriers lies in that cut. The individual-cut conclusion of Proposition 5.3 therefore removes only $o(|P_y|)$ such targets, uniformly in p . This applies to the persistent global assignment fixed before the depth bounds, including targets at physical carrier intersections.

At every remaining covered target its assigned projection is contained in V_p . It has dimension one because we retained the depth proof's deletion of every dimension-dropping incidence. An irreducible component of V_p containing this projected curve cannot have higher dimension: it would also contain the observed label, which was excluded from every such component. Hence the assigned projection is itself a member of $\mathcal{F}_p$.

The components of $\mathcal{F}_p$ are isolated components of the equations of degree T_0 defining V_p , even though that locus may also have higher-dimensional components elsewhere. Lemma 3.8 gives equations of degree at most T_1 vanishing on the whole family $\mathcal{F}_p$, with generic full rank except on components of total degree

$$O(ET_0/T_1) = O(\delta(N/A) \exp(-\eta s)). \quad (6.29)$$

These exceptional components carry negligible target mass. This assertion uses the dense-subcollection uniformity of the depth lemma: on this plateau

$$\frac{e_{d-1}}{R} = \frac{N/D_d}{B \exp(-b_d s)} = \frac{N}{A}.$$

Thus any subcollection carrying a fixed positive fraction of $|P_y|$ has projected degree at least $(N/A) \exp(-\varepsilon s)$ for the fixed tolerance selected before the depth deletion. Compare with (6.29). The ratio is $O(\delta \exp(-(\eta - \varepsilon)s)) \rightarrow 0$. Indeed, take the physical assigned carriers whose projections belong to the exceptional family. Their distinct projected degree is bounded by that family's degree, so they cannot carry a fixed positive fraction of $|P_y|$. If their mass were not $o(|P_y|)$ uniformly over typical centers, one fixed positive density along a subsequence would contradict this comparison. This applies to the family selected by the envelope and its equations, without changing the fixed target assignment or the earlier depth deletion.

The identity $RD_d = A$ explains the room in this comparison: the projection depth and the last physical cut together cost A , while regularity supplies the strict gap $\eta - \varepsilon > 0$ in the logarithmic

degree scale. One must not instead compare $(N/A) \exp(-o(s))$ directly with $\delta N/A$; the latter comparison would be unjustified for arbitrarily slowly vanishing δ .

The regularity lemma supplies one polynomial g_p of degree $O(T_1)$, a linear combination of maximal Jacobian minors, which is nonzero on every retained projected component. Its pullback to each corresponding fixed assigned physical curve is therefore a proper rational test, with cleared numerator of degree $O(T_1)$. Apply the individual-cut conclusion of Proposition 5.3, using the reserved time b_{test} , to remove the zeros of these numerators at $o(|P_y|)$ targets. This is uniform in the center. At every surviving label $g_p \neq 0$, so the common equations have full rank there.

At a surviving covered label, take a component γ of C_p through it. Since $C_p \subset V_p$, this curve lies in a component of V_p . That component cannot have larger dimension, by the preceding exclusion. Hence γ is a member of $\mathcal{F}_p$. The common equations have full rank at the label, so this member equals the projected member of $\mathcal{F}_p$ through the same label. That member is the projection of the persistent assigned terminal curve. Thus this curve projects onto $\gamma \subset C_p$, exactly the terminal incidence removed in Proposition 5.5, a contradiction.

Proof of Theorem 6.1. The profile-pair selection exhausts the possibilities. A positive core is impossible by its isolation degree. The common-window rank bound contradicts coverage except at the outermost endpoint, where the lift argument gives the contradiction. For separated windows, strict row rank propagation contradicts coverage except on the final narrow-center curve plateau, which the preceding argument resolves by common regularity and terminal deletion. Thus the proposed almost-complete sparse-cone coverage cannot exist. $\square$

7 Very rich rigid motions in three dimensions

In this section a rigid motion is an affine map $g(x) = Sx + h$, where $S \in O(3)$; both orientations are included. For finite clouds $X, Y \subset \mathbb{R}^3$, put

$$X_g = \{x \in X : g(x) \in Y\}, \quad k_g(X, Y) = |X_g|.$$

When $X = Y = P$, write simply k_g . We shall prove the following finite statement. Its hypothesis holds eventually along the contradiction sequences used in the main proof.

Theorem 7.1 (Very rich motions). *There are absolute constants $C_{\text{rich}}, K_{\text{rich}} > 0$ with the following property. Let $P \subset \mathbb{R}^3$ consist of $N \geq 2$ distinct points, and set*

$$A = N^{2/3}, \quad M = 1 + |\Delta(P)|.$$

If $M \leq A$, then the motions satisfying $k_g \geq C_{\text{rich}}A$ form a finite set and

$$\sum_{g: k_g \geq C_{\text{rich}}A} k_g^2 \leq K_{\text{rich}} \frac{N^4}{A}. \quad (7.1)$$

The same estimate holds with $k_g = |gP \cap Q|$ for any rigidly congruent copy Q of P .

In particular, the constants in (7.1) do not depend on the rate at which M/A tends to zero. The finite hypothesis $M \leq A$ will be used only through the elementary line cap

$$|P \cap L| \leq 2M \leq 2A \quad \text{for every affine line } L. \quad (7.2)$$

Indeed, after fixing a point of $P \cap L$, each nonzero distance from it occurs at most twice on L . Thus the sharper bound $1 + 2(M - 1)$ also holds.

We shall prove the same assertion for two clouds X, Y of at most N points, with $A = N^{2/3}$, whenever every line contains at most $2A$ points of either cloud: the motions with $k_g(X, Y) \geq C_{\text{rich}}A$ form a finite set, and their squared match counts satisfy (7.1) with $k_g(X, Y)$ in place of k_g . Both orientations remain included. The line cap above supplies this hypothesis for P and every congruent copy. We will divide the clouds into small regions so that each motion has matches in a controlled number of region pairs. Within each pair, a local estimate handles motions whose matches are not concentrated on a line; we count the remaining local contributions through lines in the full clouds and their possible images.

The passage from equal-distance configurations to rigid motions is the Elekes–Sharir framework [11], used by Guth and Katz in the planar problem [16]. The physical and opposite-orientation flat families below are the rigid-motion geometry developed by Tidor, Yu, and Zakharov [26, Section 3]. We give the construction and every estimate needed here. The incidence inputs are the Szemerédi–Trotter bound, Theorem B.2, and the higher-richness Guth–Katz bound, Theorem B.5.

7.1 Two elementary consequences of planar incidence bounds

We use the following form of Beck’s dichotomy [5], derived from Szemerédi–Trotter, including for real projective direction sets.

Lemma 7.2 (Many lines and many anchors). *There are absolute constants $\delta_0, \gamma_0 > 0$ such that the following holds. If a finite set of n distinct real points has at most $\delta_0 n$ points on any line, then it spans at least $\gamma_0 n^2$ distinct lines. Moreover at least $\gamma_0 n$ of its points each lie on at least $\gamma_0 n$ of these spanned lines. The assertion holds in any fixed real affine space, and in a real projective plane after choosing a generic affine chart.*

Proof. A generic real planar projection preserves the distinct points and all noncollinear triples of a finite affine configuration. It therefore suffices to work in the plane. Theorem B.2 gives

$$\#\{L : |L \cap X| \geq u\} \leq C_0 \left(\frac{n^2}{u^3} + \frac{n}{u} \right), \quad u \geq 2. \quad (7.3)$$

For a large fixed integer u_0 , sum over dyadic occupancies $u = 2^j u_0$, up to the maximum occupancy $\delta_0 n$. The number of ordered pairs of distinct points on lines in these buckets is at most

$$C_1 \sum_u \left(\frac{n^2}{u} + nu \right) \leq C_2 \left(\frac{n^2}{u_0} + \delta_0 n^2 \right).$$

Choose u_0 large and then δ_0 small, both absolute, so this is less than one half of all ordered pairs. A positive fraction of the pairs consequently lie on lines containing fewer than u_0 points. Each such line accounts for fewer than u_0^2 ordered pairs, so there are at least $c_0 n^2$ spanned lines for an absolute $c_0 > 0$. The finitely many small values of n cause no difficulty: when $\delta_0 n < 2$, the hypothesis is impossible for $n \geq 2$.

The sum, over the points, of the numbers of incident spanned lines is at least $2c_0 n^2$, whereas each point is incident to at most $n - 1$ such lines. It follows that at least $c_0 n$ points are incident to at least $c_0 n$ spanned lines, after decreasing c_0 if necessary. Take $\gamma_0 \leq c_0$. A generic projective chart puts a finite projective configuration in the affine plane without changing its incidences. $\square$

We shall also need an effective notion of richness for planes. For a finite cloud X and an occupied plane Π , define

$$e_X(\Pi) = |X \cap \Pi| - \max_{L \subset \Pi} |X \cap L|.$$

Only planes containing a noncollinear triple can have positive effective size, so the planes counted in the next lemma form a finite family.

The next lemma is motivated by the nondegenerate rich-plane bounds of Elekes and Tóth [12]. Its effective-size formulation in the high-richness range and the double-counting proof below are local; we do not invoke their theorem verbatim.

Lemma 7.3 (Planes of large effective size). *There are absolute constants $C_2, K_2 > 0$ such that, for a cloud $X \subset \mathbb{R}^3$ with $|X| \leq m$, $m \geq 1$, and $H \geq C_2 m^{2/3}$,*

$$\#\{\Pi : e_X(\Pi) \geq H\} \leq K_2 \frac{m^2}{H^2}.$$

Proof. We may assume $H \leq m$. First consider a plane for which a richest line contains at least half of $X \cap \Pi$. That line contains at least H points, and there are at least H points of $X \cap \Pi$ off the line. By (7.3), applied after a generic projection, the number of H -rich lines of X is

$$O\left(\frac{m^2}{H^3} + \frac{m}{H}\right) = O(m/H).$$

Here $H \geq \sqrt{m}$, after increasing C_2 . For any fixed line, distinct planes containing it have disjoint sets of points off that line. There are therefore at most m/H planes of the present kind through each such rich line. This part contributes $O(m^2/H^2)$.

For the remaining planes no line contains half their points. Bucket their occupancies in $[K, 2K)$, where $K = 2^j H$, and let f denote the number in one bucket. Any fixed spatial line belongs to at most $2m/K$ of these planes, since each such plane has at least $K/2$ points off the line and those off-line sets are disjoint.

For $x \in X$, let r_x be the number of bucket planes containing x , and put $I = \sum_x r_x$. Thus $Kf \leq I < 2Kf$. The sum J , over ordered distinct pairs of bucket planes, of their common sample-point counts satisfies

$$J = \sum_x r_x(r_x - 1) \geq \frac{K^2 f^2}{m} - 2Kf. \quad (7.4)$$

To bound J from above, fix one bucket plane Π . Separate its intersection lines with the other planes according as their occupancy is less than $C_3 \sqrt{K}$ or at least $C_3 \sqrt{K}$, where $C_3 \geq 2$ is a fixed constant. The first class contributes at most $C_3 \sqrt{K} f$. Within Π , the number a of distinct lines in the second class obeys, again by (7.3),

$$a \leq C_4 \left(C_3^{-3} + C_3^{-1}\right) \sqrt{K} = O(\sqrt{K}).$$

Their total summed occupancy is $O(K)$. Indeed, if a point is on $s \geq 1$ of these lines, then $s \leq 1 + \binom{s}{2}$; and two distinct lines have at most one common point. Hence the total occupancy is at most

$$|X \cap \Pi| + \binom{a}{2} = O(K).$$

Each such intersection line belongs to at most $2m/K$ bucket planes. Their contribution for the fixed Π is therefore $O(m)$. Summing over Π gives

$$J \leq C_5 f(m + \sqrt{K} f). \quad (7.5)$$

For sufficiently large C_2 , the assumption $K \geq C_2 m^{2/3}$ ensures $K^2/m \geq 2C_5 \sqrt{K}$. Combining (7.4) and (7.5), and using $K \leq m$, now gives $f = O(m^2/K^2)$. The sum over dyadic $K \geq H$ is geometric, which proves the lemma. $\square$

7.2 A local bound for motions without a rich matched line

Choose constants $0 < \eta_1 \ll \eta_2 \ll 1$ as follows. First choose η_2 so small that $4\eta_2/\gamma_0 \leq \delta_0/2$, where δ_0, γ_0 are from Lemma 7.2. Then choose

$$0 < \eta_1 \leq \min\{\eta_2/100, \delta_0/8\}.$$

These constants remain fixed throughout the section. For a bucket $\ell \leq k_g(X, Y) < 2\ell$, call a motion *line type* if some line contains at least $\eta_1\ell$ points of X_g . Among the other motions, call it *planar type* if some plane contains at least $\eta_2\ell$ points of X_g , and *general type* otherwise.

Lemma 7.4 (Local motion bound). *There are absolute constants $C_{\text{loc}}, K_{\text{loc}} > 0$ such that, if $m \geq 1$, $|X|, |Y| \leq m$, and $\ell \geq C_{\text{loc}}m^{2/3}$, the number of non-line-type motions satisfying $\ell \leq k_g(X, Y) < 2\ell$ is at most*

$$K_{\text{loc}} \frac{m^3}{\ell^{3/2}}.$$

Both orientation classes are counted.

Proof of Lemma 7.4. We first prove the general-type case by a polynomial argument. Every non-line-type match set contains a noncollinear triple. The images of an ordered noncollinear triple determine a spatial isometry up to two choices, and determine it uniquely if its orientation is specified. Thus all motion lists considered in the proof of Lemma 7.4 are finite.

Separate the two orientations. Composing the second cloud with a fixed reflection when necessary reduces either list to reversing motions $g(x) = Sx + h$, with $\det S = -1$. Next rotate the second cloud generically, applying the same rotation to every motion in the finite list. We may and do arrange that:

1. $\ker(I + S)$ is a line for each counted motion;
2. projection perpendicular to that line preserves noncollinearity for every noncollinear triple in its match set.

To justify these simultaneous choices, for any one fixed S , left multiplication by a rotation ranges over the entire negative component of $O(3)$. Outside the proper locus $S = -I$, its -1 eigenspace is a line with arbitrary generic direction. For a fixed noncollinear triple, the projection fails precisely when that line direction belongs to the two-dimensional direction space of the triple's plane. This is a proper algebraic condition. There are finitely many motions and triples, so a rotation avoiding all these proper conditions exists. Counts, distances, and line/plane occupancies are preserved by this change of coordinates.

7.2.1 The two families of null three-flats

Use coordinates $(w, c) \in \mathbb{R}^3 \oplus \mathbb{R}^3$, with $Zx = w \times x$, and equip this six-space with the nondegenerate quadratic form

$$\mathfrak{q}(w, c) = w \cdot c.$$

Its associated symmetric bilinear form is

$$\mathcal{B}((w, c), (w', c')) = w \cdot c' + w' \cdot c.$$

For a possible point pair $(x, y) \in X \times Y$, set $v = (x + y)/2$, $b = (x - y)/2$, and define

$$F_{x,y} = \{(w, c) : b = w \times v + c\}. \tag{7.6}$$

This is an affine three-flat. Its directions obey $\delta c = -\delta w \times v$, so $\mathfrak{q}(\delta w, \delta c) = 0$. Its direction space is therefore maximal totally null and equals its $\mathcal{B}$ -orthogonal space.

Fix one reversing motion, and choose positively oriented orthonormal coordinates with its axis $\ker(I + S)$ equal to $\mathbb{R}e_3$. The perpendicular block $S_\perp$ belongs to $SO(2)$, and $I + S_\perp$ is invertible. On the full graph $y = Sx + h$, there are constants $v_0, \omega \in \mathbb{R}$ and $d = (d_1, d_2) \in \mathbb{R}^2$ such that

$$v_3 = v_0, \quad b_\perp = \omega J v_\perp + d, \quad J(a_1, a_2) = (-a_2, a_1). \quad (7.7)$$

Here $v_0 = h_3/2$, and $(I - S_\perp)(I + S_\perp)^{-1} = \omega J$. The map from the physical point x to (v_1, v_2, b_3) is an invertible real affine map: its linear blocks are $(I + S_\perp)/2$ and 1.

Define the motion flat by

$$G_g = \{(w, c) : w_3 = \omega, \quad c_1 + v_0 w_2 = d_1, \quad c_2 - v_0 w_1 = d_2\}. \quad (7.8)$$

It is another affine three-flat. Its directions satisfy

$$\delta w_3 = 0, \quad \delta c_1 = -v_0 \delta w_2, \quad \delta c_2 = v_0 \delta w_1,$$

with δc_3 free, so they too are maximal totally null. If $y = g(x)$, its intersection with the pair flat is the plane

$$H_{g,x} = F_{x,g(x)} \cap G_g : \quad \tau + \xi v_1 + \eta v_2 = b_3, \quad (\xi, \eta, \tau) = (-w_2, w_1, c_3). \quad (7.9)$$

These are the usual graph-dual planes of the transformed physical points (v_1, v_2, b_3) . The orthonormal coordinates used to describe a particular G_g are only a means of writing its equations: within the normalized orientation class under consideration, every flat is regarded in the same original parameter six-space. Simultaneous rotations $(w, c) \mapsto (Rw, Rc)$ preserve $\mathfrak{q}$, so the null properties just checked hold there for the whole family.

We record the needed distinctness assertions explicitly. First, different matched physical points give different planes $H_{g,x}$, since their transformed coordinates are distinct. Second, in a fixed pair flat $F_{x,y}$, whose free coordinates are w , the incident plane from a motion with unit axis a has equation $w \cdot a = \omega$. This affine plane determines (a, ω) up to simultaneous sign. Since the quarter-turn operator $J_a : x \mapsto a \times x$ on $a^\perp$ also changes sign with a , the operator ωJ_a , and hence S , is determined. The match $g(x) = y$ then fixes h . Hence different incident motions give distinct planes in $F_{x,y}$.

Finally, the whole affine flat G_g determines g . Its direction space intersects the pure-translation subspace $\{\delta w = 0\}$ in precisely the axis line $\{\delta c \parallel a\}$, recovering that unoriented axis. Choose either orientation of it. Projection of G_g to w -space recovers ω ; its direction coupling in (7.8) recovers v_0 ; and the affine constants then recover d . Equation (7.7) gives

$$S_\perp = (I - \omega J)(I + \omega J)^{-1}, \quad h_\perp = -(I + S_\perp)d, \quad h_3 = 2v_0,$$

and $Sa = -a$. These reconstruct the physical motion, independently of the arbitrary oriented coordinates used in the reconstruction. Thus the G_g are distinct as well.

7.2.2 Polynomial construction and descent

Suppose that one orientation class contains at least

$$T = \left[K_3 \frac{m^3}{\ell^{3/2}} \right]$$

general-type motions, where the absolute constant K_3 will be chosen large. Restrict to exactly T of them. Form the bipartite incidence graph between these motions and their point-pair labels. It has at least $T\ell$ edges and at most m^2 label vertices. Repeatedly remove a motion vertex of degree below $\ell/4$, or a label vertex of degree below $T\ell/(4m^2)$. The total number of edges removed in the first way is less than $T\ell/4$, and in the second way less than $T\ell/4$. A nonempty graph remains, in which the two minimum degree bounds hold. Every retained motion has between $\ell/4$ and 2ℓ retained matches. Its retained physical cloud still has fewer than $\eta_1\ell$ points on any line and fewer than $\eta_2\ell$ on any plane.

An ambient polynomial of degree at most D has $\binom{D+6}{6}$ coefficients. Vanishing on any one affine three-flat imposes at most $\binom{D+3}{3}$ linear conditions. Thus a sufficiently large absolute multiple of $T^{1/3}$, rounded upward, is a degree D for which there is a nonzero polynomial f vanishing on all retained G_g . We may take

$$D \leq C_6 K_3^{1/3} \frac{m}{\sqrt{\ell}}. \quad (7.10)$$

We used $\ell \leq m$ here; if this fails the motion bucket is empty. Choose K_3 sufficiently large that

$$\frac{T\ell}{4m^2} > D.$$

Every retained pair flat then contains more than D distinct incident planes on which f vanishes. A nonzero polynomial of degree at most D on an affine three-space cannot vanish on more than D distinct planes: their distinct affine linear equations would all be factors. Therefore f vanishes on every retained pair flat as well.

We next show that every first partial derivative of f vanishes on every retained motion flat. For a polynomial h , define its raised gradient by

$$dh_z(u) = \mathcal{B}(\nabla_{\mathcal{B}} h(z), u).$$

If h vanishes on a maximal null affine flat, its raised gradient at a point of that flat belongs to its direction space, because that space equals its orthogonal. At an intersection of three transverse planes H_{g,x_i} in G_g , the raised gradient of f therefore belongs to the directions of G_g and of all three incident pair flats. Its direction in G_g belongs to all three plane directions, whose intersection is zero. Hence $df = 0$ there, so all ordinary first partial derivatives vanish at that point.

There are enough such points to force a derivative to vanish on the whole of G_g . Work in its transformed physical cloud of $n_g \in [\ell/4, 2\ell)$ retained points. Affine invertibility preserves all line and plane occupancy bounds. Since $\eta_1\ell \leq \delta_0 n_g$, Lemma 7.2 gives at least $\gamma_0\ell/4$ anchor points, each with at least $\gamma_0\ell/4$ distinct outgoing projective directions. For a fixed such anchor, a projective line of directions corresponds to a physical plane through the anchor and contains at most $\eta_2\ell$ of these distinct directions. The choice of η_2 permits a second application of Lemma 7.2. It gives at least $c_7\ell^2$ distinct physical planes through the anchor and two other noncollinear points, for an absolute $c_7 > 0$.

All these spanned planes are nonvertical in the coordinates (v_1, v_2, b_3) , by the generic projection chosen above. Each is the graph of a unique affine function of (v_1, v_2) . Graph duality in (7.9) therefore converts them into $c_7\ell^2$ distinct transverse triple-intersection points in the anchor plane $H_{g,x}$. They lie on the arrangement of at most 2ℓ distinct lines obtained by intersecting this anchor plane with the other incident planes. A line in this arrangement contains at most 2ℓ distinct intersection points with the other lines. Since each of the designated points lies on at least two arrangement lines, counting these incidences shows that at least $c_8\ell$ distinct lines each contain at least $c_8\ell$ designated points, for an absolute $c_8 > 0$.

Choose a fixed $\varepsilon > 0$ smaller than both $c_8/2$ and $\gamma_0/8$. After K_3 has been fixed, increase C_{loc} so that $\ell \geq C_{\text{loc}}m^{2/3}$ and (7.10) imply $D < \varepsilon\ell$. Each first derivative of f , whose degree is less than D , now vanishes on all the rich lines just found: its univariate restriction has more zeros than its degree. It consequently vanishes on the entire anchor plane, by the same distinct-factor argument in dimension two. There are at least $\gamma_0\ell/4 > D$ distinct anchor planes, so the derivative vanishes on G_g .

Every first derivative therefore vanishes on all retained motion flats, and the earlier propagation through the retained label degrees makes it vanish on all retained pair flats. The same argument applies to it, with strictly smaller degree and the same geometric configuration. Iteration implies that a nonzero constant derivative of f vanishes on a nonempty flat, a contradiction in characteristic zero. This proves the general-type bound $O(m^3/\ell^{3/2})$ for each orientation class.

7.2.3 Planar type

For two finite planar clouds of sizes a, a' , Corollary B.6 bounds the number of planar isometries giving at least u matches, for an integer $u \geq 3$, by

$$O\left(\frac{(aa')^{3/2}}{u^2}\right) \quad (u \leq \min\{a, a'\}). \quad (7.11)$$

The corollary counts both orientations. Its proof uses the higher-richness Guth–Katz line theorem [16, Theorem 4.5] with a plane cap; that theorem has no regulus hypothesis.

Now take a planar-type motion in the local bucket. Choose a plane Π with at least $\eta_2\ell$ matched source points and put $\Pi' = g\Pi$. Remove a richest ambient line from $X \cap \Pi$, and a richest ambient line from $Y \cap \Pi'$. Each deletion loses fewer than $\eta_1\ell$ matches, by the non-line-type assumption. At least

$$(\eta_2 - 2\eta_1)\ell \geq \eta_0\ell, \quad \eta_0 = \eta_2/2,$$

matches remain. In particular $e_X(\Pi), e_Y(\Pi') \geq \eta_0\ell$.

Bucket these two effective sizes in $[h, 2h)$ and $[h', 2h')$, where h, h' run dyadically from $\eta_0\ell$. Choose C_{loc} also large enough that $\eta_0\ell \geq C_2m^{2/3}$ and $\eta_0\ell \geq 3$. Lemma 7.3 supplies at most $O(m^4/(h^2(h')^2))$ ordered supporting plane pairs in these buckets. For each fixed pair, choose the two richest lines once and for all, and choose orthonormal affine coordinates on the two planes. Write a, a' for the sizes of the reduced planar clouds, so $a < 2h$ and $a' < 2h'$. A motion being counted restricts to a planar isometry with at least $k_0 = \lceil \eta_0\ell \rceil \geq 3$ matches. If $k_0 > \min\{a, a'\}$, there are no such restrictions; otherwise (7.11), applied with $u = k_0$, bounds them by $O((hh')^{3/2}/\ell^2)$. Each planar restriction extends to a spatial isometry in at most two ways, according to the image of the unit normal. The constants therefore remain absolute.

Multiplying and summing the two geometric dyadic series gives

$$\#\{\text{planar-type motions}\} \leq C \frac{m^4}{\ell^2} \sum_{h, h' \geq \eta_0\ell} (hh')^{-1/2} \leq C' \frac{m^4}{\ell^3} \leq C'' \frac{m^3}{\ell^{3/2}}.$$

The last inequality uses $\ell \geq C_{\text{loc}}m^{2/3}$. Together with the general-type proof, this completes Lemma 7.4, taking K_{loc} larger than the sum of the constants obtained for the two orientations and the planar-type estimate. The order of choices was η_2, η_1 , then K_3 , then C_{loc} , and finally K_{loc} . $\square$

7.3 A spatial subdivision with uniform overlap

We now prepare to apply Lemma 7.4 to subsets of the full clouds. For $r, m \geq 1$, suppose temporarily that each cloud has been divided into disjoint regions containing at most m of its points, and that at most Kr pairs of regions can contain matches of any fixed motion, where $K > 0$ is an absolute constant. If $n_{R,R'}(g)$ is the number of matches from points of X in region R to points of Y in region R' , these counts sum to $k_g(X, Y)$, and at most Kr are nonzero. Hence

$$\sum_{\substack{R, R' \\ n_{R,R'}(g) < b}} n_{R,R'}(g) \leq Krb \quad (b > 0). \quad (7.12)$$

Thus, for a motion with at least $k > 0$ matches, taking b to be a sufficiently small fixed multiple of k/r retains a fixed fraction of its matches in region pairs with local count at least b .

Divide the retained local counts into buckets $\ell \leq n_{R,R'}(g) < 2\ell$ with $\ell \geq b$. If the later parameter choice makes $b \geq C_{\text{loc}} m^{2/3}$, the non-line-type bound of Lemma 7.4 applies in every such bucket. A motion of line type in one local bucket instead supplies a source line and its image line, each containing at least $\eta_1 \ell \geq \eta_1 b$ points of its full cloud. We will count these local contributions through their physical lines.

For each fixed r with $1 \leq r \leq N$, the next lemma constructs the required subdivision once on each cloud, independently of the motion, with $O(r)$ regions and $m = 8N/r$. Its geometric property controls intersections of the regions themselves under every rigid motion, and hence the number of region pairs that can contain matches.

Lemma 7.5 (Subdivision into cubes and cubes with one hole). *Let $N \geq 1$ and $1 \leq r \leq N$, with r allowed to be real. For every cloud $X \subset \mathbb{R}^3$ of size at most N , there is a half-open cube containing X and a finite partition $\mathcal{R}_X$ of that cube such that:*

1. *every region is a half-open dyadic cube, or the difference $Q \setminus Q'$ of a half-open dyadic cube and one strict dyadic descendant;*
2. *$|\mathcal{R}_X| \leq 19r$, and every region contains at most $8N/r$ points of X ;*
3. *there is an absolute constant K_{part} such that, for any two partitions obtained this way and every rigid motion g ,*

$$\#\{(R, R') \in \mathcal{R}_X \times \mathcal{R}_Y : g(R) \cap R' \neq \emptyset\} \leq K_{\text{part}} r.$$

In particular the matches of any motion occur in at most $K_{\text{part}} r$ pairs of regions.

Proof. Put $\mu = N/r \geq 1$. Choose a containing half-open cube and subdivide a dyadic cube whenever it contains more than μ sample points. Call these subdivided cubes heavy. The procedure terminates: distinct points in a finite cloud have positive minimum separation, so sufficiently small cubes contain at most one point. If the root cube is light, it alone is the required partition.

Otherwise the heavy cubes form a finite rooted tree, with edges to heavy children. Let L be the number of its leaves and B the number of its vertices with at least two heavy children. The heavy leaves are disjoint and each contains more than μ points, so $L < r$. The elementary rooted-tree identity gives $B \leq L - 1$. Vertices with exactly one heavy child break into maximal chains. Each chain ends at a different branching vertex or heavy leaf, so there are at most $B + L$ such chains.

For a chain $Q_0 \supset Q_1 \supset \cdots \supset Q_a$, where each Q_{i+1} is the unique heavy child of Q_i , consider the differences

$$D_i = Q_i \setminus Q_{i+1} \quad (0 \leq i < a).$$

Each is a union of at most seven light children and contains at most 7μ points. In chain order, close a run of consecutive differences as soon as its mass reaches μ . Every completed run has mass in $[\mu, 8\mu)$, and at most one residual run per chain has smaller mass. A residual run of zero mass is retained when necessary to cover the containing cube. Every run is a difference of a cube and one strict dyadic descendant, since

$$D_i \cup \dots \cup D_j = Q_i \setminus Q_{j+1}.$$

At each branching vertex and each heavy leaf, take its light children as separate cube regions.

These regions partition the original root cube: they group the terminal cubes of the full subdivision without changing their union. The half-open convention makes the assertion exact also on boundaries. Completed runs are disjoint and have mass at least μ , so there are at most r of them. There are at most $B + L$ residual runs and at most $8(B + L)$ separate light children. Consequently

$$|\mathcal{R}_X| \leq r + 9(B + L) \leq 19r,$$

and each region has mass at most 8μ .

We prove a thickness property for these regions. If the outer cube of a region R has side S , then for every $x \in R$ and $0 < s \leq S$, there is a measurable set $E \subset R$ with

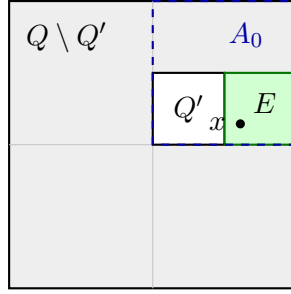
$$E \subset B(x, \sqrt{3}s), \quad \text{vol}(E) \geq (s/4)^3. \quad (7.13)$$

Choose the unique dyadic subcube A_0 of the outer cube containing x , with side a satisfying $s/2 < a \leq s$. For a cube region, take $E = A_0$. If $R = Q \setminus Q'$ and $A_0 \cap Q' = \emptyset$, the same choice works. In the remaining case dyadic nesting, together with $x \in A_0 \setminus Q'$, says that Q' is a strict descendant of A_0 . A child of A_0 other than the child leading to Q' lies in R , has side $a/2 > s/4$, and lies within distance $\sqrt{3}s$ of x . Use this child as E . This proves (7.13), including at all included half-open boundary points. The statement is unchanged by a rigid motion.

Consider the two disjoint region families $\mathcal{g}(\mathcal{R}_X)$ and $\mathcal{R}_Y$. Charge each intersecting pair to the member with smaller outer side, breaking ties by a fixed preference between the families. Fix a charged region R with outer side s , and let c_R be the center of its outer cube. For each partner R' , choose $x_{R'} \in R \cap R'$. The outer side of R' is at least s , so (7.13) gives a set

$$E_{R'} \subset R' \cap B(x_{R'}, \sqrt{3}s), \quad \text{vol}(E_{R'}) \geq (s/4)^3.$$

All these sets lie in $B(c_R, 3\sqrt{3}s/2)$. They are pairwise disjoint because the partners belong to one partition. Comparing their volumes with that ball bounds their number by an absolute constant. Summing the charges over at most $38r$ regions proves the intersection estimate. A matched point lies in a unique region of each partition, so the assertion about matches follows as well. $\square$



A region with one dyadic hole

Choose a dyadic cube A_0 containing x at scale s .

If the hole meets A_0 , nesting places it in one child.

Another child E remains in the region and provides volume comparable to s^3 in three dimensions.

Figure 3: A two-dimensional illustration of the thickness argument in Lemma 7.5. The volume estimate is proved for three-dimensional cubes. It holds near the hole as well as near the outer boundary, and survives every rigid motion. This makes the packing argument independent of the depth of the dyadic subdivision.

7.4 Counting prescribed images of rich lines

We give the two elementary geometric facts needed for the line-type contributions. In the first lemma a line-match weight is the full number of source points on the specified line mapped to target points on its specified image line.

Lemma 7.6 (Two prescribed line images). *Let (L_1, L_2) and (L'_1, L'_2) be ordered pairs of distinct affine lines in $\mathbb{R}^3$.*

1. *If the source lines are nonparallel, at most eight rigid motions map L_i onto L'_i for $i = 1, 2$.*
2. *If the source lines are parallel, the target lines must be parallel at the same positive separation. There are at most four possible orthogonal linear parts, and for each one the admissible translations form one affine line parallel to the target lines. If all four lines contain at most H points of their respective clouds, then*

$$\sum_g w_1(g)w_2(g) \leq 4H^3, \quad (7.14)$$

where only motions with positive product are included. That set of motions is finite.

Both conclusions allow both orientation classes.

Proof. For nonparallel lines choose unit directions a_1, a_2 and b_1, b_2 . The orthogonal part S of a motion must satisfy $Sa_i = \varepsilon_i b_i$, with $\varepsilon_i \in \{1, -1\}$. There are at most four sign choices. Each compatible choice determines the isometry on the two-dimensional span of a_1, a_2 , and there are at most two choices on its perpendicular line. For each resulting S , the condition on its translation h imposed by each line is an affine line in translation space with direction b_i . Those two affine lines are nonparallel, so their intersection has at most one point. This proves the bound of eight, whether the original lines intersect or are skew.

For parallel lines choose unit directions a, b . Let d be the nonzero perpendicular separation vector from L_1 to L_2 , and let e be the analogous vector for the ordered target pair. Orthogonality forces

$$Sa = \pm b, \quad Sd = e.$$

These equations determine S on a two-dimensional space, with at most two choices on its perpendicular line. There are therefore at most four orthogonal parts in total. For any fixed one, mapping the first source line onto the first target line fixes the translation modulo a scalar multiple of b ; the second-line condition is then either impossible or automatically satisfied. This proves the geometric assertion.

Parametrize one such translation family by a real number τ . Each pair of a source sample point on L_1 and a target sample point on L'_1 determines a unique τ . Thus

$$\sum_{\tau} w_1(\tau) \leq |X \cap L_1| |Y \cap L'_1| \leq H^2, \quad w_2(\tau) \leq H.$$

Only finitely many τ have positive first weight. The product sum for this linear part is at most H^3 ; summing over at most four parts proves (7.14). This argument sums the actual translations and introduces no discretization of the continuous family. $\square$

Lemma 7.7 (Parallel pairs at a fixed separation). *Let $\mathcal{L}$ be a finite family of at most L distinct lines in $\mathbb{R}^3$. For each $\rho > 0$, the number of ordered parallel pairs in $\mathcal{L}$ at separation ρ is $O(L^{3/2})$, with an absolute implicit constant.*

Proof. In one parallelism class of size q , intersect the lines with the plane through the origin perpendicular to their common direction. Their intersection points are distinct, and line separation equals the Euclidean distance of these points. Form the graph whose edges join points at distance ρ . Two different vertices have at most two common neighbors, because two distinct circles of radius ρ have at most two common points. If the graph has e edges and degrees d_z , then

$$\sum_z \binom{d_z}{2} \leq 2 \binom{q}{2}, \quad (2e)^2 \leq q \sum_z d_z^2 \leq 3q^3.$$

In the last inequality we used the preceding bound and $2e \leq q(q-1)$. Hence there are $O(q^{3/2})$ ordered pairs in this class. Sum over the classes, using $\sum_j q_j^{3/2} \leq (\sum_j q_j)^{3/2} \leq L^{3/2}$. $\square$

7.5 Proof of the very-rich-motion estimate

Proof of Theorem 7.1. We prove the stronger assertion stated at the start of the section. Let X, Y be clouds of at most N points, put $A = N^{2/3}$, and suppose that each line contains at most $2A$ points of either cloud. Write $k_g = k_g(X, Y)$ throughout this proof. We prove (7.1) with this notation. This implies the asserted result by (7.2). Alternatively, the claim for a congruent copy follows by composing the motions with the inverse of the congruence, which preserves their match counts and bijects the two orientation classes as appropriate.

Fix the absolute constants

$$\alpha = \frac{1}{14}, \quad \beta = \frac{1}{4K_{\text{part}}}, \quad \nu = \frac{\eta_1}{4}.$$

Choose C_{rich} sufficiently large that

$$C_{\text{rich}} > 4, \quad \frac{\beta}{4} C_{\text{rich}}^{9/14} \geq C_{\text{loc}}, \quad \eta_1 \beta C_{\text{rich}}^{3/7} \geq 1, \quad \nu C_{\text{rich}} \geq 4. \quad (7.15)$$

All constants on the right have already been fixed. These choices are independent of N and of the clouds.

Every motion with at least $C_{\text{rich}}A$ matches has a noncollinear matched triple, because the line cap is $2A$. The finite choice of source and target ordered triples and the at-most-two extensions per triple prove finiteness of the motion set in the sum. In particular, possible continuous families of motions fixing a line are below the threshold unless they have additional noncollinear matches, in which case this same finite triple argument applies.

Bucket the high motions by

$$k \leq k_g(X, Y) < 2k, \quad k = tA, \quad t = 2^j C_{\text{rich}} \leq N^{1/3}.$$

If no such t exists, the sum is empty. In a nonempty bucket set

$$r = t^{1+\alpha}, \quad m = 8N/r, \quad b = \beta k/r = \beta A t^{-\alpha}. \quad (7.16)$$

Since $t \leq N^{1/3}$ and $\alpha < 2$, we have $1 \leq r \leq N$, so Lemma 7.5 applies to both clouds. It supplies at most $19r$ regions in each partition, each with at most m points.

For the partitions just chosen, put $X_R = X \cap R$ and $Y_{R'} = Y \cap R'$. The local count is $n_{R,R'}(g) = k_g(X_R, Y_{R'})$. These counts sum to k_g , and at most $K_{\text{part}}r$ are nonzero. By (7.12), discarding those below b removes at most $K_{\text{part}}rb = k/4$ matches. At least $3k/4$ matches remain. Bucket each remaining local count into $\ell \leq n_{R,R'}(g) < 2\ell$, with $\ell = 2^i b$, and use the line/planar/general classification of Lemma 7.4 for the clouds $X_R, Y_{R'}$ in that local bucket. A single global motion may contribute different local types in different region pairs.

The local lower threshold is valid, since

$$\frac{b}{m^{2/3}} = \frac{\beta}{4} t^{(2-\alpha)/3} = \frac{\beta}{4} t^{9/14} \geq C_{\text{loc}}. \quad (7.17)$$

We shall also need the full-cloud rich-line threshold $q = \eta_1 b$ to be at least $\sqrt{N}$. The inequality $t \leq N^{1/3}$ gives $N^{1/6} \geq t^{1/2}$, so

$$\frac{q}{\sqrt{N}} = \eta_1 \beta N^{1/6} t^{-\alpha} \geq \eta_1 \beta t^{1/2-\alpha} = \eta_1 \beta t^{3/7} \geq 1. \quad (7.18)$$

In particular these rich lines contain at least two points in every nonempty bucket. One may also record the power margin $b \geq \beta N^{9/14}$, although only (7.18) is needed below.

Motions with substantial non-line-type mass. Let $\mathcal{G}_0$ consist of the motions in the global bucket whose retained local non-line-type counts have total at least $k/4$. For a fixed region pair and a local richness bucket, Lemma 7.4 bounds the number of non-line-type motions by $K_{\text{loc}} m^3 / \ell^{3/2}$. Their local match mass is therefore at most $2K_{\text{loc}} m^3 / \sqrt{\ell}$. Summing the geometric series over $\ell = 2^i b$ gives $O(m^3 / \sqrt{b})$ for each fixed region pair. The bound remains valid after restriction to our global bucket. There are at most $(19r)^2$ possible region pairs, whence

$$|\mathcal{G}_0| k^2 \leq C k r^2 \frac{m^3}{\sqrt{b}} \leq C' N^3 \frac{k}{r \sqrt{b}} = C'' N^3 \sqrt{A} t^{-\alpha/2} = C'' \frac{N^4}{A} t^{-1/28}. \quad (7.19)$$

The equality $N^3 \sqrt{A} = N^4 / A$ uses $A = N^{2/3}$.

Motions whose retained mass is mostly line type. For every remaining motion, at least $k/2$ retained matches belong to line-type local buckets. In each such region pair, choose one source line containing at least $\eta_1 \ell$ of its local matches, and retain those matches as assigned line matches.

Since $n_{R,R'}(g) < 2\ell$, this retains at least $\eta_1/2$ of that local count. Thus the assigned line-match mass W_g satisfies

$$W_g \geq \nu k, \quad \nu = \eta_1/4.$$

A physical source point participates in at most one region-pair count for a fixed motion. Hence no assigned match is repeated. Pool assignments belonging to the same full source line L , and write its weight as $w_{g,L}$. Its target line is necessarily gL , and

$$\sum_L w_{g,L} = W_g \geq \nu k, \quad 0 \leq w_{g,L} \leq 2A. \quad (7.20)$$

Every line appearing on either side contains at least $\eta_1 b = q$ ambient sample points, because it was chosen in a local bucket with $\ell \geq b$.

Let $\mathcal{L}_X, \mathcal{L}_Y$ be the full lists of these q -rich lines of the two clouds. These are finite lists of spanned lines. Equations (7.3) and (7.18) imply

$$|\mathcal{L}_X|, |\mathcal{L}_Y| \leq C \left(\frac{N^2}{q^3} + \frac{N}{q} \right) \leq C' \frac{N}{b}. \quad (7.21)$$

Denote a common upper bound in (7.21) by $L_* = C'N/b$.

The diagonal weights can be removed without losing a fixed fraction. Indeed, by (7.15) and (7.20), $W_g \geq \nu k \geq 4A$. Consequently

$$\sum_{L_1 \neq L_2} w_{g,L_1} w_{g,L_2} = W_g^2 - \sum_L w_{g,L}^2 \geq W_g^2 - 2AW_g \geq \frac{1}{2}\nu^2 k^2. \quad (7.22)$$

Sum this quantity over the remaining motions. For nonparallel source line pairs, there are at most L_*^4 prescriptions of an ordered source pair and its ordered target pair. By Lemma 7.6, each prescription has at most eight motions, and each product of weights is at most $(2A)^2$. Their total contribution is thus $O(A^2 L_*^4)$.

For a parallel ordered source pair, fix its positive separation ρ . Lemma 7.7 gives at most $O(L_*^{3/2})$ ordered target pairs at that separation, including all target parallelism classes. There are at most L_*^2 source pairs. For every prescription, the pooled selected weights are bounded by the full line-match weights in Lemma 7.6; with $H = 2A$, its bound is $O(A^3)$. Therefore the total parallel contribution is $O(A^3 L_*^{7/2})$.

If $\mathcal{G}_1$ denotes the remaining motions, the two cases and (7.22) give

$$|\mathcal{G}_1| k^2 \leq C \left(A^2 (N/b)^4 + A^3 (N/b)^{7/2} \right). \quad (7.23)$$

This counts both orientations. In the parallel case it sums the whole continuous translation family through its finite positive match support, rather than assuming a bounded number of motions for the two prescribed lines.

Summing the global buckets. From $b = \beta A t^{-\alpha}$, with its fixed positive β , the two terms in (7.23) are

$$A^2 (N/b)^4 = O \left(N^{8/3} t^{4\alpha} \right) = O \left(N^{8/3} t^{2/7} \right), \quad (7.24)$$

$$A^3 (N/b)^{7/2} = O \left(N^{19/6} t^{7\alpha/2} \right) = O \left(N^{19/6} t^{1/4} \right). \quad (7.25)$$

Summing these increasing geometric series up to $t \leq N^{1/3}$ gives respectively

$$O\left(N^{58/21}\right) \quad \text{and} \quad O\left(N^{13/4}\right).$$

Both are bounded by $O(N^{10/3}) = O(N^4/A)$; in fact they save the powers $N^{-4/7}$ and $N^{-1/12}$, respectively. The decreasing geometric series in (7.19) gives

$$\sum_{j \geq 0} O\left(\frac{N^4}{A} (2^j C_{\text{rich}})^{-1/28}\right) = O(N^4/A).$$

Finally $k_g < 2k$ in each global bucket, so replacing its cardinality times k^2 by the sum of the actual k_g^2 costs at most a factor of four. The two types exhaust every bucket, proving (7.1). $\square$

Remark 7.8 (Scope of the threshold and the deletion cost). The finite line cap is essential for the sum over motions to make sense. For a collinear sample without such a high threshold, infinitely many rotations about its containing line fix every point. In Theorem 7.1 each counted motion has a noncollinear matched triple, which removes this issue. For the reduction in this paper, $M = o(A)$ guarantees the finite hypothesis $M \leq A$ eventually. Deleting every existing graph edge whose endpoints are matches of the same motion above the threshold costs at most $\sum_g \binom{k_g}{2} = O(N^4/A)$. Counting an edge for several motions can only increase this upper bound. In the proof of Lemma 2.5, the real Gram-matrix calculation uses this deletion to bound the parameters in every local maximal isotropic plane.

A Uniform concentration of evaluation flats

We prove the algebraic estimate used twice in the distance argument. Its constant is uniform in the scale parameter. The distinction between the two versions below is essential: the global family has a cardinality bound, whereas a family inside one of its flats need not have one.

Throughout this section, varieties are reduced complex algebraic varieties. The degree of an affine variety means the degree of its projective closure; the degree of a pure-dimensional union is the sum of the degrees of its irreducible components. Constants denoted by C_e may change from line to line and depend only on e .

Let E be a complex vector space of dimension $e \geq 3$, and write

$$S = \bigwedge^2 E^*, \quad n = \binom{e}{2}, \quad r = e - 1, \quad s_0 = n - r = \binom{e-1}{2}.$$

An element $z \in S$ is regarded as a skew map $E \rightarrow E^*$. Set

$$\mathcal{Q} = \{[v : y] \in \mathbf{P}(E \oplus E^*) : y(v) = 0\}.$$

For $\lambda = [v : y] \in \mathcal{Q}$ with $v \neq 0$, define

$$F_\lambda = \{z \in S : zv = y\}, \quad K_v = \{z \in S : zv = 0\}. \quad (\text{A.1})$$

Evaluation at v maps S onto the r -dimensional annihilator of v . Consequently F_λ is a nonempty affine s_0 -flat, with direction $K_v = \bigwedge^2(E/\mathbb{C}v)^*$. Distinct parameters give distinct flats. Indeed, the common radical of the forms in K_v is exactly $\mathbb{C}v$: the quotient has dimension at least two, and each of its nonzero vectors is detected by an alternating form. The direction of the flat therefore recovers $[v]$, and its prescribed evaluation recovers y with the same normalization.

A maximal projective linear subspace of $\mathcal{Q}$ is a $\mathbf{P}^r$. Equivalently, it is the projectivization of an e -dimensional totally isotropic subspace for the quadratic form $(v, y) \mapsto y(v)$.

Theorem A.1 (Uniform concentration). *Fix $C_0 \geq 1$. Let $A \geq 1$ and let $\mathcal{P}$ be a finite set of distinct parameters $[v : y] \in \mathcal{Q}$, all with $v \neq 0$. Assume:*

- (i) *For every $\lambda \in \mathcal{P}$ there is a homogeneous polynomial of positive degree at most $C_0 A$ which is nonzero at λ and zero at every other member of $\mathcal{P}$.*
- (ii) *Every projective $\mathbf{P}^r \subset \mathcal{Q}$ contains at most $C_0 A^{r-1}$ members of $\mathcal{P}$.*

There is a constant $C = C(e, C_0)$ such that

$$\#\{\lambda \in \mathcal{P} : F_\lambda \subset V\} \leq C \deg(V) A^h, \quad \dim V = s_0 + h, \quad 1 \leq h < r, \quad (\text{A.2})$$

for irreducible proper affine varieties $V \subset S$, in either of these settings:

- (a) *All such V are allowed, and $|\mathcal{P}| \leq C_0 A^r$.*
- (b) *Only V contained in the zero set of some nonzero polynomial of degree at most $C_0 A$ are allowed. No bound on $|\mathcal{P}|$ is required.*

For $h = 0$, an irreducible variety containing an s_0 -flat must equal that flat, so it contains at most one member of the family. In (a) the ambient endpoint $h = r$ is supplied by the cardinality assumption. As a useful consequence of (b), a nonzero polynomial of degree $D \leq C_0 A$ contains at most CDA^{r-1} of the flats in its zero set: apply (A.2) to its irreducible hypersurface components and sum their degrees.

The proof first obtains approximate equations and regular defining equations (Theorem A.2 and Lemma A.5), then recovers controlled families of actual rulings (Lemma A.6). Proposition A.7 bounds their degree and identifies the maximal-isotropic exception. Section A.5 closes the resulting finite system of inequalities with constants independent of A .

We normalize the number of flats contained in a variety by its degree, following Walsh's concentration parameters [28, Section 1.1, definition of $\mathcal{D}_m$]. The controlled-ruling strategy and the isotropic exception are closely related to the three-dimensional rigid-motion analysis of Tidor, Yu, and Zakharov [26, Sections 5–6]. Here the ruling geometry is that of evaluation flats of alternating forms in arbitrary fixed dimension; the required degree bound and exceptional-family classification are proved in full below.

A.1 Approximate complete intersections

We prove Walsh's approximate complete-intersection theorem [27, Theorem 1.3]. The partial-degree framework and the construction of compatible cuts follow [27, Definition 2.2 and Lemma 5.3]. The comparison of restriction spaces appears in the proof of [27, Lemma 3.5]. Walsh obtains a corresponding prefix degree estimate for his chosen cuts [27, Theorem 5.5]. We prove the forms needed here; the component Hilbert bounds in Lemma 3.6 are the only quantitative algebraic input. In particular, this proof is independent of Theorem A.1.

Theorem A.2 (Approximate complete intersection). *For every n there is a constant C_n with the following property. If $V \subset \mathbb{A}_{\mathbb{C}}^n$ is irreducible of codimension $c > 0$, there are nonconstant polynomials $f_1, \dots, f_c \in I(V)$ such that V is an irreducible component of $Z(f_1, \dots, f_c)$ and*

$$\prod_{i=1}^c \deg f_i \leq C_n \deg V.$$

Here and below, a component *containing* V means that it contains the whole irreducible variety V . Set

$$I_{\leq t}(V) = I(V) \cap \mathbb{C}[x_1, \dots, x_n]_{\leq t}, \quad H_V(t) = \dim_{\mathbb{C}}(\mathbb{C}[x_1, \dots, x_n]_{\leq t} / I_{\leq t}(V)).$$

For $1 \leq i \leq c$, define δ_i to be the smallest integer t such that every irreducible component of $Z(I_{\leq t}(V))$ containing V has codimension at least i . The notation $Z(I_{\leq t}(V))$ means the zero set of a basis of this finite-dimensional vector space. Finite generation of $I(V)$ shows that these integers exist, and their definition gives

$$1 \leq \delta_1 \leq \dots \leq \delta_c. \quad (\text{A.3})$$

We also put $\delta_0 = 0$.

Lemma A.3 (Compatible partial cuts). *There are $f_i \in I_{\leq \delta_i}(V)$, $1 \leq i \leq c$, such that every component of $Z(f_1, \dots, f_i)$ containing V has codimension exactly i .*

For any such choice, if $0 \leq j < c$ and $\delta_j \leq m < \delta_{j+1}$, some component W of $Z(f_1, \dots, f_j)$ containing V satisfies

$$I_{\leq m}(W) = I_{\leq m}(V), \quad H_W(m) = H_V(m). \quad (\text{A.4})$$

When $j = 0$, the prefix zero set and its unique component are $\mathbb{A}^n$.

Proof. Suppose $f_1, \dots, f_{i-1}$ have been chosen, and list the finitely many components W of their zero set which contain V . Each has codimension $i - 1$. None is contained in $Z(I_{\leq \delta_i}(V))$: otherwise a component of that latter set would contain W , hence contain V and have codimension at most $i - 1$, contrary to the definition of δ_i . Consequently the members of $I_{\leq \delta_i}(V)$ that vanish on any fixed W form a proper linear subspace. Choose f_i outside their finite union. It cuts every such W properly. The principal ideal theorem then gives codimension exactly i for every new component containing V . This includes $i = 1$, when the preceding zero set is the ambient space, and proves the construction.

For the second assertion, suppose no component W in the indicated prefix has $I_{\leq m}(V) \subset I(W)$. The same finite-union argument provides $g \in I_{\leq m}(V)$ which vanishes identically on none of them. All components of $Z(f_1, \dots, f_j, g)$ containing V then have codimension $j + 1$. Since $m \geq \delta_j$, every one of these equations belongs to $I_{\leq m}(V)$. Thus every component of $Z(I_{\leq m}(V))$ containing V has codimension at least $j + 1$. This contradicts $m < \delta_{j+1}$. For a component W where the inclusion does hold, the reverse inclusion follows from $V \subset W$, giving (A.4). $\square$

Proof of Theorem A.2. Fix the polynomials from Lemma A.3. We will prove, for $0 \leq s \leq c$, that every component U of the first s cuts containing V satisfies

$$\deg U \geq \gamma_n^s \prod_{i=1}^s \delta_i \quad (\text{A.5})$$

for one sufficiently small $0 < \gamma_n \leq 1$ depending only on n . For $s = 0$ both sides are 1.

We record the precise consequence of Lemma 3.6 used here. There are constants $a_n > 0$ and $b_n \geq 1$ such that an irreducible k -dimensional variety X satisfies

$$H_X(m) \leq b_n \deg(X) m^k \quad (m \geq 1), \quad (\text{A.6})$$

and, if X is a component of a system whose equations have degrees at most m , then

$$H_X(m) \geq a_n \deg(X) m^k \quad (m \geq 1). \quad (\text{A.7})$$

The second assertion is the bound at the equation degree in (3.7), applied with $L = m$. It permits other components of the system to have larger dimension. The lower bound also holds for $X = \mathbb{A}^n$, after decreasing a_n , directly from $H_{\mathbb{A}^n}(m) = \binom{m+n}{n}$.

Suppose (A.5) is known for smaller indices, and let U be a component at index s . If $\delta_s = 1$, the entire product in (A.5) is 1, and $\deg U \geq 1$ proves the claim. Otherwise set

$$m = \delta_s - 1, \quad j = \max\{i : 0 \leq i < s, \delta_i < \delta_s\}.$$

Thus

$$\delta_j \leq m < \delta_{j+1} = \cdots = \delta_s \leq 2m. \quad (\text{A.8})$$

Lemma A.3 supplies a component W at index j with $H_W(m) = H_V(m)$. Its dimension is $n - j$, and the degrees of its defining prefix equations are at most $\delta_j \leq m$. The induction hypothesis and (A.7) therefore give

$$H_V(m) = H_W(m) \geq a_n \gamma_n^j \left(\prod_{i=1}^j \delta_i \right) m^{n-j}.$$

This formula also applies when $j = 0$. Since $V \subset U$, restriction from U to V is surjective on polynomial restriction spaces, so $H_U(m) \geq H_V(m)$. On the other hand, $\dim U = n - s$ and (A.6) bounds $H_U(m)$ above. Combining the bounds, then using (A.8), yields

$$\begin{aligned} \deg U &\geq \frac{a_n}{b_n} \gamma_n^j \left(\prod_{i=1}^j \delta_i \right) m^{s-j} \\ &\geq \frac{a_n}{2^n b_n} \gamma_n^j \prod_{i=1}^s \delta_i. \end{aligned}$$

Choose $\gamma_n \leq \min\{1, a_n/(2^n b_n)\}$. Since $j < s$, the last expression is at least the right side of (A.5), completing the induction. Notice that no containment relation between U and the chosen W was needed; both contain V .

At $s = c$, every prefix component containing V has the same dimension as V , so V itself is such a component. Applying (A.5) to it gives

$$\prod_{i=1}^c \deg f_i \leq \prod_{i=1}^c \delta_i \leq \gamma_n^{-c} \deg V \leq \gamma_n^{-n} \deg V.$$

This proves the theorem with $C_n = \gamma_n^{-n}$. □

The conclusion concerns an irreducible component. It does not assert that the displayed equations generate its radical ideal or have independent differentials; the next regularity argument supplies the additional control needed here.

A.2 Separators and regular equations

We use the isolated-intersection form of Bézout, with local multiplicities, and the standard local algebra and projective-bundle facts detailed in the proofs below [14, 17].

Lemma A.4 (Counting by separators). *Under the separator assumption of Theorem A.1, an irreducible projective variety $Y \subset \mathbf{P}(E \oplus E^*)$ of dimension a contains at most*

$$C_{e,C_0} \deg(Y) A^a$$

sample parameters. The variety Y need not be contained in $\mathcal{Q}$.

Proof. Induct on a . The assertion for $a = 0$ says that a point contains at most one sample parameter. Otherwise, if $Y \cap \mathcal{P} \neq \emptyset$, choose $\lambda \in Y \cap \mathcal{P}$ and a separator q for it. Since $q(\lambda) \neq 0$, the section $Y \cap Z(q)$ is proper, and it contains all the other counted parameters. Its components have dimension $a - 1$ and total degree at most $C_0 A \deg Y$. Applying the induction hypothesis to those components gives

$$|Y \cap \mathcal{P}| \leq 1 + C_{a-1} C_0 \deg(Y) A^a.$$

The initial 1 is absorbed because $A \geq 1$ and $\deg Y \geq 1$. There are only boundedly many possible dimensions. $\square$

Lemma A.5 (Regular equations of controlled degree). *Suppose that an irreducible $V \subset \mathbb{A}_{\mathbb{C}}^n$ has codimension $c > 0$ and is a component of $Z(f_1, \dots, f_c)$, where*

$$\deg f_i \leq D, \quad \prod_i \deg f_i \leq C_n \deg V, \quad D \geq 1,$$

with the dimensional constant from Theorem A.2. There are polynomials $g_1, \dots, g_c \in I(V)$ of degrees $O_n(D)$ whose differentials are independent at the generic point of V . Moreover, a polynomial H_0 of degree $O_n(D)$, nonzero on V , can be chosen so that $V \setminus Z(H_0)$ is smooth and has tangent space

$$T_z V = \bigcap_{i=1}^c \ker dg_i(z) \quad (z \in V \setminus Z(H_0)).$$

Proof. Put $P = I(V)$ and consider the regular local ring

$$R = \mathbb{C}[x_1, \dots, x_n]_P, \quad \mathfrak{m} = PR, \quad K = R/\mathfrak{m} = \mathbb{C}(V).$$

Its dimension is c . Because V is an isolated component of the common zero set at its generic point, $J = (f_1, \dots, f_c)R$ is $\mathfrak{m}$ -primary. Its generators form a parameter sequence, which is regular because R is Cohen–Macaulay [25, Tags 00OQ, 00NQ, and 02JN]. We first bound the length of R/J by a dimensional constant.

Write $k = \dim V$. A general projective linear projection of $\overline{V}$ to $\mathbf{P}^k$, with center in the hyperplane at infinity disjoint from $\overline{V}$, is finite and has degree $\deg V$. Finiteness follows because a positive-dimensional projective fiber would meet its hyperplane lying in the projection center. Restricting to the affine chart and changing affine coordinates gives base coordinates $t_1, \dots, t_k$ and complementary coordinates $w_1, \dots, w_c$. Over $F = \mathbb{C}(t_1, \dots, t_k)$, the generic fiber of V is a closed point of $\mathbb{A}_F^c$ with residue field K , where $[K : F] = \deg V$. Localizing $F[w_1, \dots, w_c]$ at that point gives the same ring R .

After extending F to an algebraic closure, this point splits into $[K : F]$ points, since the extension is separable. The sum of their local intersection multiplicities for the equations f_i is

$[K : F] \text{length}_R(R/J)$. They are isolated common zeros; other components of the system do not affect their isolated Bézout bound [14, Theorem 12.3 and Example 12.3.1]. The equations f_i define hypersurfaces in regular local ambient rings, so these factors are Cohen–Macaulay and the isolated complete-intersection multiplicity equals the quotient length; see [14, Example 12.3.7(ii)–(iii)] for this identification and its Cohen–Macaulay hypothesis. Hence

$$\text{length}_R(R/J) \leq \frac{\prod_i \deg f_i}{\deg V} \leq C_n. \quad (\text{A.9})$$

Choose an integer $L = O_n(1)$ at least this length. The maximal ideal of an Artinian local ring of length at most L has L th power zero, so

$$\mathfrak{m}^L \subset J. \quad (\text{A.10})$$

We construct the g_i one at a time. Suppose $g_1, \dots, g_u$ have independent classes in $\mathfrak{m}/\mathfrak{m}^2$, where $u < c$. They are part of a regular parameter system, and therefore

$$R_u = R/(g_1, \dots, g_u), \quad \mathfrak{m}_u = \mathfrak{m}R_u$$

is a regular local ring of positive dimension. The images of the f_i generate an ideal containing $\mathfrak{m}_u^L$. At least one image has finite $\mathfrak{m}_u$ -adic order b with $1 \leq b \leq L$. Otherwise all the images would lie in $\mathfrak{m}_u^{L+1}$, whence $\mathfrak{m}_u^L = \mathfrak{m}_u^{L+1}$ and Nakayama’s lemma would give $\mathfrak{m}_u^L = 0$, impossible in this positive-dimensional regular local ring.

We need derivatives which do not disturb the already chosen equations. For $u = 0$ use coordinate derivations. For $u > 0$, choose a u -square Jacobian block J_0 of $(g_1, \dots, g_u)$ whose determinant Δ is nonzero in K . If J_t is any remaining column, form a polynomial vector field having coefficient Δ in coordinate t and coefficients $-\text{adj}(J_0)J_t$ in the chosen coordinates. These $n - u$ fields, denoted by δ_α , satisfy

$$\delta_\alpha g_i = 0 \quad (1 \leq i \leq u)$$

as polynomial identities. Thus they extend to R and descend to R_u . If the g_i have maximum degree E_u , the coefficient degrees of these fields are $O_n(E_u)$.

At the generic point, the fields span the common kernel of the dg_i . The conormal map $\mathfrak{m}/\mathfrak{m}^2 \rightarrow K^n$, $f \mapsto df$, is injective there, as V is generically smooth in characteristic zero. It follows that the residue evaluations of the δ_α span the dual of $\mathfrak{m}_u/\mathfrak{m}_u^2$. These residue evaluations are K -linear: the extra Leibniz term from differentiating a coefficient has a factor in $\mathfrak{m}_u$.

Take an f_i of order b as above. A derivation lowers adic order by at most one, so fewer than b of these derivations still give zero residue. Some composition of b of them has nonzero residue. Indeed, the leading term of f_i belongs to the degree- b piece of the graded ring of R_u [25, Tag 00NO], namely

$$\mathfrak{m}_u^b/\mathfrak{m}_u^{b+1} \simeq \text{Sym}_K^b(\mathfrak{m}_u/\mathfrak{m}_u^2).$$

Its polarization is nonzero in characteristic zero. Since the residue functionals span the dual, some b -tuple detects it. In computing that residue, terms in which a derivation hits a coefficient retain a positive-order factor and vanish. This gives exactly the asserted iterated derivative.

Stop immediately before the last differentiation and call the resulting polynomial g_{u+1} . Its residue in R_u is zero, so it belongs to P . Its derivative has nonzero residue, so its class in $\mathfrak{m}_u/\mathfrak{m}_u^2$ is nonzero. It adds an independent conormal. This procedure uses at most $L - 1$ derivatives per stage and at most c stages. Applying a polynomial field of coefficient degree T to a degree- B polynomial gives degree at most $B + T - 1$; the bounds on L and c therefore keep every g_i of degree $O_n(D)$.

Finally choose a c -square minor H_0 of the Jacobian of the g_i which is nonzero on V . Its degree is $O_n(D)$. Where it is nonzero, the common zero set of the g_i is locally smooth of dimension $n - c$. The contained variety V has the same dimension, so its germ equals that smooth germ. This proves the assertion about tangent spaces. $\square$

A.3 Recovering actual rulings from tangent directions

In a chart $v_a = 1$, a *ruling incidence* is an irreducible set of pairs (z, v) for which $z + K_v \subset V$. Its parameter map is

$$(z, v) \longmapsto [v : zv]. \quad (\text{A.11})$$

The point of the next lemma is that both the number of such incidences and the degree of their exceptional set can be controlled.

Lemma A.6 (Controlled ruling incidences). *Let $V \subset S$ be irreducible of dimension $k = s_0 + h$, with $1 \leq h < r$. Suppose that V has $c = n - k$ approximate equations as in Lemma A.5, of maximum degree at most D . There is a polynomial H , nonzero on V and of degree $O_e(D)$, such that, on $\Omega = V \setminus Z(H)$, the following hold.*

1. *There are $O_e(1)$ smooth irreducible ruling incidences Γ . Each is closed in $\Omega \times \{v_a = 1\}$ for some a and maps smoothly and dominantly to Ω .*
2. *The fibers over Ω are either finite of bounded size or affine lines of normalized vectors v . Denote the fiber dimension by $j = 0$ or $j = 1$. The latter is possible only if $c = 1$.*
3. *Every flat $F_\lambda \subset V$ meeting Ω is represented by at least one incidence. All other contained flats lie in $V \cap Z(H)$.*
4. *If Λ is the projective closure of the parameter image of one incidence, then $\dim \Lambda = h + j$. Every nonempty parameter fiber is exactly $F_\lambda \cap \Omega$ and has dimension s_0 .*

Proof. Use Lemma A.5 to choose $g_1, \dots, g_c$ and an initial smooth open set in V . In the construction below we remove zeros of a bounded number of polynomials, each of degree $O_e(D)$ and nonzero on V . Their product, together with the initial Jacobian minor, will be H . We continue to call the current open set Ω .

First consider tangent directions. Under $S^* = \wedge^2 E$, the annihilator of K_v is $v \wedge E$. A bivector belongs to this space if and only if its wedge with v is zero. Thus the necessary condition $K_v \subset T_z V$ is the linear system

$$v \wedge dg_i(z) = 0 \quad (1 \leq i \leq c). \quad (\text{A.12})$$

Its vector solution space has dimension at most two. To see this, let a nonzero bivector w satisfy $v \wedge w = 0$ for a nonzero v . Splitting off $\mathbb{C}v$ shows that $w = v \wedge a$ for some a ; the vectors v' satisfying $v' \wedge w = 0$ are precisely $\text{span}(v, a)$. If two independent conormals had a common two-dimensional solution space, both would lie in its one-dimensional exterior square, a contradiction. A two-dimensional solution space therefore forces $c = 1$.

Work separately in each of the e normalized charts $v_a = 1$. Gaussian elimination over $\mathbb{C}(V)$, followed by removal of nonzero rank minors, puts the system throughout Ω into one of three forms: no solution, a graph $v = b(z)$, or

$$v = b(z) + \tau a_1(z). \quad (\text{A.13})$$

In the last case choose a free coordinate of v as τ , so that its coordinate in b is zero and in a_1 is one. All coefficients are regular on Ω and admit ambient rational representatives with numerator and denominator degrees $O_e(D)$, since the matrices have bounded size. Rank conditions which vanish generically vanish on all of V ; excluding the nonzero minors consequently controls all remaining fibers. Denote the resulting solution variety by Γ_0 . It is a smooth graph over Ω , or a smooth graph over $\Omega \times \mathbb{A}^1$.

We now test whether the tangent directions integrate to a contained flat. Write $\varepsilon_1, \dots, \varepsilon_e$ for a basis of E^* . In the chart $v_a = 1$, exterior products of the covectors $\varepsilon_i - v_i \varepsilon_a$, $i \neq a$, form a basis $B_1(v), \dots, B_{s_0}(v)$ of K_v . Their coefficients have degree at most two in v . Consider the horizontal fields

$$X_\alpha(z, v) = (B_\alpha(v), 0).$$

Their z -components are tangent to V , by (A.12). They are tangent to Γ_0 if and only if the derivatives of all coordinates of its graph formula vanish in direction $B_\alpha(v)$, with τ fixed in (A.13). Holding τ fixed is required because it is one of the coordinates of the normalized vector v . Call these derivative expressions T_β .

With no free coordinate the T_β are rational functions of z . With a free coordinate they are polynomials in τ of bounded degree, whose coefficients have ambient rational degree $O_e(D)$. This follows by substituting (A.13) in the quadratic expressions B_α and differentiating its rational coefficients. Differentiation along $T_z V$ is independent of the ambient representatives chosen, since it annihilates functions vanishing on V .

Every contained flat passes this first test: on a small open part of the flat, v is fixed and the direction equations continue to hold. If every T_β is identically zero, retain Γ_0 ; its horizontal fields are already tangent everywhere. If there is no free coordinate but one test is nonzero, remove the zero set of its numerator and discard this chart. No contained flat in that chart can meet the remaining open set.

It remains to treat a free coordinate and a nonidentically zero test. Let $P(\tau)$ be the monic gcd of the tests over $\mathbb{C}(V)$, and let $G(\tau)$ be its monic squarefree part. The Euclidean algorithm gives identities

$$T_\beta = PQ_\beta, \quad P = \sum_{\beta} U_\beta T_\beta, \quad G \mid P, \quad P \mid G^q$$

for a bounded integer q . Exclude the poles of all coefficients used in these identities. They then show in every remaining fiber that the common roots of the tests are exactly the roots of G . If $G = 1$, the chart contributes nothing. Otherwise exclude also the zero set of its discriminant. The root variety

$$\Sigma_G = \{(z, \tau) \in \Omega \times \mathbb{A}^1 : G(z, \tau) = 0\}$$

is now finite *étale* over Ω , with bounded degree. In particular it is smooth, and its roots form local holomorphic sheets.

All these exclusions have total degree $O_e(D)$. Indeed, the number of tests and their degrees in τ are bounded. Gcd, squarefree part, quotient, and discriminant computations therefore use a bounded number of field operations. A bounded number of operations on ambient fractions of degree $O_e(D)$ preserves that degree bound. Every excluded denominator or pivot has nonzero restriction to V . The same observation applies to the following second test.

Along a contained flat, $G(z, \tau) = 0$ persists with τ fixed. Thus a second necessary condition is

$$R_\alpha(z, \tau) = \sum_{m=1}^n (B_\alpha(v(z, \tau)))_m \frac{\partial G}{\partial z_m}(z, \tau) = 0.$$

These expressions again have bounded degree in τ and coefficient degrees $O_e(D)$. Form the monic polynomial

$$P_2 = \gcd(G, R_1, \dots, R_{s_0})$$

and exclude the poles needed to specialize its quotient and Bézout identities. Its root variety Σ_{P_2} is empty or finite *étale* over Ω . Because G remains squarefree and P_2 divides it, the factorization

$G = P_2(G/P_2)$ has disjoint root sets in every remaining fiber. Thus Σ_{P_2} is locally a union of whole sheets of Σ_G .

At a point of Σ_{P_2} , the first test makes X_α tangent to Γ_0 , where its coordinates are $(B_\alpha(v), 0)$ in (z, τ) . The second test makes it tangent to $G = 0$. Hence it is tangent to the selected sheet of Σ_{P_2} . No further sequence of tangency tests is needed. Retain its irreducible components as the incidences. A smooth variety has disjoint irreducible components. Each component of this finite *étale* cover maps openly and closedly to the irreducible base, hence surjectively; their number and fiber sizes are bounded by $\deg G$.

We verify containment rather than merely tangency. On every retained incidence the fields X_α are tangent. Their ambient flows are

$$(z, v) \mapsto (z + tB_\alpha(v), v),$$

since v is constant. Uniqueness of local holomorphic flows keeps these translations in the incidence for small t . Composing them produces an analytic open subset of $z + K_v$, because the $B_\alpha(v)$ form a basis of K_v . Every polynomial vanishing on V vanishes on that open subset, hence on the whole affine flat. Thus every retained point really represents a contained flat.

After processing all normalized charts, intersect their open sets and restrict their incidences to this common Ω . The product of the excluded polynomials is nonzero on irreducible V and has degree $O_e(D)$. Every nonempty restricted incidence remains irreducible and dominant. Its fibers are finite of bounded size, except for the retained affine-line graphs; the latter arose only in codimension one.

If a contained flat meets Ω , choose a chart for its v and a point of $F_\lambda \cap \Omega$. Both tests are necessary along that flat, and the specialized gcd identities therefore place the point in a retained incidence. This proves coverage.

Finally fix one incidence and a nonempty fiber of (A.11). Its normalized vector v is fixed, and the fiber is closed in $F_\lambda \cap \Omega$. The preceding flows preserve both v and zv and give an analytic open subset of that flat in the fiber. Such an open subset is Zariski dense in the irreducible variety $F_\lambda \cap \Omega$, so the fiber equals this whole intersection. It has dimension s_0 . The fiber-dimension theorem now gives

$$\dim \Lambda = \dim \Gamma - s_0 = (k + j) - s_0 = h + j.$$

□

A.4 Degree bounds and exceptional parameter families

The ruling construction bounds the number of incidences and the degree of the set removed from V . For an incidence with $j = 0$, its projective parameter closure Λ has dimension h , so Lemma A.4 counts at most $C \deg(\Lambda) A^h$ sample parameters. A bound $\deg \Lambda \leq C_e \deg V$ would therefore give the scale required by Theorem A.1. When $j = 1$, the parameter variety has dimension $h + 1$, and the separator bound alone would cost one extra factor of A . The next proposition proves that this case has $h = r - 1$ and that the parameter variety is a maximal isotropic $\mathbf{P}^r \subset \mathcal{Q}$, where assumption (ii) gives the required $O(A^{r-1})$ bound. The projective incidence bundle constructed below gives the degree estimate and the tangent constraint used to classify this exceptional case.

Proposition A.7. *Let $V \subsetneq S$ be irreducible of dimension $k = s_0 + h$, where $1 \leq h < r$, and let Γ be a ruling incidence provided by Lemma A.6. Write $j \in \{0, 1\}$ for the dimension of its fibers over the working open subset $\Omega \subset V$. Let Λ be the projective closure of its parameter image*

$$(z, v) \mapsto [v : zv] \quad \text{in} \quad \mathcal{Q} = \{[v : y] \in \mathbf{P}(E \oplus E^*) : y(v) = 0\}.$$

Then

$$\dim \Lambda = h + j, \quad \deg \Lambda \leq C_e \deg V.$$

If $j = 1$, then $h = r - 1$ and Λ is a linear $\mathbf{P}^r \subset \mathcal{Q}$, hence a maximal isotropic projective subspace.

Proof. The fibers of the parameter map are the nonempty sets $F_\lambda \cap \Omega$, of dimension s_0 , whereas $\dim \Gamma = k + j$. The fiber dimension theorem therefore gives $\dim \Lambda = h + j$. We prove the degree estimate and the assertion for $j = 1$ in turn.

Compactifying the parameter space. On $\mathbf{P}(E)$, let $L = \mathcal{O}_{\mathbf{P}(E)}(-1)$ be the tautological line bundle and put $Q = (E \otimes \mathcal{O})/L$. Evaluation of a skew form on the line L gives the exact sequence

$$0 \longrightarrow K \longrightarrow S \otimes \mathcal{O} \longrightarrow Q^* \otimes L^* \longrightarrow 0, \quad K = \bigwedge^2 Q^*. \quad (\text{A.14})$$

Indeed, the value of a skew form on L is a map from L to its annihilator $L^\perp \subset E^*$, every such map extends to a skew form on E , and the kernel consists of the forms on E/L . The ranks of these three bundles are s_0, n, r , respectively.

All projective bundles below parametrize lines. Consider

$$B = \mathbf{P}(\mathcal{O} \oplus Q^* \otimes L^*) \simeq \mathbf{P}(L \oplus Q^*), \quad \rho : B \longrightarrow \mathbf{P}(E). \quad (\text{A.15})$$

The second presentation is obtained by tensoring the first bundle with L . We use the line convention of [14, Section 3.1 and Example 3.1.1]. Since $L \oplus Q^*$ is a subbundle of the trivial bundle with fiber $E \oplus E^*$, it induces a morphism $\mu : B \rightarrow \mathcal{Q}$. Explicitly, write a point in the first presentation as $(L, [t : \psi])$, with $\psi : L \rightarrow L^\perp$. For any nonzero $v \in L$,

$$\mu(L, [t : \psi]) = [tv : \psi(v)]. \quad (\text{A.16})$$

Both rescaling (t, ψ) and replacing v by another generator of L leave this point unchanged. Its quadratic value is zero because $\psi(v)$ annihilates v .

Over $\mathcal{Q}^\circ = \{[v : y] \in \mathcal{Q} : v \neq 0\}$, the inverse to μ recovers $L = \mathbb{C}v$ and $[1 : \psi]$, where $\psi(v) = y$. Thus μ is an isomorphism over this open set. A boundary point $[0 : y]$ has fiber $\mathbf{P}(\ker y)$: its preimages have $t = 0$, and L can be any line annihilated by y .

Let $\mathcal{T} = \mathcal{O}_B(-1)$ be the tautological line in the *first* presentation of (A.15), and set

$$a = \rho^* c_1(\mathcal{O}_{\mathbf{P}(E)}(1)), \quad b = c_1(\mathcal{T}^*), \quad \ell = \mu^* c_1(\mathcal{O}_{\mathcal{Q}}(1)).$$

The tautological line in the second presentation is $\mathcal{T} \otimes \rho^* L$. Consequently

$$\ell = a + b. \quad (\text{A.17})$$

The classes a and ℓ are globally generated. We do not need positivity of b .

The projective incidence and its boundary. Pull back (A.14) to B . Define $\mathcal{U}$ as the inverse image of $\mathcal{T}$ under the surjective bundle map

$$(\mathbb{C} \oplus S) \otimes \mathcal{O}_B \longrightarrow \mathcal{O}_B \oplus \rho^*(Q^* \otimes L^*), \quad (x_0, z) \longmapsto (x_0, z|_L).$$

It is a vector bundle of rank $s_0 + 1$, with exact sequence

$$0 \longrightarrow \rho^* K \longrightarrow \mathcal{U} \longrightarrow \mathcal{T} \longrightarrow 0. \quad (\text{A.18})$$

Its inclusion in the trivial bundle defines a morphism $\mathbf{P}(\mathcal{U}) \rightarrow \mathbf{P}(\mathbb{C} \oplus S)$. The pullback of the hyperplane class under this morphism is

$$\xi = c_1(\mathcal{O}_{\mathbf{P}(\mathcal{U})}(1)).$$

For a parameter represented by $(L, [1 : \psi])$, choose $v \in L \setminus \{0\}$ and write $y = \psi(v)$. The projective fiber maps isomorphically to

$$\{[x_0 : z] : zv = x_0 y\} = \overline{F_{[v:y]}}. \quad (\text{A.19})$$

Over $(L, [t : \psi])$, the defining condition is instead $(x_0, z|_L) \in \mathbb{C}(t, \psi)$. In particular,

$$t = 0 \implies x_0 = 0. \quad (\text{A.20})$$

Thus no boundary parameter occurs over an affine point $[1 : z]$.

Lift the parameter image of Γ to $\tilde{B}$ using the isomorphism over $\mathcal{Q}^\circ$, and denote its reduced irreducible closure by $\tilde{\Lambda}$. The map $\mu : \tilde{\Lambda} \rightarrow \Lambda$ is birational. Put

$$X = \mathbf{P}(\mathcal{U}|_{\tilde{\Lambda}}), \quad \pi : X \longrightarrow \tilde{\Lambda}.$$

The variety X is irreducible and has dimension $s_0 + h + j = k + j$. The point coordinate and the lifted parameter give an embedding of Γ as a locally closed subset of X : in its normalized direction chart the inverse recovers the original pair (z, v) . Since the dimensions agree, this subset is dense and contains a dense open subset of X . The projective point map on X therefore has image exactly $\overline{V}$. More explicitly, its image is closed by properness; density shows that it is contained in the closure of the point image of Γ , and dominance of $\Gamma \rightarrow \Omega$ gives the opposite inclusion. We obtain a surjective morphism

$$f : X \longrightarrow \overline{V}, \quad \xi = f^* c_1(\mathcal{O}_{\overline{V}}(1)).$$

Here and below we also use a, b, ℓ for their restrictions and pullbacks to the varieties on which they occur.

The degree of a generic point fiber. Fix an affine $z \in S$. By (A.20), the full incidence over $[1 : z]$, before restriction to Λ , has parameters $[v : zv]$ with $v \neq 0$, and is isomorphic to $\mathbf{P}(E)$. On this projective space ℓ is the ordinary hyperplane class: the map $[v] \mapsto [v : zv]$ is induced by an injective linear map $E \rightarrow E \oplus E^*$.

Choose a dense open subset $X^\circ \subset X$ lying in the lift of Γ . Every component of its complement has dimension at most $k + j - 1$. A component which dominates V therefore has generic fiber dimension at most $j - 1$; a component which does not dominate V is absent over a general point. The complement cannot supply a top-dimensional component of the generic fiber of f .

When $j = 0$, the generic fiber thus consists of at most C_e points, as does the fiber of $\Gamma \rightarrow \Omega$. When $j = 1$, the generic fiber of $\Gamma \rightarrow \Omega$ is an affine line of normalized vectors. Its closure in $\mathbf{P}(E)$ is a projective line, of degree one against ℓ , and this is the only top-dimensional generic fiber component of f . These components have multiplicity one. Indeed, the fibers on the dense open ruling incidence are smooth: its projection is finite étale or a smooth affine-line family. This determines the generic multiplicity of each component, which cannot be changed by a complement of smaller fiber dimension.

If d_Γ denotes the degree of the geometric generic fiber against ℓ^j , we have proved

$$d_\Gamma \leq C_e, \quad d_\Gamma = 1 \quad \text{when } j = 1. \quad (\text{A.21})$$

Proper pushforward and the projection formula, applied to fundamental cycles, now give

$$f_*(\ell^j \cap [X]) = d_\Gamma[\bar{V}], \quad \int_X \ell^j \xi^k = d_\Gamma \deg V \leq C_e \deg V. \quad (\text{A.22})$$

The coefficient in the first identity is computed on the generic fiber. Any other component of a representative hyperplane intersection has image of dimension less than k and pushes forward to zero. These formulas do not require smoothness of X or $\bar{V}$; see [14, Sections 1.4, 2.3, 2.5].

A positive Segre-class formula. For projective bundles of lines, the projective bundle formula is

$$\pi_*(\xi^{s_0+h} \cap [X]) = s_h(\mathcal{U}|_{\tilde{\Lambda}}) \cap [\tilde{\Lambda}], \quad s_t(\mathcal{U}) = c_t(\mathcal{U})^{-1}. \quad (\text{A.23})$$

This formula applies equally to a vector bundle on a singular base [14, Sections 3.1–3.3]. We compute it before restriction to $\tilde{\Lambda}$.

Dualizing the tautological sequence on $\mathbf{P}(E)$ and tensoring by $L^* = \mathcal{O}(1)$ gives

$$0 \longrightarrow Q^*(1) \longrightarrow E^* \otimes \mathcal{O}(1) \longrightarrow \mathcal{O}(2) \longrightarrow 0.$$

Together with (A.14), this implies

$$c_t(K)^{-1} = c_t(Q^*(1)) = \frac{(1+at)^e}{1+2at}.$$

Since $c_t(\mathcal{T}) = 1 - bt$, Equations (A.18) and (A.17) yield

$$s_t(\mathcal{U}) = \frac{(1+at)^e}{(1+2at)(1-bt)} = \frac{(1+at)^e}{(1+2at)(1+(a-\ell)t)}. \quad (\text{A.24})$$

Expanding

$$\frac{1}{1+(a-\ell)t} = \sum_{i \geq 0} \frac{\ell^i t^i}{(1+at)^{i+1}}$$

and taking the coefficient of t^h gives

$$s_h(\mathcal{U}) = \sum_{i=0}^h \ell^i a^{h-i} c(e-1-i, h-i), \quad c(p, u) = [z^u] \frac{(1+z)^p}{1+2z}. \quad (\text{A.25})$$

For integers $p \geq u+1$ and $u \geq 0$, these scalar coefficients are nonnegative. Indeed, $c(p, 0) = 1$, and $c(p, 1) = p-2 \geq 0$ in its asserted range. For $u \geq 2$, the identity $(1+z)^2 = (1+2z) + z^2$ gives

$$c(p, u) = \binom{p-2}{u} + c(p-2, u-2).$$

The second term lies in the same range, because $p-2 \geq (u-2)+1$. Induction proves the claim, with a binomial coefficient equal to zero when its lower index exceeds its upper index. In (A.25), the required inequality is $e-1-i \geq h-i+1$, which follows from $h \leq r-1 = e-2$. The coefficient of ℓ^h is $c(e-1-h, 0) = 1$.

All mixed top intersections of a and ℓ on $\tilde{\Lambda}$ are nonnegative. They can be computed by successive general members of globally generated linear systems, whose intersections with the fundamental cycle are effective or empty. Birationality of $\tilde{\Lambda} \rightarrow \Lambda$, followed by (A.25), (A.23), and (A.22), therefore gives

$$\deg \Lambda = \int_{\tilde{\Lambda}} \ell^{h+j} \leq \int_{\tilde{\Lambda}} \ell^j s_h(\mathcal{U}) = \int_X \ell^j \xi^k \leq C_e \deg V. \quad (\text{A.26})$$

The tangent constraint in the affine-line case. Suppose now that $j = 1$. Lemma A.6 gives $\text{codim}_S V = 1$, so $h = r - 1$ and $\dim \Lambda = r$. We first show that the tangent space of Λ at a general smooth point is maximal isotropic for the tangent quadratic form of $\mathcal{Q}$.

Choose a general smooth parameter $\lambda = [v : y] \in \Lambda \cap \mathcal{Q}^\circ$, fix $z_0 \in F_\lambda$, and let $U = E/\mathbb{C}v$, of dimension r . With a representative (v, y) fixed, tangent vectors to $\mathcal{Q}$ are pairs $(\dot{v}, \dot{y})$ satisfying $\dot{y}(v) + y(\dot{v}) = 0$, modulo $\mathbb{C}(v, y)$. There is an isomorphism

$$T_\lambda \mathcal{Q} \longrightarrow U \oplus U^*, \quad (\dot{v}, \dot{y}) \longmapsto (\alpha, \beta) = (\dot{v} \bmod v, \dot{y} - z_0 \dot{v}). \quad (\text{A.27})$$

The covector β annihilates v , by skew symmetry and $z_0 v = y$. Adding a multiple of (v, y) changes neither coordinate, and the kernel before quotienting is exactly that line, proving the isomorphism. The tangent quadratic form, defined up to scale on $\lambda^\perp/\lambda$, becomes $\beta(\alpha)$, because

$$(\dot{y} - z_0 \dot{v})(\dot{v}) = \dot{y}(\dot{v}).$$

Let $\Theta \subset U \oplus U^*$ be the resulting r -dimensional image of $T_\lambda \Lambda$.

The affine incidence over the smooth part of $\Lambda \cap \mathcal{Q}^\circ$ is an affine bundle with fibers F_λ , and its point image lies in V by the preceding compactification. At a point $z = z_0 + D \in F_\lambda$, with $D \in K_v = \bigwedge^2 U^*$, differentiating $zv = y$ gives

$$\dot{z}v = \dot{y} - z\dot{v} = \beta - D\alpha.$$

The vertical tangent space maps onto K_v . Evaluation at v identifies S/K_v with U^* , so modulo these vertical directions the differential is

$$\Theta \longrightarrow U^*, \quad (\alpha, \beta) \longmapsto \beta - D\alpha. \quad (\text{A.28})$$

Every parameter tangent vector lifts because the incidence is an affine bundle. At general points its differential has rank at most $\dim V = n - 1$. Hence (A.28) has rank less than r for general D and general λ . Its determinant is polynomial in the entries of D ; for each general λ it therefore vanishes for every skew form D on U .

Linear algebra of the singular skew pencil. We prove that this determinant condition forces Θ to be isotropic. Put

$$B_* = \Theta \cap U^*, \quad A_* = \text{pr}_U(\Theta), \quad H_* = B_*^\perp \subset U.$$

If $p = \dim A_*$, then $\dim B_* = r - p$ and $\dim H_* = p$. The plane Θ defines a linear map

$$C : A_* \longrightarrow U^*/B_* \simeq H_*^*.$$

The map in (A.28) is the identity on B_* . On taking the quotient by this subspace, the determinant condition is precisely

$$\det(C - D|_{A_* \times H_*}) = 0 \quad \text{for every } D \in \bigwedge^2 U^*. \quad (\text{A.29})$$

Here the restriction is regarded as a map $A_* \rightarrow H_*^*$. If $p = 0$, the original map is the identity on U^* , a contradiction. Thus $p > 0$.

We claim that $A_* = H_*$. Otherwise let $q = \dim(A_* \cap H_*) < p$, and choose bases

$$\begin{aligned} A_* &= \text{span}(c_1, \dots, c_q, a_1, \dots, a_{p-q}), \\ H_* &= \text{span}(c_1, \dots, c_q, h_1, \dots, h_{p-q}). \end{aligned}$$

All vectors displayed here are jointly linearly independent when the common vectors are listed only once. A restriction matrix of an ambient skew form on $A_* \times H_*$ may therefore be chosen arbitrarily except that its common q -by- q block must be skew. Each such prescription extends to a skew form on $A_* + H_*$, and then to U .

There is an invertible restriction matrix of this kind. For even q , use a nondegenerate skew block on the common indices and the identity between the remaining row and column indices. For odd q , use a nondegenerate skew block on $c_1, \dots, c_{q-1}$, the block

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

with rows (c_q, a_1) and columns (c_q, h_1) , and an identity matrix on the unused extra indices. Its two off-diagonal entries pair different independent vectors and so can both be chosen equal to one without violating skew symmetry. In either case we obtain an ambient skew form D_0 with invertible restriction. The polynomial $\det(C - tD_0|_{A_* \times H_*})$ then has a nonzero leading coefficient, contradicting (A.29). This proves $A_* = H_*$.

If p were even, a nondegenerate skew form on A_* , extended to U , would give the same contradiction. Hence p is odd. We may now regard C as a bilinear form on A_* . If it is not skew, polarization in characteristic zero gives $x \in A_*$ with $C(x, x) \neq 0$. Choose a complement to $\mathbb{C}x$, and choose a skew form D_0 which annihilates x and is nondegenerate on this even-dimensional complement. Extend it to U . In a basis beginning with x , the coefficient of t^{p-1} in $\det(C - tD_0|_{A_*})$ is

$$(-1)^{p-1} C(x, x) \det(D_0|_{A_*/\mathbb{C}x}) \neq 0.$$

To obtain this power of t , the determinant must use all remaining rows and columns from D_0 , leaving the entry $C(x, x)$. This also covers $p = 1$, with the empty determinant equal to one. Again we contradict (A.29); therefore C is skew.

Finally $B_* = A_*^\perp$. For every $(\alpha, \beta) \in \Theta$ we consequently have $\beta(\alpha) = C(\alpha, \alpha) = 0$. Polarization shows that Θ is totally isotropic. Its dimension is r in the nondegenerate $2r$ -dimensional space $U \oplus U^*$, so it is maximal isotropic. We have proved this tangent property on a dense open subset of the smooth locus of Λ .

Local rigidity of maximal null tangent spaces. We finish by showing directly that this tangent property makes Λ projectively linear. In the vector space defining $\mathcal{Q}$, choose a hyperbolic pair e_0, f_0 , with orthogonal complement W , and normalize the quadratic form as

$$q(se_0 + x + tf_0) = st + q_0(x), \quad x \in W.$$

Choose the pair so that a general smooth point of Λ lies in the chart

$$W \longrightarrow \mathcal{Q}, \quad x \longmapsto [e_0 + x - q_0(x)f_0]. \quad (\text{A.30})$$

The tangent quadratic form in this chart is $q_0(dx)$: the derivative of the displayed lift is $dx - (dq_0)_x(dx)f_0$, with quadratic value $q_0(dx)$. Shrink to a complex analytic neighborhood in the smooth locus on which the maximal null tangent property holds everywhere.

Linear coordinates $(u, w) \in \mathbb{C}^r \oplus \mathbb{C}^r$ on W can be chosen so that

$$q_0(u, w) = u \cdot u - w \cdot w, \quad T_{x_*} \Lambda = \{(\zeta, \zeta) : \zeta \in \mathbb{C}^r\}.$$

Here the dot denotes the standard complex bilinear pairing. For justification, choose a basis of the null tangent plane and a dual basis in a complementary plane. Subtracting half the Gram matrix

in the first set of directions makes the complementary plane null as well. Sums and differences of these bases, with rescaling over $\mathbb{C}$, give the displayed model. The projection to the u coordinates is invertible on the tangent space at x_* , so the holomorphic inverse function theorem writes the local submanifold as $w = F(u)$.

Write $F_i = \partial_i F$ and $F_{ij} = \partial_i \partial_j F$. Null tangency says

$$g_{ij} := F_i \cdot F_j = \delta_{ij}.$$

Differentiating these identities and commuting mixed partials gives

$$2F_{ij} \cdot F_l = \partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij} = 0.$$

The vectors $F_1, \dots, F_r$ are a basis, since their Gram matrix is the identity. Thus $F_{ij} = 0$ for all i, j , and F is affine on a smaller connected neighborhood. The local submanifold in W is an open subset of an affine plane $x_* + L_0$, where L_0 is an r -dimensional totally isotropic linear subspace.

Let $B_0(x, w) = q_0(x + w) - q_0(x) - q_0(w)$ be the polarization. For $w \in L_0$, $q_0(x_* + w) = q_0(x_*) + B_0(x_*, w)$, so the lift in (A.30) is affine-linear on this plane. Its projective closure is the projectivization of

$$\mathbb{C}(e_0 + x_* - q_0(x_*)f_0) + \{w - B_0(x_*, w)f_0 : w \in L_0\}.$$

This vector space has dimension $r + 1$: the first summand has nonzero e_0 coordinate, and the second maps injectively to L_0 . Its quadratic form vanishes identically. Indeed, it vanishes on the displayed affine lift, and homogeneity extends this identity to its linear span. We have obtained a projective linear $\mathbf{P}^r \subset \mathcal{Q}$ containing a nonempty analytic open subset of the smooth irreducible variety Λ .

Such an open subset is Zariski dense, since a proper algebraic subset has smaller dimension and cannot contain a smooth neighborhood. Hence Λ is contained in this $\mathbf{P}^r$, and equality follows from $\dim \Lambda = r$. This proves the exceptional assertion and completes the proof. $\square$

A.5 Closing the estimates with uniform constants

We now prove Theorem A.1. The preceding results control flats away from a low-degree exceptional section when V has a low-degree approximate description. A description containing a large degree instead produces a larger carrier whose degree is sufficiently smaller. A finite system of inequalities combines the two alternatives.

Proof of Theorem A.1. Fix the sample, A , and one of Specifications (a) and (b). For $0 \leq h < r$, let R_h be the supremum of

$$\frac{\#\{\lambda \in \mathcal{P} : F_\lambda \subset V\}}{\deg(V)A^h}$$

over the allowed irreducible varieties of dimension $s_0 + h$. An empty supremum is zero. These suprema are finite for the fixed sample: each numerator is at most $|\mathcal{P}|$, and each denominator is at least one. As noted after the theorem, $R_0 \leq 1$. In (a) also put

$$R_r = |\mathcal{P}|/A^r \leq C_0.$$

In (b) the index r is unavailable and will never be used.

We first refine the choice of approximate equations. Take the $c = n - \dim V$ equations from Theorem A.2, ordering their degrees as $d_1 \leq \dots \leq d_c$. They form a parameter sequence in the regular local ring $\mathbb{C}[S]_{I(V)}$, so every initial segment of length i has height i [25, Tags 00NQ, 02JN, and 00NA].

Choose a polynomial q of smallest positive degree in $I(V)$. We may use q as the first equation and preserve the upper bounds d_i on the remaining ones. Indeed, suppose a replacement prefix of length $i - 1$ has height $i - 1$. Its minimal primes have that height, because a partial parameter sequence in this regular local ring is a regular sequence. None can contain the ideal generated by the first i original equations, whose height is i . A generic constant linear combination of those i equations avoids all these minimal primes and extends the replacement prefix to height i . Its degree is at most d_i . Starting with q and continuing gives a parameter system again. Consequently V remains a component of its global common zero set. Replace d_1 by $\deg q$. The nondecreasing upper bounds still satisfy

$$\prod_{i=1}^c d_i \leq C_n \deg V. \quad (\text{A.31})$$

The later actual degrees may be smaller than their upper bounds; this only strengthens the relevant Bézout estimates.

Choose constants

$$K_1 > K_2 > \cdots > K_{r-1} > C_0,$$

to be fixed below in terms of e, C_0 . Consider an allowed carrier V of dimension $s_0 + h$, with $1 \leq h < r$.

Suppose first that every description degree d_i is at most $K_h A$. Lemma A.6 supplies a proper exceptional section $V \cap Z(H)$ with

$$\deg H \leq C_e K_h A.$$

Outside it, there are boundedly many ruling incidences. For one with $j = 0$, Proposition A.7 gives a parameter variety of dimension h and degree at most $C_e \deg V$; the separator bound then counts at most $C_{e,C_0} \deg(V) A^h$ parameters on it. For one with $j = 1$, that proposition says $h = r - 1$ and the parameter variety is a maximal isotropic $\mathbf{P}^r$. Assumption (ii) counts at most $C_0 A^h$ parameters on it. Thus all represented flats together cost at most $C_{e,C_0} \deg(V) A^h$.

The remaining flats are in $V \cap Z(H)$. Its components, if any, have dimension $s_0 + h - 1$ and total degree at most $C_e K_h A \deg V$. They remain allowed in (b), since they are subsets of the original allowed carrier. The definition of R_{h-1} gives

$$\frac{\#\{\lambda : F_\lambda \subset V\}}{\deg(V) A^h} \leq C(1 + K_h R_{h-1}). \quad (\text{A.32})$$

Here and below C depends only on e, C_0 , not on the K_i .

Suppose instead that some $d_i > K_h A$. Let $p < c$ be the length of the prefix with $d_i \leq K_h A$. The local prefix is a regular sequence of height p . A minimal prime of it in $\mathbb{C}[S]_{I(V)}$ contracts to a global component W of that prefix, containing V and of codimension p . Isolated-component Bézout gives

$$\deg W \leq \prod_{i \leq p} d_i.$$

For $p = 0$, take $W = S$ and the empty product to be one. Put $i = r - p > h$, so $\dim W = s_0 + i$. From (A.31),

$$\deg V \geq C_n^{-1} \deg W (K_h A)^{i-h}. \quad (\text{A.33})$$

In Case (b), the minimal first equation has degree at most $C_0 A$. Since $K_h > C_0$, it belongs to the prefix. Therefore $p \geq 1$, $i < r$, and W lies in that same hypersurface of degree at most $C_0 A$. Thus W is allowed in (b). In (a) the ambient possibility $i = r$ is already controlled. In either case, counting first on W yields

$$\frac{\#\{\lambda : F_\lambda \subset V\}}{\deg(V) A^h} \leq \frac{C R_i}{K_h^{i-h}}. \quad (\text{A.34})$$

Taking suprema in the two alternatives gives the finite system

$$R_h \leq \max \left\{ C(1 + K_h R_{h-1}), \max_{\substack{i > h \\ i \text{ available}}} \frac{C R_i}{K_h^{i-h}} \right\}, \quad 1 \leq h < r. \quad (\text{A.35})$$

An empty inner maximum is omitted. We close this system explicitly, so that no constant depends on the description degrees or on the size of the sample.

Enlarge C to at least one. Choose

$$B_0 \geq \max(1, C_0), \quad c_1 > 4C + 4, \quad \kappa > 4C c_1.$$

Take $K_{r-1} \geq \max(2, C_0 + 1)$, and successively set $K_h = \kappa K_{h+1}$ for $h = r - 2, \dots, 1$. Define barriers

$$B_h = c_1 K_h B_{h-1} \quad (1 \leq h < r), \quad B_r = B_0 \quad \text{in Case (a).}$$

For interior indices $h < i < r$,

$$\frac{C B_i}{B_h K_h^{i-h}} = C c_1^{i-h} \prod_{j=h+1}^i \frac{K_j}{K_h} \leq C (c_1/\kappa)^{i-h} < \frac{1}{4}.$$

If the ambient index is available, $B_h \geq c_1 K_h B_0$ gives

$$\frac{C B_r}{B_h K_h^{r-h}} \leq \frac{C}{c_1 K_h^{r-h+1}} < \frac{1}{2}.$$

The other transition satisfies

$$\frac{C(1 + K_h B_{h-1})}{B_h} \leq \frac{2C}{c_1} < \frac{1}{2}.$$

All choices so far depend only on e, C_0 .

If the maximum $T = \max_h R_h/B_h$ were greater than one, a maximizing index would be interior: $R_0 \leq B_0$, and the ambient endpoint, when present, obeys the same bound. Substitute $R_j \leq T B_j$ into (A.35). Since $T > 1$, each term in its right side is strictly less than $T B_h$, contradicting maximality. Hence $R_h \leq B_h$ for every available index. This is (A.2), with a constant depending only on e, C_0 . $\square$

B Real incidence estimates

We give the incidence arguments used in the proof. All constants in this section are absolute. The elementary algebraic facts used below are the following forms of Bézout's Theorem: two relatively prime polynomials of degrees a, b in three variables have at most ab distinct common line components, and a polynomial of degree a has at most a zeros on a line unless it vanishes identically on that line. We also use unique factorization, the implicit function theorem, and the Borsuk–Ulam Theorem in its standard form that every continuous odd map $S^q \rightarrow \mathbb{R}^q$ has a zero [18, Corollary 2B.7]. In the ruled-surface argument we use the basic projective facts that a projective morphism has closed image, that varieties have finitely many irreducible components, and that a proper closed subset of an irreducible curve is finite. We use the dimension and generic-fiber-dimension theorems, and extension of derivations across separable algebraic field extensions. The incidence theorems and the special assertions about ruled surfaces are proved below. The historical sources for the two principal incidence bounds are [24, 16].

B.1 The planar incidence bound

We use the crossing inequality of Ajtai, Chvátal, Newborn, and Szemerédi and of Leighton [1, 20], following Székely's crossing-number proof of the incidence estimate [23].

Lemma B.1 (Crossings). *Suppose a simple graph with $v \geq 1$ vertices and $e \geq 4v$ edges is drawn in the plane, no edge goes through a vertex other than its endpoints, and two edges with a common endpoint have no other intersection. If X counts intersecting pairs of edges away from their endpoints, then $X \geq e^3/(64v^2)$.*

Proof. Deleting at most one edge for each intersecting pair produces a planar simple graph. Euler's formula therefore gives $e \leq 3v + X$. Select each vertex independently with probability p and keep the induced drawing. An edge survives with probability p^2 and an intersecting pair with probability p^4 , because its four endpoints are different. Taking expectations in the same inequality gives $p^2e \leq 3pv + p^4X$. Set $p = 4v/e$ and rearrange. $\square$

Theorem B.2 (Szemerédi–Trotter). *For finite sets Q of n points and $\mathcal{L}$ of L distinct real lines in the plane,*

$$I(Q, \mathcal{L}) \leq C(n^{2/3}L^{2/3} + n + L).$$

Consequently the number of points incident to at least $k \geq 2$ lines is at most $C(L^2/k^3 + L/k)$, and the number of lines containing at least $k \geq 2$ points of an n -point set is at most $C(n^2/k^3 + n/k)$. The same assertions apply to finite point–line configurations in $\mathbb{R}^s$ and in real projective space.

Proof. The assertion is immediate if $n = 0$ or $L = 0$, so assume both are positive. Join consecutive points of Q on each line of $\mathcal{L}$. The resulting graph is simple: a pair of distinct points determines just one line. If e is its number of edges, then $e \geq I - L$. Edges on the same line have disjoint interiors. Edges on two different lines cross at most once, so their crossing count is at most $\binom{L}{2}$. Lemma B.1 gives either $e < 4n$ or $e^3 \leq 32n^2L^2$, proving the incidence bound. Concurrent crossings cause no problem: count pairs of intersecting edges, or separate the crossings in a sufficiently small disk without changing the bound.

For a set of s points each incident to at least k lines, the bound gives $ks \leq C(s^{2/3}L^{2/3} + s + L)$. If k exceeds a fixed constant, absorb the term Cs and compare the two remaining terms, obtaining $s \leq C'(L^2/k^3 + L/k)$. For bounded $k \geq 2$, count pairs of lines: $s \leq \binom{L}{2}$, which gives the same assertion after increasing the constant. Interchanging points and lines by planar projective duality proves the rich-line statement. One can instead apply the incidence bound directly to the finite collection of rich lines, using that two points determine at most one line for bounded k .

For a finite configuration in higher-dimensional affine space, choose a linear projection to $\mathbb{R}^2$ which is injective on its points and sends its lines to distinct lines. Each forbidden coincidence is a proper algebraic condition on the projection, and finitely many such conditions can be avoided. Existing incidences survive, which is all that an upper bound requires. For projective data first choose an affine chart containing every specified point and in which no specified line is contained in the hyperplane at infinity. Apply the affine argument. $\square$

B.2 Polynomial bisection and flat lines

Polynomial bisection is the Stone–Tukey ham-sandwich construction [22], as used in Guth–Katz polynomial partitioning [16].

Lemma B.3 (Polynomial bisection). *Given a finite set $S \subset \mathbb{R}^3$ and an integer $r \geq 1$, there is a nonzero real polynomial f of degree at most Cr such that $\mathbb{R}^3 \setminus Z(f)$ is partitioned into at most Cr^3 open sets, each containing at most $C|S|/r^3$ points of S . Every line not contained in $Z(f)$ meets at most $\deg f + 1$ of these sets. The sets need not be connected.*

Proof. For $m < \binom{D+3}{3}$ finite-volume bounded measurable sets in $\mathbb{R}^3$, normalize the coefficients of a degree-at-most- D polynomial to lie on their unit sphere. Its signed-volume imbalances on the m sets form a continuous odd map to $\mathbb{R}^m$. Continuity follows by dominated convergence, since the zero set of a nonzero polynomial has Lebesgue measure zero. The latter fact follows by induction on the number of variables and Fubini’s Theorem. Restrict the coefficient sphere to an $(m+1)$ -dimensional linear subspace and apply Borsuk–Ulam. We obtain a polynomial bisecting all m sets.

Replace each point in each of m finite sets by a disjoint small ball of the same radius within that set, apply this assertion, and let the radius tend to zero. A convergent subsequence of normalized coefficient vectors gives a nonzero polynomial for which each strict sign contains at most half the points of each finite set. Indeed, a strict positive value at a point persists on its small ball and for nearby coefficient vectors; more than half of the points of either sign would contradict the volume bisection.

Iterate this finite bisection J times, bisecting all the at most 2^{j-1} sign classes at stage j . A degree $C2^{j/3}$ suffices at that stage. The product f of these J polynomials has degree at most $C2^{J/3}$, and every strict sign class has at most $2^{-J}|S|$ points. Choose 2^J comparable to r^3 . A line outside $Z(f)$ is cut into at most $\deg f + 1$ intervals by its intersections with $Z(f)$; all the factors have constant sign on each interval. This proves the last assertion as well. $\square$

The critical-line and flat-line tests below are part of the algebraic incidence method of Guth–Katz and Elekes–Kaplan–Sharir [15, 10]. For a square-free real polynomial f of degree D , call a point of $Z(f)$ *critical* if $\nabla f = 0$. A noncritical point is *flat* if its second fundamental form vanishes. A line in $Z(f)$ is critical if all its points are critical, and is flat if it is not critical and all its noncritical points are flat.

Lemma B.4 (Critical and flat lines). *For a square-free polynomial f of degree D :*

1. *A point lying on three distinct lines contained in $Z(f)$ is critical or flat.*
2. *There are at most $D(D-1)$ critical lines.*
3. *A line containing more than D critical points is critical. A line containing more than $3D$ flat points is flat.*
4. *At most $3D^2$ flat lines fail to lie in a plane component of $Z(f)$.*

Proof. At a noncritical point all contained lines are tangent to the smooth surface. The second fundamental form vanishes in their directions. A homogeneous quadratic on a two-dimensional vector space which vanishes in three distinct projective directions vanishes identically. This proves the first assertion.

Square-freeness in characteristic zero implies that a generic constant linear combination of the first derivatives of f shares no irreducible factor with f . All critical lines are common line components

of their zero sets, so Bézout gives the second assertion. Restriction of f and its derivatives to a line proves the assertion about many critical points.

Here is an explicit polynomial test for flatness. With H_f denoting the Hessian matrix and e_1, e_2, e_3 the coordinate vectors, put

$$T_j = (H_f(e_j \times \nabla f)) \times \nabla f, \quad j = 1, 2, 3.$$

The nine scalar components have degree at most $3D$. At a regular point the vectors $e_j \times \nabla f$ span the tangent plane. Thus all T_j vanish exactly when H_f maps every tangent vector into the normal line, which is exactly vanishing of the second fundamental form. If a line contains more than $3D$ flat points, f and all these components vanish identically on it. It has a regular point, so it is a flat line.

It remains to bound flat lines on nonplanar components. Fix an irreducible real factor g of f which contains a flat line and is not linear. That line has a point regular for f , and consequently g has a smooth real point. If every component of every T_j were divisible by g , the second fundamental form would vanish on a neighborhood of that real point in $Z(g)$. The differential of the unit normal would vanish there. The normal would be constant on a smaller connected neighborhood, and that neighborhood would be contained in its tangent plane. The implicit function theorem then makes it an open subset of that plane. Restricting g to the plane shows that its plane equation divides g , contrary to irreducibility and $\deg g > 1$. Therefore some component T is not divisible by g , and Bézout bounds their common lines by $3D \deg g$.

Every flat line belongs to exactly one irreducible factor of f : membership in two factors would make the entire line critical. Summing the preceding bound over the nonplanar factors gives at most $3D^2$ flat lines. Factors having no smooth real point contain no flat line and do not need a separate argument. $\square$

B.3 The higher-richness line theorem

Theorem B.5 (Guth–Katz, higher richness). *Let $\mathcal{L}$ be L distinct lines in $\mathbb{R}^3$, with at most B_0 in any affine plane. For every integer $k \geq 3$, the number R_k of points lying on at least k lines satisfies*

$$R_k \leq C \left(\frac{L^{3/2}}{k^2} + \frac{LB_0}{k^3} + \frac{L}{k} \right).$$

There is no assumption about lines on quadrics in this theorem.

Proof. We first prove a uniform version. Let S be a set of s points each incident to between k and $2k$ lines. Suppose at least $L/100$ lines each contain at least $sk/(100L)$ points of S . We will prove the stated bound for s .

Fix a sufficiently small absolute $\varepsilon > 0$. We may assume

$$s > A \left(L^{3/2} k^{-2} + L k^{-1} \right), \tag{B.1}$$

where A will be chosen sufficiently large. By Theorem B.2, this assumption implies $s \leq CL^2 k^{-3}$. Choose

$$r = \left\lfloor \theta L^2 s^{-1} k^{-3} \right\rfloor,$$

where the constant θ is sufficiently large. Thus $r \geq 1$ and r is comparable to the expression inside the floor. Lemma B.3 gives a polynomial f of degree $D \leq Cr$. We claim that at most εs points of S lie outside $Z(f)$.

Otherwise, among its $O(r^3)$ sign cells, at least $c_\varepsilon r^3$ contain at least $c_\varepsilon s/r^3$ points of S . To verify this, discard cells with fewer than a small fixed multiple of $\varepsilon s/r^3$ points and use the uniform upper bound Cs/r^3 for the others. Each line meets at most $D+1 = O(r)$ cells. One of the retained cells is therefore met by at most $C_\varepsilon L/r^2$ lines. Apply Theorem B.2 inside it:

$$\frac{s}{r^3} \leq C_\varepsilon \left(\frac{L^2}{r^4 k^3} + \frac{L}{r^2 k} \right).$$

Substitution of r gives

$$s \leq C_\varepsilon \left(\theta^{-1} s + \theta L^3 s^{-1} k^{-4} \right).$$

Choose θ first so that the first term can be absorbed. The resulting inequality $s \leq C_\varepsilon \sqrt{\theta} L^{3/2} k^{-2}$ contradicts (B.1) when A is sufficiently large. This proves the claim.

Replacing f by its square-free part does not change its zero set and can only lower its degree. The same estimates and (B.1) give

$$D \leq C\theta L^2 s^{-1} k^{-3}, \quad \frac{D}{sk/L} \leq \frac{C\theta}{A^2}, \quad \frac{D}{\sqrt{L}} \leq \frac{C\theta}{Ak}. \quad (\text{B.2})$$

Write $\mathcal{L}_Z$ for the lines contained in $Z(f)$, and S_* for the points of S lying on at least three lines of $\mathcal{L}_Z$. A point of $(S \cap Z(f)) \setminus S_*$ lies on at least $k-2$ lines outside $\mathcal{L}_Z$. Each such line has at most D points on $Z(f)$. Hence

$$|S \setminus S_*| \leq \varepsilon s + \frac{LD}{k-2} \leq \left(\varepsilon + \frac{3C\theta}{A^2} \right) s.$$

Choose A so large that this is at most $2\varepsilon s$.

Let $\mathcal{L}_0$ be the at least $L/100$ lines in the uniformity assumption. If such a line contains fewer than $sk/(200L)$ points of S_* , it contains at least $sk/(200L)$ points of $S \setminus S_*$. But this latter set has at most $2k|S \setminus S_*| \leq 4\varepsilon sk$ incidences. Thus at most $800\varepsilon L$ lines of $\mathcal{L}_0$ fail to contain $sk/(200L)$ points of S_* . With ε small enough, at least $L/200$ lines succeed.

All points of S_* are critical or flat by Lemma B.4. On each successful line one of these two kinds occurs at least $sk/(400L)$ times. By (B.2) this is more than $3D$, on enlarging A . Every successful line is consequently a critical line or a flat line. Lemma B.4 and the last bound in (B.2) show that fewer than $L/400$ of them are critical or flat on nonplanar components. At least $L/400$ therefore lie in plane components of $Z(f)$. There are at most D such components, so one contains at least $L/(400D)$ lines. It follows that

$$B_0 \geq \frac{L}{400D} \geq c \frac{sk^3}{L}, \quad s \leq CLB_0 k^{-3}.$$

Together with the alternative to (B.1), this proves the uniform version with an absolute constant C_0 .

We remove the uniformity assumption by induction on L , keeping this same constant C_0 . Let $\mathcal{L}_1$ consist of the lines containing at least $sk/(100L)$ points of S . If $|\mathcal{L}_1| \geq L/100$, we have already finished. Otherwise the other lines supply fewer than $sk/100$ incidences, and at least $9s/10$ points of S lie on at least $9k/10$ lines of $\mathcal{L}_1$. Partition these points according as their multiplicity in $\mathcal{L}_1$ is at least k or less than k . One class has at least $9s/20$ points. Its multiplicities lie between k_1 and $2k_1$, where either $k_1 = k$ or $k_1 = \lceil 9k/10 \rceil \geq 3$. The induction hypothesis applies with $L_1 = |\mathcal{L}_1| < L/100$ and the unchanged plane cap B_0 . Consequently

$$s \leq \frac{20}{9} C_0 \left(\frac{L_1^{3/2}}{k_1^2} + \frac{L_1 B_0}{k_1^3} + \frac{L_1}{k_1} \right) \leq C_0 \left(\frac{L^{3/2}}{k^2} + \frac{L B_0}{k^3} + \frac{L}{k} \right),$$

because $(20/9)(1/100)(10/9)^3 < 1$. Empty configurations start the induction. Finally divide all k -rich points into multiplicity intervals $[2^j k, 2^{j+1} k)$ and sum this uniform-multiplicity bound. The three resulting geometric series converge. $\square$

Corollary B.6 (Planar motions with many matches). *Let $U, V \subset \mathbb{R}^2$ have respective cardinalities u, v . For $3 \leq k \leq \min(u, v)$, at most $C(uv)^{3/2}/k^2$ Euclidean isometries g have $|g(U) \cap V| \geq k$.*

Proof. There are finitely many isometries with at least two matches: two distinct ordered matches determine at most two isometries. Separate the two orientation classes and reflect one cloud for the reversing class. A generic rotation of V makes $I + R$ invertible for the linear part R of every counted direct isometry. Write $J(x_1, x_2) = (-x_2, x_1)$. In its planar Cayley chart a direct isometry is represented by a point $(t, c) \in \mathbb{R} \times \mathbb{R}^2$, and a match $x \mapsto y$ is represented by the line

$$c = \frac{x - y}{2} - tJ\frac{x + y}{2}. \quad (\text{B.3})$$

Indeed, putting $Z = tJ$, the match equation is equivalent to $(I + Z)y = (I - Z)x - 2c$, and $(I + Z)^{-1}(I - Z)$ is orthogonal of determinant one. Conversely every direct isometry with $I + R$ invertible has this representation.

The uv lines are distinct: their intercept and slope determine $x - y$ and $x + y$, hence both endpoints. A plane in (t, c) -space has an equation $a \cdot c + bt = d$. If $a = 0$ it contains no such line. Otherwise, for a line (B.3) contained in the plane,

$$a \cdot (x - y) = 2d, \quad a \cdot J(x + y) = 2b.$$

With x fixed these determine y uniquely, because a and $J^T a$ are independent; the same holds with y fixed. Thus the plane cap is $B_0 \leq \min(u, v)$. Apply Theorem B.5. Its middle term is bounded by its first because $B_0 \leq \sqrt{uv} \leq k\sqrt{uv}$, and its last is bounded by its first because $k \leq \min(u, v) \leq \sqrt{uv}$. The bounded orientation split completes the proof. $\square$

B.4 A ruled-surface criterion with a degree bound

This is the flecnode approach to ruledness used in [16, Section 3, Proposition 3.2 and Corollary 3.3]. The direct resultant calculation below gives degree at most $17d - 24$; the sharper flecnode degree used in that source is not needed here.

All varieties in the next two lemmas are over $\mathbb{C}$, unless a real structure is explicitly mentioned. A projective surface is *ruled* if every point belongs to a projective line contained in the surface. The statement is geometric: a real sphere is ruled over $\mathbb{C}$ even though it contains no real line. A *regulus* in the real incidence statements is the affine part of a smooth real projective quadric with real lines.

Lemma B.7 (A polynomial detecting ruledness). *An irreducible surface of degree d in $\mathbb{C}^3$ which is not geometrically ruled contains at most $17d^2$ affine lines.*

Proof. Planes are ruled. An irreducible complex projective quadric has rank three or four: rank three gives a cone over a conic, and in rank four a linear coordinate change gives $X_0X_3 = X_1X_2$, whose two rulings follow by fixing the ratio in either pair of factors. Hence we may assume $d \geq 3$. Write its irreducible defining polynomial as F . After a linear coordinate change, $A = F_z$ is not identically zero. Put $B = F_x$, $C = F_y$ and form the tangent vector

$$V(u, v) = (Au, Av, -Bu - Cv).$$

The binary forms

$$q(u, v) = D^2 F[V, V], \quad c(u, v) = D^3 F[V, V, V]$$

are obtained by differentiating F only, with the displayed vector held fixed. Their coefficient degrees are at most $3d - 4$ and $4d - 6$. Let R be their homogeneous binary resultant. The Sylvester determinant is homogeneous of degree three in the quadratic coefficients and degree two in the cubic coefficients; therefore

$$\deg R \leq 3(3d - 4) + 2(4d - 6) = 17d - 24. \quad (\text{B.4})$$

The resultant vanishes precisely when the two binary forms have a common projective root; this includes identically zero forms.

Every affine line in $Z(F)$ lies in $Z(R)$. At a point where $A \neq 0$, use its tangent direction, for which all directional derivatives vanish. At a point where $A = 0$, the two forms become

$$q = F_{zz}(Bu + Cv)^2, \quad c = -F_{zzz}(Bu + Cv)^3,$$

so they again have a common projective root. This also handles a line lying entirely in $Z(F_z)$.

We next prove that $F \mid R$ implies ruledness. Let $K = \mathbb{C}(Z(F))$. This field is a finite separable extension of $\mathbb{C}(x, y)$. Extend ∂_x, ∂_y to an algebraic closure of K , where they commute, and regard z as an algebraic function of x, y . Write

$$r = z_{xx}, \quad s = z_{xy}, \quad t = z_{yy}.$$

If this Hessian is zero, $D = \partial_x$ satisfies $D^2 x = D^2 y = D^2 z = 0$, and the formal argument below already gives a line. Otherwise a constant invertible linear change of x, y ensures $t \neq 0$. Thus the binary quadratic of second derivatives does not vanish in the infinite-slope direction.

Since $F \mid R$, there is a finite common tangent slope a in the algebraic closure of K . Put $w = z_x + az_y$. Implicit differentiation, keeping a fixed in the directional derivatives, gives

$$\begin{aligned} D^2 F[(1, a, w), (1, a, w)] &= -F_z Q(a), \\ D^3 F[(1, a, w), (1, a, w), (1, a, w)] &= -F_z P(a) - 3Q(a)(F_{xz} + aF_{yz} + wF_{zz}), \end{aligned}$$

where

$$Q(a) = r + 2sa + ta^2, \quad P(a) = z_{xxx} + 3az_{xxy} + 3a^2 z_{xyy} + a^3 z_{yyy}.$$

Consequently $Q(a) = P(a) = 0$.

Set $D = \partial_x + a\partial_y$. If a is a simple root of Q , differentiating $Q(a) = 0$ gives

$$0 = D(Q(a)) = P(a) + 2(s + ta)Da,$$

so $Da = 0$. If it is a repeated root, then $s = -ta$ and $r = ta^2$. Use the mixed-partial identities $r_y = s_x$, $s_y = t_x$:

$$t_x = -t_y a - ta_y, \quad t_y a^2 + 2taa_y = -t_x a - ta_x = t_y a^2 + taa_y - ta_x.$$

Their difference gives $t(a_x + aa_y) = 0$, so $Da = 0$ in this case as well. It follows that

$$Dx = 1, \quad Dy = a, \quad Dz = w, \quad D^2 x = D^2 y = D^2 z = 0.$$

The formal Taylor map $\exp(TD) = \sum_{j \geq 0} T^j D^j / j!$ is an algebra homomorphism in characteristic zero: the product rule proves this coefficient by coefficient. Apply it to $F(x, y, z) = 0$ to obtain

$$F(x + T, y + aT, z + wT) = 0.$$

Thus the geometric generic point of the surface lies on a contained line. The set of lines on its projective closure is a closed subset of the Grassmannian: substituting the line parameterization in the homogenized equation gives polynomial equations for it. The projective point–line incidence variety has closed image in the surface. Since its image contains the geometric generic point, it is the whole surface. This proves ruledness.

If the surface is not ruled, F and R are therefore relatively prime, and Bézout with (B.4) bounds their common lines by $17d^2$. $\square$

B.5 Intersections among generators

Lemma B.8 (Ruled-surface structure). *Let S be an irreducible ruled projective surface over $\mathbb{C}$ which is neither a plane nor a smooth quadric. Its lines have a unique one-dimensional irreducible family Γ , called the generator family, whose members cover S . A generic point of S belongs to exactly one generator. There is at most one point lying on infinitely many contained lines, and at most two contained lines meeting infinitely many other contained lines away from such points. Every line outside Γ is one of these at most two lines.*

Proof. The algebraic set of lines on a nonplanar irreducible surface has dimension at most one. Otherwise its incidence variety would have dimension at least three, and through a generic smooth point of S there would be infinitely many contained lines. All would lie in the tangent plane. Infinitely many distinct lines through that point fill the tangent plane algebraically, making this plane a component of S , a contradiction. Each curve component of the line variety covers S : its union is closed by projectivity, and infinitely many distinct lines cannot be contained in a proper closed subset of an irreducible surface. Indeed that subset has finitely many curve components. All other components of the line variety are isolated points and thus represent finitely many lines.

Suppose a generic point had at least two generators. Fix a generic generator ℓ from a component Γ_1 . A dense set of points of ℓ then belongs to a second generator distinct from ℓ . These second generators are infinitely many, since each meets ℓ in only one point. Some curve component Γ_j therefore has infinitely many members meeting ℓ , and closedness makes every member meet ℓ . For each j , the set of members of Γ_1 meeting every member of Γ_j is closed: it is the intersection of the closed meeting loci for the individual lines. These finitely many closed sets cover a dense subset of Γ_1 . Irreducibility makes one of them all of Γ_1 . Call the corresponding component Γ_2 . This includes the case $\Gamma_2 = \Gamma_1$; the diagonal alone was not counted as a second generator.

If three generic members of Γ_1 are pairwise skew, they determine a unique smooth quadric, and their common transversals are exactly its opposite ruling. To see this directly, take the first two lines as complementary coordinate two-planes in $\mathbb{C}^4$ and the third as the graph of an invertible linear map between them. A transversal is the projectivization of the span of u and its image, with $[u] \in \mathbb{P}^1$; their union is a nonsingular quadric. Hence Γ_2 would cover that quadric and S would equal it.

If generic pairs in Γ_1 are not skew, every pair meets, because the meeting condition is closed in $\Gamma_1 \times \Gamma_1$. A family of pairwise meeting lines is coplanar or concurrent. Indeed, fix two meeting at p and spanning a plane H . Every line meeting both is contained in H or passes through p ; irreducibility of the family forces one alternative throughout. The first makes S a plane. The second makes it a cone with vertex p . Such a nonplanar cone contains no line ℓ avoiding p : all the joins from p to the points of ℓ would give the entire plane spanned by p and ℓ in S . It consequently has a unique generator through a generic point, contradicting the assumption of a second generator. This proves generic uniqueness. It also proves uniqueness of Γ , since every curve component covers S .

Call a point or line *exceptional* if it has the corresponding infinite-incidence property in the statement. If p is an exceptional point, the closed subfamily of lines through it contains the whole

generator curve. Thus all generators pass through p . There cannot be two such points: a generator through both would be their unique joining line. Hence there is at most one exceptional point.

An exceptional line meets every generator, since infinitely many generators meet it and Γ is irreducible. Three pairwise skew exceptional lines would force S into their quadric as above. Two exceptional lines cannot meet: all generators meeting both would lie in their plane or pass through their common point. The former makes S planar, while the latter makes the common point exceptional and prevents either line from meeting infinitely many generators away from it. Thus there are at most two exceptional lines. Finally, a contained line outside Γ has all its points covered by generators. Infinitely many generators meet it at distinct points, and removing the possible one exceptional point does not change this fact. It is therefore exceptional. $\square$

Lemma B.9 (Counting on ruled surfaces). *Let a union of real absolutely irreducible geometrically ruled surfaces have total degree $d \geq 1$, and let $\mathcal{L}$ consist of $M \geq 1$ distinct real affine lines contained in that union. If at most s lines lie in any plane or regulus, then the number of points on at least two lines of $\mathcal{L}$ is at most $5dM + sM/2$.*

Proof. First suppose the projective closure of every irreducible component is neither a plane nor a smooth projective quadric. Across all the components there are at most d exceptional points and at most $2d$ exceptional lines by Lemma B.8. Intersections involving an exceptional line cost at most $2dM$ points. The exceptional points themselves cost at most d .

Fix any remaining line ℓ . Choose a generic complex plane H through it and restrict the square-free equation of the entire union to H . Divide out the entire power of the linear factor defining ℓ . The residual plane curve C_H has degree at most $d - 1$ and does not contain ℓ . Dividing out the entire power is necessary when ℓ is a singular line of a component.

We claim that every remaining intersection $q = \ell \cap \ell'$ belongs to C_H . The nonexceptional line ℓ' belongs to the generator curve of some component S_j . Take H not containing ℓ' . If only finitely many generators of S_j meet ℓ , then the map sending a generator to its intersection with H is regular near ℓ' . On a dense open subset its values lie in C_H . As C_H is closed, its value at ℓ' is also in C_H , which says $q \in C_H$. This argument applies even at a singular point of the generator curve.

The alternative of infinitely many generators meeting ℓ cannot occur at the present q . If $\ell \subset S_j$, either these meetings range through infinitely many nonexceptional points, making ℓ exceptional, or all generators pass through a single exceptional point on ℓ . In the second case every generator other than ℓ meets it only at that point. If $\ell \not\subset S_j$, the finite set $\ell \cap S_j$ contains all the meetings; irreducibility then forces all generators through one point of this finite set, which is exceptional. Both cases contradict the choice of q . This proves the claim. Bézout in H now bounds the number of such q on ℓ by $d - 1$. Summing and restoring the exceptions bounds the total by $(d - 1)M + 2dM + d \leq 4dM$.

For the general union, extract all lines in each plane component or component with smooth projective quadric closure, one component at a time. A smooth real projective quadric having real lines gives a regulus; a quadric without real lines contributes nothing. If an extracted group has a_j lines, then $a_j \leq s$ and its internal intersections number at most $a_j^2/2 \leq sa_j/2$. Every line left after its extraction is not contained in that component, so it meets it in at most the degree of the component points. All such cross intersections cost at most dM . The remaining components and lines are covered by the preceding $4dM$ estimate. Since the extracted groups are disjoint, their internal contributions sum to at most $sM/2$. $\square$

B.6 The two-rich line theorem

The sampling and degree-reduction argument follows the architecture of the two-rich-point proof in [16, Section 3].

Theorem B.10 (Two-rich points). *For m distinct real affine lines, with at most $s \geq 1$ in any plane or regulus, the number P_2 of points lying on at least two lines satisfies*

$$P_2 \leq K(m^{3/2} + ms)$$

for an absolute constant K .

Proof. Write $H(m, s) = m^{3/2} + ms$ and induct on m , keeping the same parameter s throughout. Choose K larger than all the absolute constants required below. The case $m \leq 1$ is immediate. Suppose, towards a contradiction, that $P_2 > KH(m, s)$. Repeatedly delete a line having fewer than

$$T = \frac{K}{4}(\sqrt{m} + s)$$

distinct intersection points with other currently retained lines. Each deletion loses fewer than T two-rich points. The remaining set of $M \leq m$ lines therefore has more than $3KH(m, s)/4$ two-rich points and is nonempty. Every remaining line has at least T distinct intersections with the other remaining lines.

Select each remaining line independently with probability $64/K$. For each fixed remaining line choose one witness line at every distinct intersection point. These witnesses are distinct. The number X of selected witnesses is a binomial random variable with mean at least $16\sqrt{m}$. The elementary binomial tail bound $\Pr(X < \mathbb{E}X/2) \leq \exp(-\mathbb{E}X/8)$ gives

$$\Pr(X < 8\sqrt{m}) \leq e^{-2\sqrt{m}}.$$

For completeness, this tail bound follows by applying Markov's inequality to $e^{-\lambda X}$ and the inequality $1 + p(e^{-\lambda} - 1) \leq \exp(p(e^{-\lambda} - 1))$, then taking $\lambda = \log 2$; the resulting exponent is at most $-\mathbb{E}X/8$. A union bound over all remaining lines costs at most $me^{-2\sqrt{m}} \leq e^{-2} < 1/4$. Markov's inequality also shows that with probability at least $1/2$ the number q of selected lines is at most $128M/K$. There is therefore a choice with both properties. It necessarily has $q \geq 1$.

Set $D = \lceil \sqrt{6q} \rceil$. A polynomial of degree at most D has $\binom{D+3}{3}$ coefficients, and vanishing on a line imposes at most $D + 1$ linear conditions. Since $(D + 3)(D + 2) > 6q$, there is a nonzero real polynomial vanishing on every selected line. Moreover

$$D < 4\sqrt{q}, \quad D^2 \leq 2048m/K,$$

and $D < 8\sqrt{m}$ for our choice of K . Each remaining line contains at least $8\sqrt{m}$ distinct points on the selected lines, so the polynomial vanishes identically on all M remaining lines. Replace it by its square-free part.

Factor the polynomial over $\mathbb{C}$. Retain the real, absolutely irreducible, geometrically ruled factors. At most $b \leq 18D^2$ of the M lines fail to lie on their union. Indeed, Lemma B.7 bounds the lines on each nonruled factor of degree d_i by $17d_i^2$. A real line on a factor which is not real up to a scalar also lies on its distinct conjugate; Bézout bounds such lines by d_i^2 . Summing these bounds and using $\sum d_i \leq D$ proves the assertion. Thus

$$b \leq 36864m/K \leq m/4.$$

Every one of these b lines meets the retained ruled union in at most D points, since it is not contained in that union. Lemma B.9 bounds intersections among the ruled lines by $5Dm + sm/2$. The induction hypothesis, with the original cap s , bounds intersections among the b other lines by $KH(b, s)$. Altogether the remaining configuration has at most

$$6Dm + sm/2 + KH(b, s) \leq (49 + K/4)H(m, s)$$

two-rich points. Here $D < 8\sqrt{m}$ and $H(b, s) \leq H(m, s)/4$ were used. Taking $K > 98$, in addition to the preceding requirements, contradicts the lower bound $3KH(m, s)/4$. This completes the induction. $\square$

B.7 The planar distinct-distance estimate

Lemma B.11 (Concentration of planar Cayley lines). *For two finite planar sets U, V of sizes u, v , the uv lines (B.3) have at most $\min(u, v)$ members in a plane and at most $6u + 2v$ members in a regulus.*

Proof. Distinctness and the plane bound were proved in Corollary B.6. Fix $x \in U$ and allow the other endpoint y to range over all of $\mathbb{R}^2$. These lines partition (t, c) -space: at every point their equation uniquely determines

$$y = (I + tJ)^{-1}((I - tJ)x - 2c).$$

Their nonparallel directions and lack of intersections mean that they are pairwise skew even projectively. The polynomial vector field

$$W_x(t, c) = (1 + t^2, (tI + J)(c - x))$$

is nowhere zero over $\mathbb{R}$ and tangent to this family. Indeed along its line through (t, c) the derivative of c with respect to t is $(tI + J)(c - x)/(1 + t^2) = -J(x + y)/2$.

Let a regulus have irreducible quadratic equation $f = 0$. Every line in this fixed- x family contained in the regulus is a common line of f and $W_x \cdot \nabla f$. The latter has degree at most three. If there are at least seven such lines, Bézout forces

$$W_x \cdot \nabla f = hf$$

for a polynomial h . Every member of the fixed- x family through a real point of the regulus is then entirely contained in it. To verify this without an appeal to invariance of flows, restrict to that line and write $g(t) = f(t, c(t))$. The identity becomes

$$(1 + t^2)g'(t) = h(t, c(t))g(t).$$

A nonzero polynomial g cannot have a real root: at a root its derivative has strictly smaller vanishing order, while $1 + t^2$ does not vanish. The chosen line has such a root, so g is identically zero.

Thus this fixed- x family supplies a full real affine ruling of the regulus. Its lines are pairwise projectively skew, so they all belong to the same ruling. Conversely, coverage of every real affine point implies that no real affine line of that ruling is missing. A ruling line contained in the plane at infinity is irrelevant here. Different fixed- x families share no line, because the slope and intercept recover both endpoints. There are only two rulings, so at most two values of x contribute seven or more lines. They contribute at most $2v$ lines in all; the others contribute at most $6u$. This proves the claimed cap. $\square$

Theorem B.12 (Planar distinct distances). *Every finite set $Q \subset \mathbb{R}^2$ of $n \geq 2$ points determines at least $cn/\log(2n)$ distinct positive distances.*

Proof. Two distinct ordered matches determine a unique orientation-preserving planar isometry whenever the matched segments have the same positive length. There are consequently finitely many direct isometries with at least two matches from Q to itself. Rotate a second copy of Q by a generic rotation R_0 to form $Q' = R_0Q$. We may ensure that none of the direct isometries with two matches from Q to Q' has linear part $-I$: each is R_0g for one of the finitely many just described, and each excludes only one rotation R_0 . All these motions therefore lie in the Cayley chart used in (B.3). Translations lie in this chart as well.

Use the n^2 lines for the pairs $(x, y) \in Q \times Q'$. Lemma B.11 gives plane cap n and regulus cap $8n$. Theorem B.10 bounds the two-rich points by Cn^3 . For $3 \leq k \leq n$, Theorem B.5 bounds the k -rich points by

$$C \left(\frac{n^3}{k^2} + \frac{n^3}{k^3} + \frac{n^2}{k} \right) \leq C' \frac{n^3}{k^2}.$$

Multiplicity equals the number of matches, by the bijective Cayley representation, and it is at most n .

For each positive distance ρ let a_ρ be the number of ordered pairs of distinct points of Q at distance ρ . The corresponding number for Q' is also a_ρ . Every ordered pair of equally long nonzero segments, one from each cloud, gives the unique direct isometry described above. A motion with k matches contributes exactly $k(k-1)$ such pairs. Splitting multiplicities into intervals $[2^j, 2^{j+1})$, $j \geq 1$, and using the preceding bounds yields

$$\sum_{\rho} a_{\rho}^2 = \sum_g k_g(k_g - 1) \leq Cn^3 \log(2n).$$

Since $\sum_{\rho} a_{\rho} = n(n-1)$, Cauchy-Schwarz gives

$$|\Delta(Q)| \geq \frac{n^2(n-1)^2}{Cn^3 \log(2n)} \geq c \frac{n}{\log(2n)}.$$

□

Corollary B.13 (Planar occupancy). *If a finite set in Euclidean space determines at most $M \geq 1$ positive distances, then any affine two-plane contains at most $CM \log(2M)$ of its points. In particular, along a sequence with $B \rightarrow \infty$ and $M = o(B^2)$, every two-plane has at most $B^{2+o(1)}$ points, uniformly in the plane.*

Proof. For an occupancy $n \geq 2$, Theorem B.12 gives $n \leq CM \log(2n)$. The elementary inequality $\log(2n) \leq C\sqrt{n}$ first yields $n \leq C'M^2$; substitute this back into the logarithm to get $n \leq C''M \log(2M)$. Occupancies zero and one are harmless. If $M = o(B^2)$, eventually $M \leq B^2$, so the uniform bound is $O(B^2 \log(2B))$, which is $B^{2+o(1)}$. □

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